Tunable VUV Photoionization and Slow-Light Enhanced Third-Harmonic Generation in TiO_2 for Selective VOC Sensing

PhD Thesis

to obtain the degree of

Doktor der Naturwissenschaften (Dr. rer. nat.)

Author:

Diogo Poeta

Supervisors:

Prof. Dr. Thomas Huser
Dr. Pieter A. A. De Beule

Department of Physics

Universität Bielefeld

August 2026

Referees:
Prof. Dr. Thomas Dahm
Prof. Dr. Walter Pfeiffer

Abstract

Indoor air-quality (IAQ) monitoring needs fast, selective detection of volatile organic compounds (VOCs). Conventional photoionization detectors (PIDs) cannot provide it: their vacuum-ultraviolet (VUV) discharge lamps emit at fixed wavelengths, which limits both sensitivity and molecular selectivity. This thesis develops nonlinear-optics routes to compact, wavelength-tunable VUV sources and integrates them into photoionization sensing.

A table-top femtosecond source was built from an optical parametric amplifier (OPA), second-harmonic generation (SHG) in bulk β-barium borate and third-harmonic generation (THG) in suspended SiO_2 nanomembranes. It produces quasi-monochromatic VUV light tunable across 120–180 nm, with generation demonstrated experimentally at 130 nm and at 120 nm. Coupled to a custom photoionization detector, the source ionized representative VOCs at ambient saturated vapor pressure (SVP). The same source was then used for sample ionization potential spectroscopy (SIPS), which resolved individual BTEX compounds in a complex VOC background at concentrations set by occupational exposure limits. To raise the nonlinear conversion efficiency, the work turns to dielectric TiO_2. TiO_2 films grown by atomic layer deposition (ALD) were first characterized as a dielectric medium. The thesis then identifies energy velocity, the ratio of cycle-averaged Poynting flux to electromagnetic energy density, as the slow-light metric that stays well-defined in dispersive, weakly absorbing TiO_2, where group velocity fails; in ozone-grown ALD films, a strongly reduced energy velocity correlates with the measured rise in THG. Finally, metasurfaces supporting non-radiating anapole resonances were designed and analyzed by electromagnetic simulation; their fabrication is left for future work. These results connect nonlinear nanophotonics, slow-light design, and photoionization sensing, and point toward compact VUV gas sensors for real-time monitoring.

Contents

Abstract **iii**

1 Introduction **1**

2 Theoretical background **4**

2.1 Introduction to IAQ monitoring 4

2.2 Photoionization detection fundamentals 5

2.2.1 Photoionization process and ionization threshold 5

2.2.2 PID architecture and signal formation 6

2.2.3 Lamp emission lines, window cutoffs, and practical selectivity . . 6

2.2.4 Limitations of fixed energy lamps and motivation for tunable VUV 7

2.2.5 Photodissociation in MS/MS (context; not used in PIDs) 8

2.3 Motivation for tunable VUV photoionization 8

2.3.1 VUV source classes and practical suitability 9

2.3.2 Approach of this project: nonlinear frequency conversion 9

2.3.3 nonlinear optics considerations in the VUV 9

2.3.4 Resonant thin films for enhanced THG in compact geometries . . 11

2.3.5 Alternative approaches not pursued 11

2.4 Nanophotonic field enhancement 12

2.5 Dielectric metasurfaces 13

2.5.1 Scope of present work (limitations of plasmonic approach) 15

2.6 ALD TiO_2 for nanophotonic materials and future nanostructuring 15

2.6.1 Why ALD is particularly useful for TiO_2 nanophotonics 15

2.6.2 TiO_2 ALD oxidizing agents and implications for film quality . . . 15

2.6.3 Cycle timing: pulse, exposure and purge 16

2.6.4 Substrate and nucleation compatibility 16

2.6.5 Target properties for UV nano-structuring 16

2.6.6 Defect chemistry and "Black-TiO_2" 17

2.6.7 Alternative TiO_2 deposition methods not pursued in this work . . 17

2.7 Slow-light metric for dispersive TiO_2 17

2.7.1 Why group velocity can be misleading in lossy or strongly dispersive media 17

2.7.2 Energy velocity, Brillouin energy density 18

2.8 Anapole and other high-Q non-radiating modes for field enhancement . . 19

2.8.1 Historical context and terminology 21

3 Tunable VUV Source for VOC Photoionization **23**
3.1 Publication context and author contributions . . . 23
3.2 Experimental platform . . . 23
3.2.1 Suspended SiO_2 nanomembrane device . . . 23
3.2.2 Device fabrication . . . 25
3.2.3 Ionization chamber and gas handling . . . 26
3.2.4 Photo-current acquisition chain . . . 27
3.3 Third-harmonic generation in suspended SiO_2 nanomembranes . . . 29
3.3.1 Analytical model . . . 29
3.3.2 FDTD verification . . . 30
3.3.3 Experimental THG verification . . . 32
3.4 Proof-of-concept VOC photoionization . . . 33
3.4.1 Ethanolamine . . . 33
3.4.2 Eucalyptol and bounding the ionization potential . . . 34
3.4.3 Laser-induced background and the multiphoton regime . . . 35
3.5 Conclusion of the chapter . . . 36

4 Real-Time Molecular Specific BTEX Detection at TLV-TWA Concentration Levels **37**
4.1 Introduction . . . 37
4.1.1 Limits of fixed-photon-energy photoionization detection . . . 38
4.1.2 Threshold scanning and the SIPS derivative . . . 38
4.1.3 Two-wavelength switching as a finite-difference approximation . . . 39
4.2 Second-generation experimental system . . . 40
4.3 SIPS acquisition protocol . . . 41
4.4 Results . . . 42
4.4.1 BTEX SIPS traces . . . 43
4.4.2 Reconstructed spectrograph and per-compound calibration . . . 43
4.5 Discussion and limitations . . . 45
4.6 Conclusion of the chapter . . . 46

5 Material Study of ALD-grown TiO_2 Thin Films for UV Nanophotonics **47**
5.1 Publication context and author contributions . . . 47
5.2 Scope of the chapter . . . 47
5.3 Thin-film growth . . . 47
5.3.1 ALD process and oxidant selection . . . 47
5.3.2 Growth calibration and thickness control . . . 49
5.4 Optical characterization . . . 49
5.4.1 Ellipsometric modelling and extraction of optical constants . . . 49

5.4.2 Depolarization and near-UV absorption . . . 50
5.4.3 UV–Vis–NIR reflectance . . . 51
5.5 Surface and structural characterization . . . 52
5.5.1 AFM morphology and roughness . . . 52
5.5.2 XRD and partial anatase crystallization . . . 54
5.6 Chemical and electronic characterization . . . 55
5.6.1 XPS stoichiometry and defect states . . . 55
5.6.2 REELS and UPS analysis of bandgap and defect features . . . 57
5.7 Conclusions and implications for Chapter 6 . . . 58

6 Energy Velocity and Third-Harmonic Generation in UV TiO_2 Thin-Films 60
6.1 Publication context and author contributions . . . 60
6.2 Introduction . . . 60
6.3 Methods . . . 63
6.3.1 Energy velocity . . . 63
6.3.2 Third-harmonic generation . . . 67
6.4 Results . . . 67
6.4.1 Complementary simulations . . . 71
6.4.2 Third-harmonic response . . . 72
6.5 Discussion . . . 74
6.6 Conclusion . . . 76

7 Design and Simulation of a TiO_2 Anapole Metasurface 77
7.1 Publication context and author contributions . . . 77
7.2 Introduction . . . 77
7.3 In-depth analysis of the resonance profile . . . 79
7.4 Material selection and geometry definition . . . 80
7.5 Effect of the incident angle on the formation of the anapole state . . . 82
7.6 Temporal evolution of the Anapole State in the UV Spectrum . . . 83
7.7 Geometry variation and multipole analysis . . . 85
7.8 Comparison between nano-rectangle and nano-ellipsoid . . . 87
7.9 Metasurface . . . 88
7.10 Conclusions . . . 92

8 Conclusions and Discussion 94
8.1 Summary of main findings . . . 94
8.2 Comparative discussion of the two approaches . . . 95
8.3 Outlook and future work . . . 96

Bibliography **99**

A List of Publications **122**

B Conference Contributions **123**

C Acronyms **124**

D Materials and Methods **126**

D.1 Materials and methods 126

D.1.1 Photoionization detection platform 126

D.2 Simulations 131

D.2.1 Simulation setup 131

D.2.2 Field behavior in metallic and dielectric nanorectangles 131

D.2.3 Anatase and amorphous TiO_2 137

E Statutory Declaration **140**

F Acknowledgements **141**

List of Figures

1.1 Concept for real-time air-toxicity monitoring using VUV photoionization. 2

2.1 Lamp PID schematic (VUV lamp → ionization chamber → transimpedance amplifier) and TVOC voltage readout. 6
2.2 VUV approach summary. 10

3.1 Schematic of the experimental setup. A femtosecond Yb laser pumps an OPA, the output is frequency-doubled in a BBO crystal, and the resulting near-UV beam is focused onto a suspended SiO_2 nanomembrane integrated within an ionization chamber. THG in the membrane generates VUV light that ionizes VOC molecules delivered by the gas sampling unit. The photo-ionization current is collected on AlSiCu electrodes patterned on the membrane device, amplified by an in-house transimpedance amplifier, and acquired through a NI DAQ under LabVIEW control. Dashed lines correspond to electrical connections; arrows indicate gas flow. Adapted from [1]. 24
3.2 Schematic of the device cross-section and process flow for the suspended SiO_2 nanomembranes: (1) PECVD deposition of 3 µm SiO_2 hard mask on the back side; (2) photoresist patterning by DWL lithography; (3) ICP–RIE opening of the hard mask; (4) spin-coat and cure of a sacrificial polyimide layer; (5) PECVD deposition of the 600 nm membrane SiO_2; (6) LOR/photoresist bi-layer and DWL lithography for metal-pad definition; (7) sputter deposition of 250 nm AlSiCu; (8) lift-off; (9) deep RIE silicon etch through the back side to the polyimide etch-stop; (10) plasma-ashing release of the free-standing membrane. Adapted from [1]. 26
3.3 Air-sampling system for VOC generation. A nitrogen flow is split at a tee into a carrier line and a dilution line, each metered by a mass flow controller (MFC, MFC_N_2; Aalborg). The carrier line is bubbled through the headspace of the liquid VOC to entrain its saturated vapor, recombined with the dilution line at a second tee, and delivered to the device. The valves select the active line; the delivered concentration follows from the Antoine saturated-vapor pressure and the dilution ratio. Adapted from [1]. 27

3.4 Switched-integrator transimpedance front-end. Top: the custom PCB (top and bottom faces). Bottom left: the circuit around the IVC102 integrator, with selectable feedback capacitors ($C_1 = 10\,\text{pF}$, $C_2 = 30\,\text{pF}$, $C_3 = 60\,\text{pF}$), the integrate/reset switches S_1 and S_2, and the $\pm 15\,\text{V}$ supply with $0.1\,\mu\text{F}$ decoupling; the input current I_{IN} is integrated onto the selected capacitor and read out as V_O. Bottom right: the board mounted on the ionization chamber. The front-end was validated against an external current source (Keithley 2400) over the pA–nA range. Adapted from [1]. 28

3.5 Analytical THG intensity from Eqs. (3.3) and (3.4). **(Left)** $I^{3\omega}$ as a function of membrane thickness l and pump wavelength λ_ω. **(Centre)** 3-D view of the same data, with one trace per simulated pump wavelength in 340–440 nm. **(Right)** Coherence length L_c as a function of pump wavelength (data points) and Cauchy-style fit (dashed line). The red dashed line in the left panel marks the 600 nm operating thickness used in this work. . . 30

3.6 FDTD verification of the THG intensity model. **(Top)** Simulation domain: (1) light source, (2) FDTD region, (3) planar nanomembrane, (4) field monitor, (5) refractive-index monitor. **(Bottom)** $I^{3\omega}$ versus membrane thickness at six pump wavelengths in 340–440 nm. Solid orange curves: analytical expression Eq. (3.3); scattered blue points: FDTD simulation. Adapted from [1]. 31

3.7 Near-UV THG verification on the 600 nm-thick SiO_2 nanomembranes. **(a)** Alignment setup. **(b)** CMOS image of the focused laser spot centred on a single suspended membrane. **(c)** Optical layout used for spectrally resolved THG detection. **(d)** THG emission spectra recorded for pump wavelengths from 1190 nm to 1380 nm in 10 nm steps. Adapted from [1]. 32

3.8 Photoionization signals in the in-house ionization chamber. **(a)** Four-step acquisition protocol: laser OFF / laser ON / VOC ON / VOC OFF. **(b,c)** Chemical structures of ethanolamine and eucalyptol. **(d,e)** Current variation for the laser background (blue) and with VOC present (black): **(d)** 178 ppm ethanolamine at $\lambda_\omega = 390\,\text{nm}$ ($\lambda_{3\omega} = 130\,\text{nm}$, 9.5 eV); **(e)** 711 ppm eucalyptol at $\lambda_\omega = 360\,\text{nm}$ ($\lambda_{3\omega} = 120\,\text{nm}$, 10.3 eV). **(f)** Concentration linearity for ethanolamine over 94–234 ppm. **(g)** Eucalyptol response at $\lambda_\omega = 390\,\text{nm}$ (no photoionization) and $\lambda_\omega = 430\,\text{nm}$ (no photoionization), bracketing the eucalyptol ionization threshold. Adapted from [1]. 34

4.1 Traditional lamp-based PID compared with the SIPS PID. The traditional architecture uses a fixed-energy Kr/Xe lamp and reports the total photoionization current: compounds with $E_{IP} < h\nu$ all contribute, with no molecular selectivity. The SIPS architecture replaces the lamp with the femtosecond-laser + OPA system of Chapter 3, producing quasi-monochromatic VUV that can be tuned across molecular ionization thresholds, enabling molecular-specific detection. Adapted from [2]. 38
4.2 SIPS principle. **(a)** The tunable VUV source delivers quasi-monochromatic photons that are scanned (decreasing λ, equivalently increasing E_{ph}) across the VUV range 160–105 nm, covering most VOC ionization potentials (7.75–11.80 eV). **(b)** For a multi-component mixture, the photoionization current rises monotonically with E_{ph}, each compound contributing as E_{ph} crosses its $E_{IP,i}$. **(c)** The derivative $\partial I/\partial E_{ph}$ produces peaks at each $E_{IP,i}$, with amplitude proportional to $r_i C_i$, providing a molecular spectrograph of the mixture. Adapted from [2]. 39
4.3 Sensor integration. (a) Membrane device: SiO_2 suspended nanomembranes (purple), metal electrodes (grey), and the silicon substrate (green); the focused laser is aligned to the centre of one nanomembrane. (b) Transimpedance board (left) and the small carrier PCB that contacts the membrane with silver paste and is soldered to the transimpedance board (right). (c) Assembly of the ionization chamber, electronics, and membrane device. Adapted from [2]. 41
4.4 Gas delivery for the BTEX measurements: the BTEX mixture is supplied from a certified gas cylinder, regulated through a shut-off valve and a mass flow controller (MFC), and delivered to the SIPS prototype. Adapted from [2]. 41
4.5 Acquisition timing for one SIPS measurement. The laser is switched on after a N_2 purge; the BTEX mixture is admitted (VOCs ON); the OPA wavelength is alternated twice between $\lambda_{\mathrm{VUV-H}}$ (blue band) and $\lambda_{\mathrm{VUV-L}}$ (orange band) bracketing the ionization potential of the target compound; the laser is switched off (Laser OFF) at the end of the sequence. The benzene measurement uses doubled hold times because of the lower target concentration. Adapted from [2]. 42
4.6 SIPS current traces for the five BTEX compounds at the wavelength pairs listed in Table 4.1. Blue bands: $\lambda_{\mathrm{VUV-H}}$; orange bands: $\lambda_{\mathrm{VUV-L}}$. Top row: (a) benzene at 0.5 ppm, (b) toluene at 50 ppm. Bottom row: (c) ethylbenzene at 20 ppm, (d) o-xylene at 50 ppm, (e) p-xylene at 50 ppm. In each panel a clear SIPS step is observed when the wavelength crosses the corresponding ionization potential. Adapted from [2]. 44

4.7 Reconstructed sample ionization potential spectrograph for the BTEX mixture. **(a)** Cumulative photoionization current as a function of photon energy, obtained from the SIPS steps of Fig. 4.6. **(b)** Derivative $\delta I/\delta E_{ph}$ at each BTEX ionization potential; bar colours indicate the contributing compound, and labels above each bar give the mixture concentration. Adapted from [2]. 44

5.1 Photographs of ALD-grown TiO_2 coatings deposited using different oxidants and cycle numbers: H_2O (1250 cycles), O_3 (1250 cycles), H_2O_2 (1250 cycles), and O_3 (2135 cycles), from left to right. 48

5.2 Depolarization percentage as a function of wavelength at different incident angles for TiO_2 films grown with **a)** O_3-grown (2135 ALD cycles) and **b)** H_2O_2-grown (1250 cycles). 51

5.3 UV–Vis–NIR specular reflectance measurement of ALD-grown TiO_2 films. **a)** Comparison of films deposited with different oxidants, all measured at a 45° incidence angle. The O_3-grown film (2135 cycles, orange) shows reduced reflectance across the visible range compared to the O_3-grown film (1250 cycles, green), H_2O_2-grown film (1250 cycles, blue), and H_2O-grown film (1250 cycles, yellow). **b)** Angle-dependent reflectance of the O_3-grown TiO_2 film (2135 cycles, ≈ 98 nm thick) measured at 30° (yellow), 45° (blue), and 60° (green). 52

5.4 AFM representative topography images of TiO_2 films deposited via ALD using different oxidizing agents. **a)** H_2O-grown (1250 cycles, thickness ≈ 42 nm), **b)** H_2O_2-grown (1250 cycles, thickness ≈ 53 nm), **c)** O_3-grown (1250 cycles, thickness ≈ 59 nm), and **d)** O_3-grown (2135 cycles, thickness ≈ 98 nm). The color bars indicate height variations across each scan. . . 53

5.5 (**Left**) XRD patterns of TiO_2 thin films grown using different oxidizing agents via ALD. Samples include O_3-grown films with 2135 cycles (pink) and 1250 cycles (green), H_2O_2-grown (yellow), H_2O-grown (orange), and black-TiO_2 (blue). (**Right**) Magnified view of the (101) anatase reflection near 25°, highlighting differences in peak intensity and crystallinity among the samples. 55

5.6 High-resolution XPS spectra of TiO_2 films grown with different oxidants. **a)** Ti *2p* region for O_3-grown (≈ 98 nm, orange), O_3-grown (≈ 59 nm, yellow), H_2O_2-grown (≈ 53 nm, blue), H_2O-grown (≈ 42 nm, green), and black TiO_2 (red) films. **b)** O *1s* spectra of the same films reveal differences in the oxygen bonding states. 56

6.1 Experimental setup for third-harmonic measurements. A tunable infrared source (1110–1350 nm) is directed by protected-silver mirrors ($E1$,$E2$) onto a reflective neutral density filter (M),then focused by an uncoated N-BK7 lens (L) onto the sample on an XYZ stage. The backscattered third-harmonic is collected by (L), spectrally filtered by a bandpass filter (BP), coupled through an Olympus 4× objective (O), and recorded by an AvaSpec ULS2048×64 EVO spectrometer. 66

6.2 **a)** Simulation layout for the 2D FDTD model (Meep): a 100 nm TiO_2 film is centered at $y = 0$, surrounded by air and terminated by PMLs. A TE_z line source in the lower air region launches a normally-incident plane wave in the $+y$ direction. **b)** Energy velocity map and streamlines at $\lambda_0 = 380$ nm. Colors show $|v_e|/c$ computed as $v_e = \langle \mathrm{S} \rangle / \langle u \rangle$. Streamlines indicate energy flowing upward from the source through the slab into the upper half space. The reduced $|v_e|/c$ near the left and right edges of the map corresponds to the 1 µm absorbing PML at the lateral boundaries and is not physically significant. 69

6.3 Normalized velocities for 100 nm slab under TE_z excitation at normal incidence, evaluated at $\lambda = 380$ nm, 400 nm and 800 nm. The three panels show (top) phase velocity v_p/c, (middle) group velocity v_g/c, and (bottom) energy velocity v_e/c. The horizontal axis is wavelength λ (nm) and the vertical axis is the corresponding normalized velocity. Bar colour and hatching indicate the material: solid blue = TiO_2, hatched blue = Si, and orange cross = SiO_2. Numeric labels give the values of v/c. 70

6.4 **a)** Energy velocity map and streamlines $\lambda_0 = 800$ nm. Colors show $|v_e|/c$ computed as $v_e = \langle \mathrm{S} \rangle / \langle u \rangle$. Streamlines indicate energy flowing upward from the source through the slab into the upper half space. 71

6.5 Energy velocity maps at $\lambda_0 = 380$ nm. Maps show $|v_e|/c$ (color) and energy-flow streamlines ($\langle \mathbf{S} \rangle$) computed from Fourier-domain fields; $v_e = \langle \mathbf{S} \rangle / \langle u \rangle$ with Brillouin energy density u. **a)** Air (reference): uniform $|v_e|/c \approx 1$. **b)** SiO_2 slab (100 nm thick) in air. 72

6.6 Fitted ellipsometry data for the O_3-grown TiO_2 film. **(a)** Real (n) (in blue) and imaginary (k) (in red) parts of the refractive index. 73

6.7 Reflected third-harmonic spectra recorded from bare silicon and from TiO_2 coated samples at $\lambda_{3\omega} = 380$ nm and 400 nm. **a)** Spectra centered at 380 nm. **b)** Spectra centered at 400 nm. The comparison includes TiO_2 films grown with H_2O and H_2O_2 at 1250 cycles and with O_3 at 1250 and 2135 cycles. 74

7.1 Illustration of some geometries that are proven of exciting anapoles for wavelengths between 550–2265 nm. Our goal is to excite anapole states in the UV regime. 78
7.2 **a)** Dispersion profile of the amorphous TiO_2 refractive index (n) and extinction coefficient (k). **b)** Schematic of a free-standing TiO_2 nano-rectangle with dimensions: length L = 190 nm, width w = 160 nm, and height h = 100 nm, showing light propagation direction ($\vec{k}$) and polarization direction ($\vec{E}$) directions. **c)** Contributions of various multipole moments: C_p (blue), C_T (red), C_m (green), C_{Qe} (yellow), C_{Qm} (cyan blue) and total contribution C_{total} (black). **d)** Phase difference between the electric and toroidal dipoles (allusive to Eq.7.1). 79
7.3 **a)** Electric and **b)** magnetic field intensities in the free-standing nano-rectangle structure. **c)** Shift of the anapole wavelength when varying the width (w) of the nano-rectangle. The electric and magnetic fields are normalized relative to the source. 81
7.4 Simulated scattering cross-section for several angles of incidence θ. . . . 83
7.5 Temporal evolution of the electric and magnetic fields during the excitation of the anapole state in the UV spectrum. 84
7.6 This plot illustrates the radiated power as a function of time during the transient phase of the anapole state. 85
7.7 Contributions of various multipole moments: C_p (blue), C_T (red), C_m (green), C_{Qe} (yellow), C_{Qm} (cyan blue) and C_{total} (black) for different geometries that support anapole resonances within the UV-A region, using amorphous TiO_2: **a)** Nano-hexagon, **b)** Nano-octagon, **c)** Nano-decagon, **d)** Nano-ellipsoid. The arrow indicates the corresponding anapole state in λ (nm). 86
7.8 Phase difference between the electric and toroidal dipole moments for different geometries that support anapole resonances within the UV-A region, using amorphous TiO_2: **a)** Nano-hexagon,**b)** Nano-octagon, **c)** Nano-decagon, **d)** Nano-ellipsoid. The arrow indicates the anapole state. 87
7.9 Simulation results of the anapole resonance for different nano-structures, illustrating how varying dimensions affect the resonance behavior. The wavelengths corresponding to the observed resonances are 353.8 nm, 360.2 nm, 353.7 nm, and 353.9 nm for nano-hexagon **a)**, nano-octagon **b)**, nano-decagon **c)** and nano-ellipsoid **d)** (with enlarged dimensions), respectively. 88

7.10 **a)** The transmittance spectrum shows a dip at 370 nm (indicated with an arrow), corresponding to the anapole state for a period of 733 nm (blue curve), indicating strong localization of the electromagnetic field. **b)** and **c)** Field intensity maps showing the electric and magnetic fields respectively, at the resonant anapole condition (P = 733 nm). The electric and magnetic fields are normalized relative to the source. 89
7.11 (**Top**) This image compares the sizes of a nano-ellipsoid and a nano-rectangle. The nano-rectangle, with dimensions L = 190 nm, w = 160 nm, and h = 100 nm, supports the anapole state. The nano-ellipsoid is shown in dark blue and has similar dimensions but does not support the anapole state. (**Bottom**) By enlarging the nano-ellipsoid to dimensions L = 200 nm, w = 180 nm, and h = 130 nm, the anapole state is achieved. 90
7.12 Metasurface of TiO_2 nano-rectangles deposited atop a fused silica substrate, illuminated by UV light incident perpendicular to the metasurface plane. 91
7.13 **a)** and **b)** Field intensity maps showing the electric and magnetic fields respectively, at the resonant anapole condition (P = 733 nm) in the x-z plane cross-section of the nano-rectangle. 91
7.14 Transmittance spectrum for other periods: **a)** 500 nm, **b)** 600 nm, **c)** 700 nm and **d)** 800 nm. 92

D.1 Krypton photoionization lamp (typical emission lines near 10.0 and 10.6 eV) in our designed chamber. 128
D.2 2D top view illustrating the interaction of light with the rectangular nanostructure. The plane wave propagates with the wave vector (**k**) along the z-axis (normal to the surface). The electric field (**E**) is aligned parallel to the rectangle's length (x-axis), while the magnetic field (**H**) is aligned parallel to its width (y-axis). Both (**E**) and (**H**) are orthogonal to the propagation vector (**k**). 132
D.3 Dispersion profile of **a)** Ag, **b)** Au, **c)** low index TiO_2 and **d)** amorphous TiO_2 [3] refractive indices (n) and extinctions (k). 133
D.4 Electric field intensity distribution at 370 nm λ. **a)** Ag, **b)** Au, **c)** low index TiO_2, **d)** TiO_2 amorphous. The monitor is positioned 5 nm from the nanorectangle surface, and the fields are simulated under plane wave excitation with the electric field (**E**) aligned along the rectangle's length. 134
D.5 Electric field intensity distribution at 450 nm λ. **a)** Ag, **b)** Au, **c)** low index TiO_2, **d)** TiO_2 amorphous. The monitor is positioned 5 nm from the nanorectangle surface, and the fields are simulated under plane wave excitation with the electric field (**E**) aligned along the rectangle's length. 134

D.6 Electric field intensity distribution at 520 nm λ. **a)** Ag, **b)** Au, **c)** low index TiO_2, **d)** TiO_2 amorphous. The monitor is positioned 5 nm from the nanorectangle surface, and the fields are simulated under plane wave excitation with the electric field (**E**) aligned along the rectangle's length. 135
D.7 Electric field intensity distribution at 370 nm λ. **a)** Ag, **b)** Au, **c)** low index TiO_2, **d)** TiO_2 amorphous. The monitor is positioned at the geometric center of the nanorectangle. The fields are simulated under plane wave excitation with the electric field (**E**) aligned along the rectangle's length. 136
D.8 Electric field intensity distribution at 450 nm λ. **a)** Ag, **b)** Au, **c)** low index TiO_2, **d)** TiO_2 amorphous. The monitor is positioned at the geometric center of the nanorectangle. The fields are simulated under plane wave excitation with the electric field (**E**) aligned along the rectangle's length. 136
D.9 Electric field intensity distribution at 520 nm λ. **a)** Ag, **b)** Au, **c)** low index TiO_2, **d)** TiO_2 amorphous. The monitor is positioned at the geometric center of the nanorectangle. The fields are simulated under plane wave excitation with the electric field (**E**) aligned along the rectangle's length. 137
D.10 Comparison of optical properties of TiO_2 in anatase and amorphous phases. The refractive index n (orange) and extinction coefficient k (blue) as functions of wavelength. Solid lines represent the anatase phase, while dashed lines represent the amorphous phase. The refractive index (n) and extinction coefficient (k) are higher for the anatase phase at shorter wavelengths, while differences diminish as the wavelength increases. . . . 138
D.11 Study for anatase TiO_2. **a)** Contributions of various multipole moments: C_p (blue), C_T (red), C_m (green), C_{Qe} (yellow), C_{Qm} (cyan blue) and total contribution C_{total} (black). **b)** Phase of electric dipole moment (blue) and toroidal dipole moment (red). **c)** Electric and **d)** magnetic field intensities in the free-standing nano-rectangle structure. The electric and magnetic fields are normalized relative to the source. 138

List of Tables

2.1 Characteristics of common PID lamps used for VOC detection. 7

4.1 BTEX target compounds: ionization potentials, corresponding VUV wavelengths, the two-point switching wavelengths used in this work, and the mixture concentrations. Ionization potentials taken from [4]. 43
4.2 SIPS measurement summary for the five BTEX compounds. $\Delta I_1, \Delta I_2$: first and second SIPS steps. SIPS sensitivity S_{BTEX}: SIPS step normalised by mixture concentration. Limit of detection $\mathrm{LOD}_{\mathrm{VOC}}$ defined as in Eq. (4.4). 45

5.1 Thickness and uniformity of Al_2O_3 films deposited at different O_3 flow rates (725 μm Si substrates). 49
5.2 Ellipsometry fit parameters for the O_3-grown TiO_2 thin film (2135 cycles) using the Cauchy + Urbach absorption model. 50
5.3 Roughness parameters for TiO_2 films deposited by ALD using different oxidizing agents and cycle counts. Values are mean ± standard deviation from three measurements over a $2 \times 2\,\mu$m area. 54
5.4 XPS parameters for TiO_2 samples. Binding energies (BE) for Ti *2p3/2* and O *1s* peaks are reported. 57
5.5 High-resolution XPS peak-fit parameters for the black H_2O-grown TiO_2 film. Binding energies (BE) and FWHM are in eV, atomic percentages are derived from area analysis. 57
5.6 High-resolution XPS peak parameters for O_3-grown TiO_2 film. Binding energies (BE) and FWHM are in eV; atomic percentages are derived from area analysis. 57
5.7 Bandgap energies estimated from REELS spectra of TiO_2 films. 58

6.1 Complex refractive index $n + i\,k$ at selected wavelengths (interpolated from the provided datasets). 66
6.2 Optical constants of the ≈ 98 nm TiO_2 film grown using O_3 ($\approx 200\,^{\circ}$C), the numerical derivative of the refractive index (computed from spectroscopic ellipsometry), the resulting group index, and the estimated energy velocity. The ratio v_e/c highlights how much the energy velocity is reduced relative to c in vacuum. 68
6.3 Material characterization results for the TiO_2 films used in the third-harmonic-generation study. Film thickness values are approximate. . . . 74

D.1 Antoine parameters and SVP values for VOCs (as used in this work). . . 127

D.2 Transmission values for sapphire windows and interpolated estimate at 3 mm thickness. 128
D.3 Five “shadow compounds” selected for SVP validation (Antoine parameters, conditions, SVP, ionization potential and threshold wavelength). 130

1 Introduction

Indoor air is dynamic. In schools and other occupied buildings, pollutant concentrations can vary on time scales of minutes to hours as ventilation rates, occupancy, and routine activities such as cleaning change. Because people spend about 87 % of their time indoors [5], exposure in schools can dominate daily intake for children even when outdoor concentrations are moderate [6]. These fluctuations are particularly relevant in educational environments, where children spend a substantial fraction of the day indoors and where insufficient ventilation can coincide with elevated pollutant levels [7, 8]. Because indoor volatile organic compound (VOC) concentrations can evolve on minute-to-hour time scales, a practical indoor air quality (IAQ) instrument must combine high temporal resolution with molecular selectivity.

Monitoring VOCs, however, usually involves a trade-off between chemical specificity and temporal resolution: highly specific analysis is often performed offline in the laboratory, whereas compact sensors typically provide faster measurements with more limited selectivity. Photoionization detectors (PIDs) occupy an intermediate position. They offer high sensitivity and rapid response, but conventional instruments based on vacuum-ultraviolet (VUV) lamps provide limited chemical discrimination. Because the VUV photon energy is fixed and the detector reports a total ion-current signal, the response is an aggregate measurement whose magnitude depends on compound-specific response factors [9, 10]. This limitation is fundamentally photonic: with the photon energy fixed, the sensor cannot probe ionization thresholds, and improving molecular selectivity therefore depends less on the ion-current readout itself than on providing a compact photon source whose wavelength, and thus photon energy, can be tuned across those thresholds.

Exploiting ionization thresholds with wavelength-tunable VUV photons immediately faces two practical constraints. First, VUV light is strongly absorbed by air and many optical materials, so beam paths typically require vacuum or controlled-gas enclosures and specialized optics. Second, compact tunable VUV sources remain challenging because deep-UV and VUV frequency conversion is limited by material transparency and phase-matching requirements, and many established VUV generation approaches are experimentally complex [11, 12, 13]. This mismatch between how fast indoor air changes and how we are typically able to measure it motivates the central question of this thesis: how can a compact, real-time VOC sensor achieve high molecular selectivity using wavelength-tunable VUV photoionization, for example through threshold scanning or sample ionization potential spectroscopy (SIPS), while avoiding the fixed-photon-energy limitations of discharge lamps and minimizing vacuum infrastructure? Answering this question requires connecting the

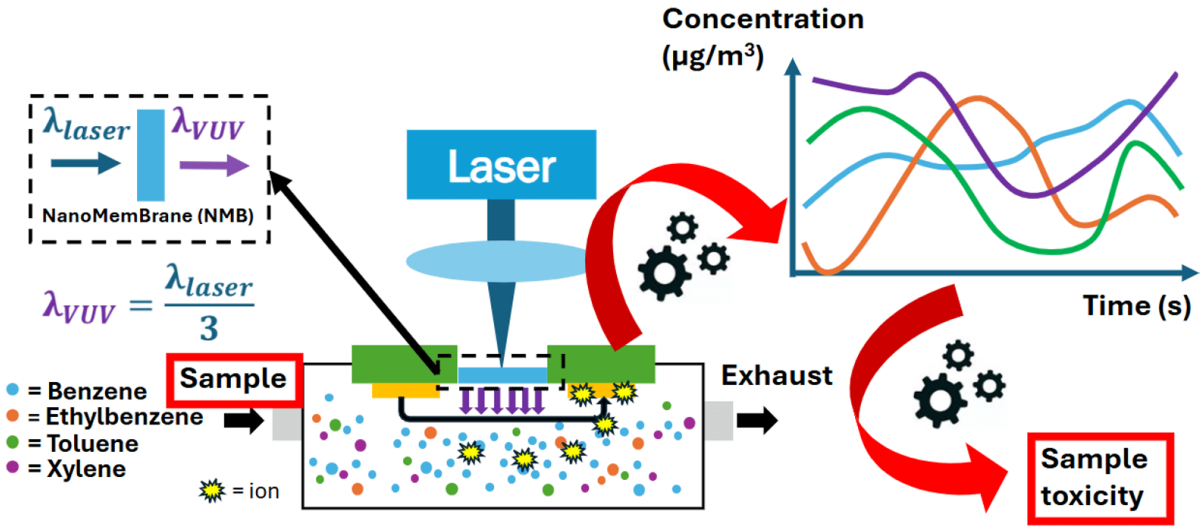


Figure 1.1: Concept for real-time air-toxicity monitoring using VUV photoionization.

constraints of IAQ and VOC monitoring constraints to the photoionization physics principles and to a practical, compact source of tunable VUV light based on nonlinear optics and nanophotonics.

In response to these limitations, this thesis develops photonic methods for more efficient generation and control of VUV radiation in compact optical layouts. A central theme is the use of resonant dielectric nanostructures, particularly anapole states, to confine electromagnetic energy and enhance nonlinear processes such as third-harmonic generation (THG), including pathways that extend into the VUV [14, 15, 16]. The work was carried out within the Horizon Europe project LEARN (grant agreement No. 101057510), which provides the application context of IAQ monitoring in schools and defines relevant performance targets [17]; the thesis itself focuses on the underlying photonics and nonlinear optics required to generate tunable VUV radiation and to apply it to photoionization-based sensing concepts.

The objective of this research is therefore to advance wavelength-tunable VUV photoionization for improved molecular selectivity in VOC sensing, pursued along two complementary directions. The first develops and validates a laser-driven frequency-conversion approach that generates quasi-monochromatic tunable VUV radiation and applies it to threshold-based photoionization and SIPS. The second studies TiO_2 thin films, nanostructures, and metasurface concepts as candidate systems for enhancing third-harmonic generation and for assessing alternatives toward more compact UV/VUV generation. Motivated by reports of enhanced third-harmonic generation in dielectric resonators operating in anapole regimes [18, 19, 20, 21], the thesis presents the electromagnetic design and numerical analysis of a TiO_2 metasurface supporting an anapole resonance in the UV-A,

estimating the achievable field enhancement and third-harmonic response and asking in particular whether a simpler geometry can perform comparably to more demanding shapes. The scope is thus focused on nonlinear optics strategies for UV/VUV generation, their coupling to photoionization sensing and selectivity concepts, and field enhancement in TiO_2 enabled by resonant and dispersive effects, including weakly radiating states such as anapole excitations and the slow-light framework developed later in the thesis. Feasibility is established through simulation, thin-film characterization, and proof-of-concept experimental validation; for the metasurface approach the work is limited to design and numerical analysis, with fabrication left for future implementation.

This dissertation is cumulative and includes peer-reviewed publications. It is structured around four main contributions. The first is anapole field confinement in TiO_2: the design and numerical analysis of UV-A anapole resonances in TiO_2 nanostructures, including the dependence of the resonance wavelength on geometry and its implications for wavelength targeting and nonlinear enhancement (main author; *Optics Letters*, 2024, DOI:10.1364/OL.536925). The second is table-top tunable VUV generation by nonlinear frequency conversion: the development and characterization of a laser system for generating quasi-monochromatic vacuum-ultraviolet radiation, tunable down to the deep-VUV range, and its integration with photoionization detection (co-author; *Optics Express*, 2025, DOI:10.1364/OE.564577). The third concerns selectivity concepts for real-time sensing: the application of tunable photoionization to threshold-based selectivity and SIPS for molecular-specific detection in complex VOC backgrounds (co-author). The fourth addresses slow-light and energy-transport metrics in TiO_2: the validation of energy velocity as a physically appropriate slow-light metric in dispersive, weakly absorbing TiO_2, and its connection to enhanced nonlinear response (main author). These contributions are motivated by the need for fast, selective VOC sensing in compact instrumentation, and they examine both VUV source development and nanophotonic structures as complementary routes toward tunable photoionization. Fabrication tolerances and material losses are discussed throughout as practical constraints on future implementation.

2 Theoretical background

2.1 Introduction to IAQ monitoring

Volatile organic compounds (VOCs), a primary category of indoor air contaminants, are emitted by building materials, cleaning agents, and outdoor pollutants that enter indoor environments. VOC exposure has been associated with many health issues, including respiratory conditions, allergies, and potential long-term effects on development in children [22, 23, 24].

Early evidence that building materials can contribute to indoor contaminant exposures, such as aldehydes, is provided by Andersen et al. (1975), who reported elevated formaldehyde in Danish dwellings where chipboard was used as a construction material [25].

Historically, indoor VOCs were predominantly quantified using offline, time-integrated sampling (hours to weeks) based on active multi-sorbent tubes or passive diffusive samplers, followed by laboratory thermal desorption gas chromatography–mass spectrometry or canister gas chromatography–mass spectrometry analysis, as demonstrated in large residential and office surveys and exposure campaigns [26, 27, 28, 29]. While these approaches provide accurate compound identification and quantification, they typically require sample collection, storage and transport, laboratory analysis, and post-processing.

As a result, offline workflows often lead to delays of days to weeks before results are available.

Because offline methods are time integrated, they report an average over the sampling window, which can attenuate brief high emission events. For example, activities such as cleaning or disinfection may create sharp VOC peaks that are highly relevant for exposure assessment during occupancy, yet these peaks can be attenuated in long averaging periods [30].

These limitations have motivated a shift toward *online* (real time) VOC monitoring approaches that provide frequent measurements and allow for a stronger connection between concentration changes and activities, ventilation conditions, and sources. Early "online" VOC monitoring appeared through automated preconcentration gas chromatography (GC) systems capable of repeated field measurements without continuous operator intervention [31].

More recently, advanced mass spectrometry techniques have enabled real-time VOC

monitoring with seconds to minutes temporal resolution, supporting source attribution and exposure assessment for brief emission episodes [32].

Because VOC mixtures are chemically diverse and measurement methods are inherently selective, no single approach can quantify all relevant VOCs with equal accuracy; consequently, hybrid strategies increasingly combine continuous monitoring with targeted speciation.

The quantitative performance of low-cost VOC sensors is limited by non-specific responses to multiple chemical species (cross-sensitivity), dependence of the signal on ambient humidity and temperature, and slow changes in sensor response over time (long-term drift). Field calibration methods have been developed to compensate for these effects, and support scalable monitoring in schools [33].

2.2 Photoionization detection fundamentals

This thesis focuses on photoionization for VOC sensing with PIDs. Photodissociation is introduced briefly at the end of this section as context for interactions between photons and ions in tandem (two-stage) mass spectrometry, but it is not the focus of the present thesis.

2.2.1 Photoionization process and ionization threshold

Photoionization is the formation of gas-phase ions following absorption of a photon [34]. In PID operation, vacuum ultraviolet (VUV) photons ionize molecules with ionization energy (IE) lower than the photon energy delivered to the sample [35].

In its simplest form, the process is written as:

$$\mathrm{VOC} + h\nu \rightarrow \mathrm{VOC}^{*} \rightarrow \mathrm{VOC}^{+} + e^{-} \tag{2.1}$$

where $h\nu$ is the photon energy, VOC^{*} is an electronically excited neutral, VOC^{+} is the radical cation, and e^{-} is the photoelectron. In Eq. 2.1 an absorbed photon excites an electron and, if the energy is sufficient, ejects it from the molecule, producing a radical cation and a free electron (see Ch. 2, p. 16 in Ref. [34]).

The ionization energy (ionization potential) is the minimum photon energy required to remove an electron from a molecule. In practical PID sensing, selectivity follows a threshold condition: compounds with $\mathrm{IE} < h\nu$ are ionized and contribute to the signal, while those with $\mathrm{IE} > h\nu$ do not. Lower energy lamps improve selectivity by excluding

higher-IE interferents, whereas higher energy lamps extend the detectable set of VOCs but typically offer lower output and shorter operating life [34].

2.2.2 PID architecture and signal formation

A PID uses a VUV lamp to illuminate an ionization cell through which the sampled gas flows (Fig. D.1). Within the concentration range where the response is approximately linear the collected ion current is approximately proportional to the analyte concentration. PIDs are usually calibrated to isobutylene. To report concentrations for other VOCs, the PID reading is corrected using a VOC-specific response factor, defined as the ratio of the PID sensitivity for isobutylene to its sensitivity for the target compound [36].

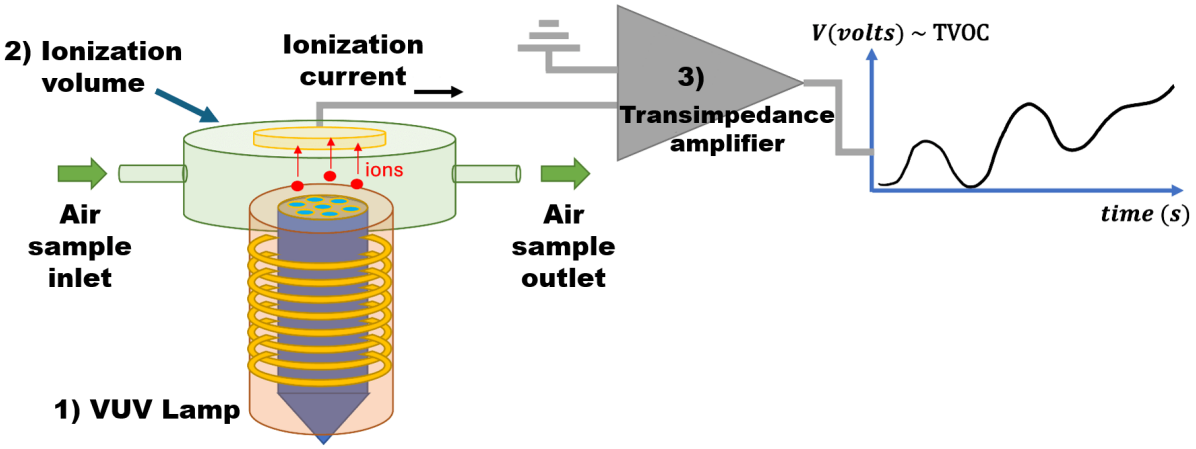


Figure 2.1: Lamp PID schematic (VUV lamp → ionization chamber → transimpedance amplifier) and TVOC voltage readout.

The intrinsic photoionization step is instantaneous; practical response time (typically a few to several seconds) is controlled primarily by chamber flow and sampling path effects (residual volume and adsorption/desorption), which can strongly slow response for low volatility compounds [34].

2.2.3 Lamp emission lines, window cutoffs, and practical selectivity

In commercial PIDs, the available photon energies are set primarily by the lamp fill gas (typically Xenon, Krypton, or Argon) and by the VUV transmitting window material. Table 2.1 summarizes the commonly used lamp types and their characteristic photon energies and wavelengths. As a practical example of selectivity, formaldehyde requires photon energies above 10.6 eV and therefore cannot be ionized with xenon or krypton lamps, but can be detected with an argon lamp emitting near 11.6 eV [34].

Window transmission sets a short wavelength cutoff that can suppress higher energy lamp lines. As a result, the maximum photon energy delivered to the sample is determined by the lamp emission spectrum and the window transmission.

Table 2.1: Characteristics of common PID lamps used for VOC detection.

Lamp Fill Gas	Main Emission Lines (eV)	Window Material
Xenon (Xe)	8.4, 9.6	CaF_2
Krypton (Kr)	10.0, 10.6	MgF_2
Argon (Ar)	11.6, 11.8	LiF

In addition to MgF_2, CaF_2, and LiF, sapphire (Al_2O_3) is sometimes used as a window or filter material; its longer wavelength cutoff removes deeper VUV components and can therefore define an additional cutoff ionization threshold, improving selectivity. In practice, the onset of ionization is not always a perfectly sharp cutoff because gas purity, leakage, window contamination, and VUV-induced window darkening (solarization) can modify the delivered spectrum over time [34].

Beyond selecting the lamp fill gas and window material, PID performance can also be improved by engineering the optical and electrode interface that impacts how efficiently VUV photons generate and sustain the measurable ion current. For example, Ref. [37] demonstrated a proof-of-concept atmospheric PID electrode based on a chromium/gold nanolayer thin film at the lamp window, reporting a substantial increase in photon utilization compared to a conventional metal-grid approach. In parallel, photonic structure engineering enhances light-matter interactions via resonant field redistribution; Ref. [38] reported tunable hybrid photonic–plasmonic crystals with strong field enhancement and mode shifts induced by controlled geometric modification. A strategy of extracting chemical contrast from controlled light-matter interaction is through label free spectroscopic methods such as Raman spectroscopy of cells and tissues [39].

2.2.4 Limitations of fixed energy lamps and motivation for tunable VUV

Modern PIDs can achieve low detection limits with compact, low power instruments and near real-time readout, making them well suited to IAQ monitoring and screening applications [40, 9, 37].

However, the fixed photon energies available from discharge lamps constrain both the detectable compound set and the achievable selectivity: any mixture component with IE $< h\nu$ contributes to the same aggregate ion current, so the PID signal is not specific to certain compounds without additional separation or discrimination mechanisms [34]. See Fig. 4.2 in the included article in Chapter 4, which illustrates how wavelength tunability enables selectivity in SIPS.

Extending PID capability toward higher-IE targets and improving molecular selectivity therefore motivates VUV sources that provide higher photon flux and tunable photon

energy so that ionization thresholds can be selectively addressed. Recent initiatives reflect the growing recognition of this issue. One example is the *IDEAL Cluster* (*Indoor Air Pollution and Health*), a consortium of seven EU-funded projects under the *Horizon Europe* [41], specifically the call *HORIZON-HLTH-2021-ENVHLTH-02-02*, which aims to characterize indoor pollutants and develop actionable monitoring tools.

2.2.5 Photodissociation in MS/MS (context; not used in PIDs)

Photodissociation is an ion activation method in tandem mass spectrometry in which a mass selected precursor ion absorbs photon(s) and fragments to form product ions. In contrast to photoionization, which creates ions from neutrals, photodissociation typically begins with an already charged precursor and redistributes internal energy into specific dissociation channels. Reviews of photodissociation MS and UV photodissociation are available in the Refs. [42, 43]. In its simplest form:

$$\mathrm{M}^+ + h\nu \rightarrow \mathrm{M}^{+*} \rightarrow \sum_i \mathrm{F}_i^+ + \sum_j \mathrm{N}_j \tag{2.2}$$

where M^+ is the precursor ion, F_i^+ are product ions, and N_j are neutral losses.

UV photodissociation (e.g., 193 nm) can produce rich fragment ion series that support structural elucidation [44], while infrared multiphoton dissociation transfers energy to the ion through repeated IR absorption and is sometimes implemented in ion traps [45]. Although photodissociation is effective for MS/MS structural characterization, it is generally not used in portable PID VOC sensors, which target threshold selective ionization of neutrals rather than fragmentation of mass selected ions under vacuum [34].

2.3 Motivation for tunable VUV photoionization

The photon energies available from discharge lamps are limited to fixed emission lines, which restricts both the detectable compound set and the achievable selectivity in photoionization detectors.

A tunable VUV source can overcome this limitation by allowing the photon energy to be adjusted relative to ionization thresholds of interest, allowing discrimination by operating near threshold, which cannot be obtained with discharge lamps. Tunability does not imply accelerator scale infrastructure: while synchrotrons and free-electron lasers provide VUV tunability at large facilities, the focus here is on a compact laser frequency conversion system, which can be implemented in a laboratory optical setup scale.

2.3.1 VUV source classes and practical suitability

Several technologies can provide VUV photons.

Large accelerator facilities, including synchrotrons and free-electron lasers, offer high photon flux with wide wavelength tunability, but their scale and infrastructure requirements are incompatible with compact instrumentation [46, 47].

Excimer lasers deliver high pulse energies at fixed ultraviolet wavelengths (e.g., 193 nm for argon fluoride and 248 nm for krypton fluoride), and fluorine lasers operate at 157 nm; however, these photon energies (approximately 6–8 eV) remain below the ionization thresholds of many VOCs. In addition, radiation near 157 nm is strongly absorbed in ambient air, so propagation typically requires purged enclosures (e.g., nitrogen) or a vacuum beam path [48, 49].

Recent developments in tabletop ultrafast sources have demonstrated continuously tunable VUV pulses through high-field harmonic generation, but these systems are currently implemented primarily in laboratory settings and require careful control of beam delivery and the propagation environment [50].

A field deployment of the system in a primary school in Portugal, including enclosure, safety measures, and on-site operation constraints, is described in Appendix D.

2.3.2 Approach of this project: nonlinear frequency conversion

This thesis develops tunable VUV generation using nonlinear frequency conversion. A tunable laser field is converted to shorter wavelengths through harmonic generation, increasing the photon energy while providing wavelength tunability beyond conventional PID lamps. In the present implementation, a tunable pump field is frequency-doubled to the visible and then converted to the VUV through THG in a thin membrane. This configuration reaches photon energies relevant for VOC ionization while maintaining continuous wavelength control.

The objective is to enable photoionization with photon energies selected with respect to the ionization thresholds of target VOCs. Although the current implementation is a laboratory prototype, it demonstrates selective photoionization capabilities that are not available with discharge lamps operating at discrete emission lines.

2.3.3 nonlinear optics considerations in the VUV

nonlinear frequency conversion relies on the nonlinear polarization response of matter. In the perturbative regime, SHG depends on $\chi^{(2)}$ and therefore requires non-centrosymmetric

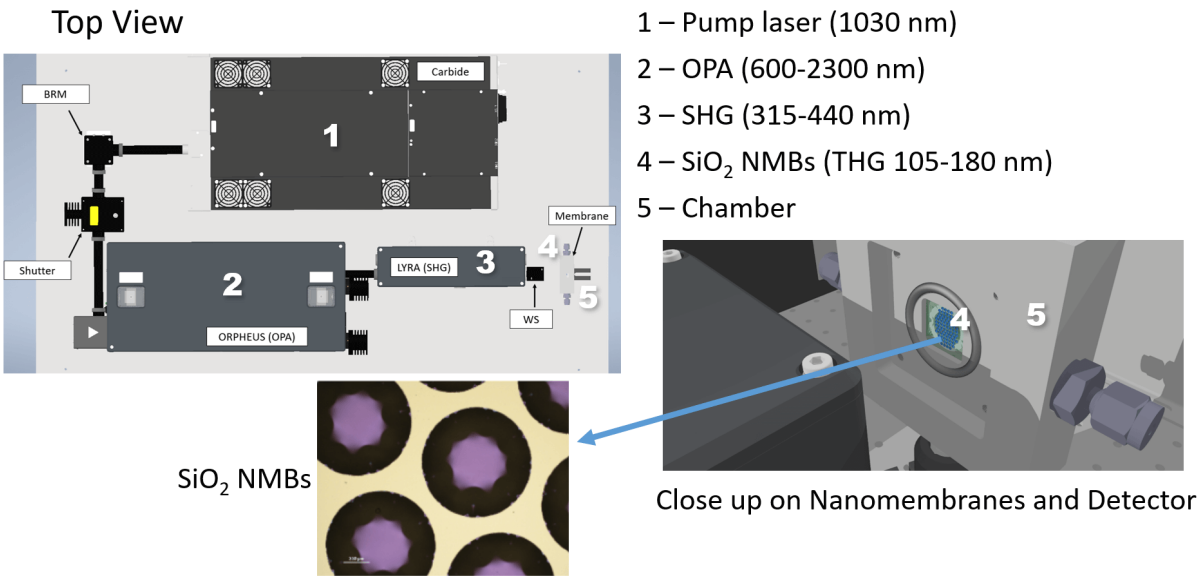


Figure 2.2: VUV approach summary.

media, whereas THG depends on $\chi^{(3)}$ and can occur in centrosymmetric materials [51].

Efficient conversion in bulk media usually requires phase matching so that the interacting waves remain synchronized over the interaction length. When the phase mismatch $\Delta k \neq 0$, the relative phase between the nonlinear polarization and the generated harmonic field accumulates along the propagation direction. As a consequence, contributions generated at different positions interfere partially destructively, reducing the overall conversion efficiency. In bulk nonlinear crystals, this is sometimes mitigated by using birefringent phase matching, temperature tuning, or quasi-phase matching [52].

The VUV regime imposes additional practical and material limitations: below $\approx 200\,\text{nm}$, many optical media exhibit strong absorption and high dispersion, which reduces usable transparency and makes phase matching increasingly difficult. Widely used nonlinear crystals, such as BBO and LBO, remain effective in the near-UV, but their applicability diminishes toward shorter wavelengths due to transmission cut-offs and less favourable dispersion for phase matching [53].

Gas-phase approaches (e.g., four-wave mixing in rare gases or high-order harmonic generation) can extend further into the VUV; however, they usually require controlled environments (vacuum or purged beam paths), careful management of phase matching and absorption, and laser systems with sufficiently high peak intensity, which complicates compact and field deployable implementations.

2.3.4 Resonant thin films for enhanced THG in compact geometries

To avoid relying on conventional phase-matching in bulk crystals, this thesis investigates nonlinear conversion in subwavelength resonant structures. In particular, we consider an air/TiO_2/air membrane, which is accurately modeled as a plane-parallel, *low-finesse Fabry–Pérot* formed by the two TiO_2/air interfaces. In this configuration, the resonance is not imposed by numerical boundaries (the PMLs suppress reflections at the simulation boundaries), but results from multiple Fresnel reflections inside the film, yielding an Airy-type interference response [54]. At normal incidence, resonance peaks occur approximately when

$$2\,n(\omega)\,d \simeq m\,\lambda \qquad (m \in \mathbb{Z}), \tag{2.3}$$

so the spectral position of the enhancement is set primarily by the film thickness d (for a given dispersion and incidence angle). Even with moderate interface reflectivity (i.e., low finesse), the resonance increases the internal field intensity and strengthens the interaction time in the nonlinear medium, thereby enhancing THG in films only a few hundred nanometers thick, where phase-matching is less restrictive than in bulk materials [13].

In the current experimental setup, a suspended SiO_2 membrane serves as the experimentally used nonlinear medium. Within the scope of this thesis, TiO_2 is used to assess the potential of resonant thin-film and membrane designs, rather than to demonstrate a fabricated nonlinear device.

The target emission range is 110–180 nm, within ionization thresholds of many organic molecules (see Tab. D.1). Unlike discharge lamps, which emit fixed spectral lines and therefore provide only discrete selectivity levels, a tunable nonlinear source can allow wavelength selective photoionization with improved discrimination between compounds.

These developments align with the goals of the Horizon Europe project LEARN, which aims to advance IAQ monitoring by enabling precise, real-time detection of pollutants in sensitive environments [17].

2.3.5 Alternative approaches not pursued

While SHG is an essential step in many UV/VUV source setups, the metasurface investigated in this thesis was not designed to provide SHG output.

Using AlGaAs (a high-index, non-centrosymmetric material with strong second-order nonlinearity), large backward-to-forward SHG emission ratios and vector beam patterns

at the second harmonic were demonstrated, illustrating how dielectric metasurfaces can control SHG radiation and directionality [55].

In a more recent study [56], bound states in the continuum were used in a silicon metasurface to enhance THG and degenerate four-wave mixing, demonstrating how resonance engineering can enable strong nonlinear mixing in compact devices.

More broadly, nonlinear dielectric metasurfaces have been investigated for SHG [57, 58] and sum-frequency mixing [59, 60]. This thesis, however, focuses on THG as the primary route toward shorter wavelengths.

Nonlinear frequency mixing approaches, including sum- and difference-frequency generation and four-wave mixing, can provide tunable UV output by combining multiple pump frequencies and can support processes of interest in quantum optics [61, 62]. These multiple wavelength mixing and oscillator approaches were outside the scope of the present work.

Finally, higher-order harmonic generation (e.g., fifth and seventh harmonic) in nanostructures can access deep-UV and even extreme-UV wavelengths and has been demonstrated in other material systems, including AlGaAs nanoresonators under mid-IR pumping [63]. Such higher-harmonic generation was not pursued here due to increased constraints associated with phase synchronization, dispersion control, and material damage, and the thesis remains focused on THG as the main nonlinear process.

Given the poor efficiency of VUV conversion in bulk/compact geometries, the next sections focus on nanophotonic strategies to enhance the local field and thus the nonlinear yield.

2.4 Nanophotonic field enhancement

Metasurfaces are planar arrays of subwavelength nanostructures that enable local control of reflection, transmission, phase, and polarization beyond what is achievable with bulk optics [64]. The geometry and periodicity of the meta-atoms determine the resonant modes of the array, enabling control of the amplitude, phase and polarization of the scattered light, while confining electromagnetic energy in the near-field [65, 66].

In this work we focus on dielectric metasurfaces. By a dielectric we mean a low-loss, electrically insulating medium whose optical response at the operating wavelengths is dominated by bound-charge polarization, i.e. displacement currents, rather than by the ohmic dissipation associated with free-carrier conduction in metals [67]. In the spectral range of interest, this means materials with weak absorption, where the extinction coefficient is much smaller than the real part of the refractive index, $k \ll n$. Dielectric

metasurfaces, in particular, rely on high-index, low-loss materials to support electric and magnetic Mie-type resonances with minimal absorption, making them ideal for short wavelengths (within the VUV and UV range) [68] where standard plasmonic metals exhibit strong interband losses [69]. For comparison, plasmonic metasurfaces use metallic nanoantennas to excite localized surface plasmon resonances, providing extreme field confinement but at the expense of significant absorption losses, especially in the visible and ultraviolet spectral ranges.

Weakly radiating modes (also termed "dark modes" or "dark states"), such as anapole excitations, can strongly enhance the internal field and are therefore well suited to nonlinear frequency conversion.

In the following, we treat metasurfaces as ultra-thin optical layers, made of two-dimensional arrays of subwavelength nanostructures ("meta-atoms"). As described in early works [65, 66], such arrays can modify the transmitted and/or reflected wavefront by designing the local phase, amplitude, and polarization response of the meta-atom.

In this thesis, metasurfaces are treated primarily as simulated design systems for understanding field confinement, resonance tuning, and nonlinear enhancement, rather than as fabricated devices.

2.5 Dielectric metasurfaces

Dielectric metasurfaces implement the metasurface concept using high-index, low-loss dielectric meta-atoms (e.g., TiO_2). Their optical response differs fundamentally from metallic metasurfaces: instead of plasmonic free-electron oscillations, dielectric resonances are determined by displacement currents. As a result, absorptive (ohmic) losses are reduced, enabling strong electromagnetic energy confinement and improved radiation efficiency compared with plasmonic metasurfaces at short wavelengths. A comparative simulation study of field propagation in TiO_2 versus silver and gold as a function of wavelength is provided in Appendix D.2.

Strong confinement in dielectric metasurfaces is often associated with *Mie-type* multipolar resonances. While Mie theory is strictly derived for spherical particles [70, 68], the term *Mie-type* is commonly used to describe analogous electric and magnetic multipole resonances in non-spherical subwavelength resonators (e.g., disks, cylinders, cuboids and pillar geometries) [71, 72]. Depending on geometry and refractive index, individual dielectric resonators (meta-atoms) can support electric dipole (ED), magnetic dipole (MD), and higher-order multipoles [73]. The resonance wavelength and Q-factor are sensitive to the resonator dimensions (including aspect ratio), material dispersion, and

the surrounding environment, therefore, small variations in size or material parameters can produce spectral shifts.

The scattering and absorption response is determined by the complex refractive index ($\tilde{n} = n + ik$) together with the resonator geometry. The real part n sets the phase velocity and optical path length, whereas k quantifies absorption [74, Ch. 11, pp. 286–288]. These parameters become particularly important in the UV and VUV, where material dispersion and absorption can change rapidly with wavelength.

By tuning multipolar resonances in dielectric nanostructures, it has been shown that nonlinear conversion efficiencies, including THG, can be significantly increased [15, 75, 16]. In particular, Ref. [16] demonstrates VUV THG from an anapole resonance in a TiO_2 metasurface and discusses the potential relevance of compact VUV sources for photochemical applications such as photodissociation. While photodissociation and photoionization are different processes, this result provides important evidence that TiO_2 metasurfaces can support efficient nonlinear conversion into the VUV spectral range.

Traditional PIDs rely on UV lamps to photoionize volatile organic compounds (VOCs). These sources provide sufficient photon energy for a broad range of VOCs but offer limited tunability, fixed emission spectra, and constrained operational lifetimes [40, 76]. Alternative strategies to reach deeper-UV photon energies, such as resonance enhanced multiphoton ionization or high-harmonic generation (HHG), can access extreme UV photon energies [77, 78], however, they require ultrafast laser systems, high optical intensities, and complex optical setups that are not compatible with compact, field deployable sensing instruments. Likewise, UV generation using bulk nonlinear crystals (e.g., BBO, LBO) is well established [53, 51], but phase matching and material transparency become increasingly challenging below $\approx 200\,\text{nm}$, limiting scalability toward shorter wavelengths.

The work presented in this thesis addresses these limitations by evaluating two photonic strategies for UV generation: resonant TiO_2 meta-atoms and thin-film Fabry–Pérot resonators. The aim is to design and compare these platforms computationally, and to relate the results to photoionization requirements, rather than to demonstrate a fabricated metasurface device within the scope of this thesis.

TiO_2 is selected as the dielectric platform due to its high refractive index and low absorption in the UV-A; its optical constants and fabrication are discussed in Section 2.6. Toward the deep-UV and VUV, absorption increases near the material absorption edge, which limits operation at shorter wavelengths and motivates resonant field enhancement to sustain high internal fields and nonlinear conversion efficiency. The precise optical constants depend on growth conditions and phase (amorphous, anatase, or rutile), but these characteristics

make TiO_2 a strong candidate for confined resonances targeting UV/VUV nonlinear generation.

2.5.1 Scope of present work (limitations of plasmonic approach)

Angle dependent SHG can be achieved using wavevector selective nonlinear plasmonic metasurfaces based on tilted gold nanopillars [79].

More recent devices exhibit strongly asymmetric SHG [80] and exploit monocrystalline gold metasurfaces to control anisotropic SHG via the crystal symmetry of 111 surfaces [81].

However, plasmonic metasurfaces more generally are fundamentally limited by ohmic and interband losses in metals [82], and their performance is strongly affected by film quality, surface roughness, and grain structure, which can reduce resonance quality factors and nonlinear conversion efficiencies [83]. This motivates the exploration of high-index dielectric metasurfaces as a low-loss alternative for efficient linear and nonlinear wavefront control, especially at short wavelengths [73].

2.6 ALD TiO_2 for nanophotonic materials and future nanostructuring

2.6.1 Why ALD is particularly useful for TiO_2 nanophotonics

ALD is valuable for nanophotonic TiO_2 because each reaction step saturates at the surface, which enables very precise thickness control and conformal coating on high-aspect-ratio nanostructures. This is important for obtaining uniform optical response across metasurfaces, nano-pillar arrays and other nanostructures [84]).

Amorphous TiO_2 is often preferred when low scattering is critical: grain boundaries in polycrystalline anatase/rutile can increase roughness and act as light scattering centers [85]. This motivates using low temperature ALD (or other low crystallinity approaches) when low optical loss is required.

2.6.2 TiO_2 ALD oxidizing agents and implications for film quality

The properties of ALD TiO_2 films depend strongly on the oxidizing agent, precursor chemistry, and deposition temperature. These factors determine impurity content, defect density (e.g., Ti^{3+}/oxygen vacancies), phase (amorphous/anatase/rutile), and thus refractive index n, extinction coefficient k, and scattering [84]. The most common precursor classes for TiO_2 ALD are halides (e.g., $TiCl_4$), alkoxides (e.g., TTIP), and alkylamides (e.g., TDMAT).

A recent report shows that H_2O-grown films can exhibit Ti^{3+}/oxygen vacancies and

visible-NIR absorption, while stronger oxidants (H_2O_2, O_3 and O_2 plasma) result in more stoichiometric films with fewer Ti^{3+} states and no visible-NIR absorption below the band edge [86].

2.6.3 Cycle timing: pulse, exposure and purge

An ALD cycle comprises (i) precursor exposure, (ii) purge, (iii) oxidant exposure, (iv) purge. The required timing is set by the need to reach surface saturation in each half reaction and by transport/adsorption kinetics, which can change with temperature and reactor design [87].

Timing is set to reach saturation, which is identified by each half-reaction: beyond a threshold exposure, additional pulse time does not increase growth rate because the surface is already saturated. Timing is dependent on reactor volume, flow regime, pumping speed, substrate loading, and topology, recipes vary depending on the machine.

2.6.4 Substrate and nucleation compatibility

For metasurfaces and UV optics, fused silica and other SiO_2 substrates are common choices. In Ref. [85] amorphous TiO_2 ALD films deposited on fused silica show root mean square (RMS) roughness $\approx 0.738\,\text{nm}$, attributed primarily to the roughness of the underlying substrate.

Ref. [84] reports that the nature and density of functional groups (–OH on oxide surfaces) influence initial nucleation and early cycle growth. As a result, substrate choice can increase the growth-per-cycle value or introduce a nucleation delay.

Ref. [88] motivates ozone (highly volatile, strong oxidizing agent) as a better alternative for low temperature coatings on thermally sensitive materials, they note that H_2O can physisorb strongly on reactor surfaces, therefore leading to longer purge times required for removing the physisorbed H_2O sufficiently.

2.6.5 Target properties for UV nano-structuring

For metasurfaces and meta-atoms, we want high n, low k and low scattering. Achieving this combination is challenging and depends not only on deposition temperature, but also on oxidant chemistry, stoichiometry, defect density (including oxygen vacancies and Ti^{3+} states), crystallinity, grain size, surface roughness, thickness, and any post-deposition thermal treatment [84, 86]. These parameters are strongly coupled, and improving one property may degrade another.

Ref. [84] details how crystalline nucleation and grain evolution depend on deposition

temperature and thickness, with phase and morphology growing from nucleation processes that can be dependent on substrate choice. They also note that polycrystalline anatase/rutile often exhibit RMS roughness values as high as 5–10 nm, which is optically significant for UV and visible wavelengths.

2.6.6 Defect chemistry and "Black-TiO_2"

In ALD-grown TiO_2 using TDMAT and H_2O, the limited oxidizing strength of H_2O can prevent complete oxidation of the precursor, promoting the formation of oxygen vacancies and Ti^{3+} states ("black TiO_2"). The formation of Ti^{3+} has been associated to oxygen displacement within TiO_2, producing coordinated Ti^{4+} ions (5–8 coordinated) together with Ti^{3+} species. These defects introduce visible–NIR absorption (non-zero extinction coefficient k). Therefore, when low optical loss (minimized k) is required, using stronger oxidizing agents such as H_2O_2, O_3, or O_2-plasma is preferred to obtain more stoichiometric TiO_2 with fewer defects [86].

2.6.7 Alternative TiO_2 deposition methods not pursued in this work

TiO_2 could be fabricated by chemical methods (include sol-gel and dip coating [89, 90], chemical vapor deposition [91, 92], hydrothermal synthesis [93]), physical processes (such as sputtering [94, 95], laser ablation [96]), and green methods also called biological synthesis [97].

These methods were not prioritized in the present work because the target application requires nanophotonic quality films with excellent conformality, thickness control, and low roughness on nanostructured substrates. In addition, the available ALD setup was more suitable for process optimization within the scope of this thesis. While methods such as sputtering may be advantageous in other contexts (e.g., deposition rate), ALD was selected here based on the combination of film quality requirements and process controllability.

2.7 Slow-light metric for dispersive TiO_2

2.7.1 Why group velocity can be misleading in lossy or strongly dispersive media

In transparent, weakly dispersive media, the group velocity

$$v_g = \partial\omega/\partial k \tag{2.4}$$

often provides a good estimate of the propagation speed of a narrowband pulse envelope.

In absorbing or amplifying media, however, the wave vector is sometimes complex, $k(\omega) = k'(\omega) + ik''(\omega)$. Near resonances, definitions that consider only the dispersion of $k'(\omega)$ can produce superluminal or even negative values of v_g without implying superluminal transport of energy or information [98].

To make this distinction practical, we can define the propagation delay in terms of what a detector measures, rather than referring to an "envelope speed". Following Ref. [99], the pulse arrival time at a given position can be defined as the first temporal moment of the detected Poynting flux. With this definition, the delay between two positions separates into (i) a group delay term, given by a spectral average of the group delay across the pulse bandwidth, and (ii) a reshaping term which includes the effects of frequency varying absorption or gain. This decomposition clarifies how spectral filtering can advance or retard the pulse peak even though the overall energy transport remains consistent with causality [99].

2.7.2 Energy velocity, Brillouin energy density

A useful measure of energy transport is the *energy velocity*, defined as the ratio of the cycle-averaged Poynting flux to the cycle-averaged electromagnetic energy density,

$$v_e = \frac{\langle \mathbf{S} \rangle}{\langle u \rangle}. \tag{2.5}$$

In dispersive media the definition of the appropriate effective energy density is subtle: expressions that are valid in nondispersive materials (e.g. $u \propto \epsilon|E|^2$) do not correctly account for energy stored in the medium's polarization and magnetization response when $\epsilon(\omega)$ and $\mu(\omega)$ vary strongly with frequency [99].

For monochromatic fields in dispersive, lossless media, the time-averaged stored energy density reduces to the Brillouin formula,

$$u_{\mathrm{B}} = \frac{1}{4}\left[\frac{d(\omega\epsilon)}{d\omega}\,|E|^2 + \frac{d(\omega\mu)}{d\omega}\,|H|^2\right], \tag{2.6}$$

where $\epsilon(\omega)$ and $\mu(\omega)$ are real in the lossless limit. A derivation and a careful discussion of the underlying assumptions can be found in Ref. [102]. Using a consistent expression for $\langle u \rangle$ makes v_e a meaningful measure of energy transport in strongly dispersive systems and makes it straightforward to establish subluminal bounds in causal, passive media.

In homogeneous free space, this definition reproduces the expected result. For a monochro-

matic plane wave, it follows

$$\langle S \rangle = c \langle u \rangle, \tag{2.7}$$

and therefore $v_e = c$.

In homogeneous passive linear media satisfying causality, the energy transport speed does not exceed c [103].

In guided and periodic structures with strong dispersion (such as waveguides near cutoff, photonic crystals, resonators, or metamaterials) the energy velocity can become much smaller than c. When dispersion is strong and energy is temporarily stored in resonant modes, v_e provides a more physically correct measure than v_g of how rapidly electromagnetic energy propagates through such structures. A reduced energy velocity increases the interaction time between the field and the material and, together with field localization, can enhance nonlinear processes such as high-harmonic generation.

2.8 Anapole and other high-Q non-radiating modes for field enhancement

An anapole mode is a weakly radiating (radiationless) state that occurs when the fields emitted by co-located Cartesian electric and toroidal dipole moments destructively interfere. As a result, far-field scattering is strongly suppressed while electromagnetic energy remains concentrated within the resonator and its near-field, leading to large internal field amplitudes [14]. Such strong energy confinement is particularly attractive for enhancing nonlinear optical processes, including THG [15].

In subwavelength dielectric resonators this behavior is conveniently described in a Cartesian multipole expansion, where different current configurations can produce identical far-field angular patterns but with controllable relative phase [105]. Of particular relevance here is the cancellation between the Cartesian electric dipole and toroidal dipole terms, which strongly suppresses far-field dipolar scattering while maintaining strong energy confinement in the resonator and its near-field [106].

The definition used for our simulations and the anapole criterion are introduced in Section 7.2 (see Eq. (7.1)). This condition implies that even though both dipole contributions exist and oscillate strongly, the radiation in the far-field is highly suppressed. The toroidal dipole can be pictured as a ring shaped loop of oscillating current inside a nanoparticle. Its far-field pattern has the same angular dependence as an electric dipole, but shifted in

phase. The two interfere destructively, suppressing radiation while storing energy in the near-field.

At resonance, the scattering cross-section drops sharply, indicating strongly suppressed radiation, while the electromagnetic field energy remains trapped inside the nanoparticle.

Anapole states have been experimentally observed in high-index dielectric nanoparticles [107], such as silicon spheres [14] and germanium disks [15], where the geometry is carefully tuned so that the electric and toroidal dipole moments spectrally overlap and destructively interfere.

We have shown that several geometries (e.g., nanorectangular cuboids and polygonal prisms) can support anapole states at UV wavelengths. In general, the excitation and strength of an anapole resonance are a consequence of the material dispersion and losses (n, k) together with geometry dependent multipolar interference. UV-range anapoles remain relatively uncommon [108]. These considerations, together with the LEARN project objectives, motivated our choice of TiO_2 and the different geometries explored in Chapter 7.

It should also be noted that a single TiO_2 nanostructure can support multiple modes simultaneously. A nanorectangular cuboid, for example, might simultaneously exhibit both electric dipole (ED) and magnetic dipole (MD) modes. However, under specific conditions, the electric dipole and toroidal dipole (TD) can destructively interfere in the far-field, generating a non-radiating state known as the anapole mode [108].

The main applications of the anapole mode are boosting the THG, or other nonlinear optical phenomena [15, 75, 21, 18, 16, 106, 109]. Because of the strong internal field created by the anapole, nanoparticles excited at the anapole resonance can show orders of magnitude higher THG efficiency compared to those operating at conventional radiative resonances.

Compared to radiative resonances, anapole modes provide stronger near-field intensity and longer interaction times, both of which scale THG [109].

When we say that a nanostructure “supports" an anapole mode, we mean that its geometry and material are suitable for the mode to exist; this is a passive property of the nanostructure. When we say we “excite” an anapole, we refer to the process of illuminating the nanostructure with light to activate that mode.

Motivated by these reports, we performed a numerical study of TiO_2 rectangular cuboids (referred to as nano-rectangles in Ref. [108]) targeting UV-A anapole excitation. By

choosing TiO_2, whose refractive index is high and extinction coefficient very low below 400 nm, and introducing in-plane asymmetry (length $\neq$ width), we found the resonance could be tuned between 350–380 nm. The anapole remains observable for incident angles up to $\approx 25°$, and time-domain simulations show a short transient radiative response followed by a steady regime with strongly suppressed scattering [108]. These results justify the design choices in Chapter 7.

2.8.1 Historical context and terminology

A useful narrative overview is given in "Anapole: its birth, life and death" [111].

J.J. Thomson, in 1897, demonstrated that cathode rays consist of negatively charged particles (electrons). By measuring their charge-to-mass ratio, Thomson showed that these particles are much smaller than atoms [112].

In 1910, physicist and mathematician Paul Ehrenfest demonstrated that certain extended charge distributions can accelerate without emitting radiation, a concept now central to the theory of non-radiating modes [113]. This result goes directly against the classical expectation that any accelerating charge must radiate.

In 1913, Bohr introduced his model of the atom. It explained why electrons in atoms don't radiate continuously: they only emit energy when jumping between quantized orbits. The model was later replaced by quantum mechanics, but it introduced the idea that a bound charge can occupy a stationary, non-radiating state [114].

Understanding non-radiating states advanced considerably in 1957 with Yakov Zel'dovich's work on parity violation in weak interactions, which introduced the concept of a toroidal dipole moment, eventually referred to as the anapole moment [115, 116, 117]. [1] Subsequent experiments on caesium atoms measured atomic parity non-conservation and, with it, the nuclear anapole moment of the caesium nucleus [121]; this is the parity violating moment Zel'dovich had predicted, and it is conceptually distinct from the optical anapole modes of dielectric resonators discussed below.

It is also worthwhile to include Chien-Shiung Wu's experiment here. Wu led beta decay measurements on cobalt-60 that directly showed that the weak force violates mirror symmetry. The history of how this experiment was explained to different audiences is analyzed by Bertozzi (2017) [122]. The story of parity violation is also discussed in Leon

[1] Non-radiating or weakly radiating states also appear in other areas of physics. For example, Weyl (1929) and Majorana (1937) fermions motivate quasiparticle descriptions in certain condensed-matter systems; in superconducting platforms, Majorana-like modes are discussed in the context of charge-neutral excitations and protected states [118, 119, 120]. This connection is conceptual rather than direct: it is not the same phenomenon as optical anapoles in dielectric resonators.

Lederman's book *The God Particle* [123], in the interlude titled "How we violated parity in a weekend... and discovered God", where Lederman describes how the experiment showed that the universe distinguishes left from right.

3 Tunable VUV Source for VOC Photoionization

3.1 Publication context and author contributions

This chapter is based on the peer-reviewed article *Volatile organic compound photoionization utilizing tunable quasi-monochromatic vacuum ultraviolet down to 120 nm obtained by third harmonic generation from a suspended dielectric nanomembrane* (R. Vilarinho, H. Mahmoodnia, C. Maibohm, D. Poeta, J. Fernandes, C. Calaza, A. Gómez-Varela, M. López-García, A. Miranda, P. De Beule), *Optics Express* **33**(12), 26342 (2025), DOI: 10.1364/OE.564577 [1], on which I am a co-author. My main responsibility was the gas-handling subsystem (Section 3.2.3) and the low-leakage transimpedance acquisition chain (Section 3.2.4), which I designed, built, and characterized together with the LabVIEW data-acquisition pipeline. I also contributed to the ionization-chamber CAD modifications, to the THG simulations supporting Section 3.3.2, and to the photoionization measurements carried out with the lead authors, as well as to procurement, logistics, and team training. The earlier PID test-platform work from which this development grew is documented in Section D.1.1.

3.2 Experimental platform

Fig. 3.1 shows the overall experimental layout. A high-power ultrafast Ytterbium laser (Carbide, Light Conversion, Lithuania) generates femtosecond pulses at kilohertz rates and pumps an optical parametric amplifier (OPA; Orpheus, Light Conversion, Lithuania). The tunable quasi-monochromatic OPA output is frequency-doubled by a β-barium-borate crystal (BBO; Lyra, Light Conversion, Lithuania) via second-harmonic generation (SHG). This chain delivers femtosecond pulses across the 315–540 nm range with pulse energies of $\sim$4 µJ at 60 kHz repetition rate and $\sim$230 fs duration. The SHG pulses are focused by a lens (Thorlabs LA1509–A, AR-coated 300–480 nm) onto a suspended SiO_2 nanomembrane integrated into the ionization chamber. Third-harmonic generation (THG) in the nanomembrane down-converts the pump light to the VUV (120–180 nm). The VUV photons ionize VOC molecules diluted in a nitrogen carrier flow inside the chamber, and the resulting photo-ionization current is amplified by an in-house low-leakage transimpedance front-end and read out by a National Instruments DAQ controlled in LabVIEW.

3.2.1 Suspended SiO_2 nanomembrane device

Suspended SiO_2 was selected as the THG-generating medium because of its high laser-induced damage threshold and its commercial availability as a thin-film stack [13]. Each

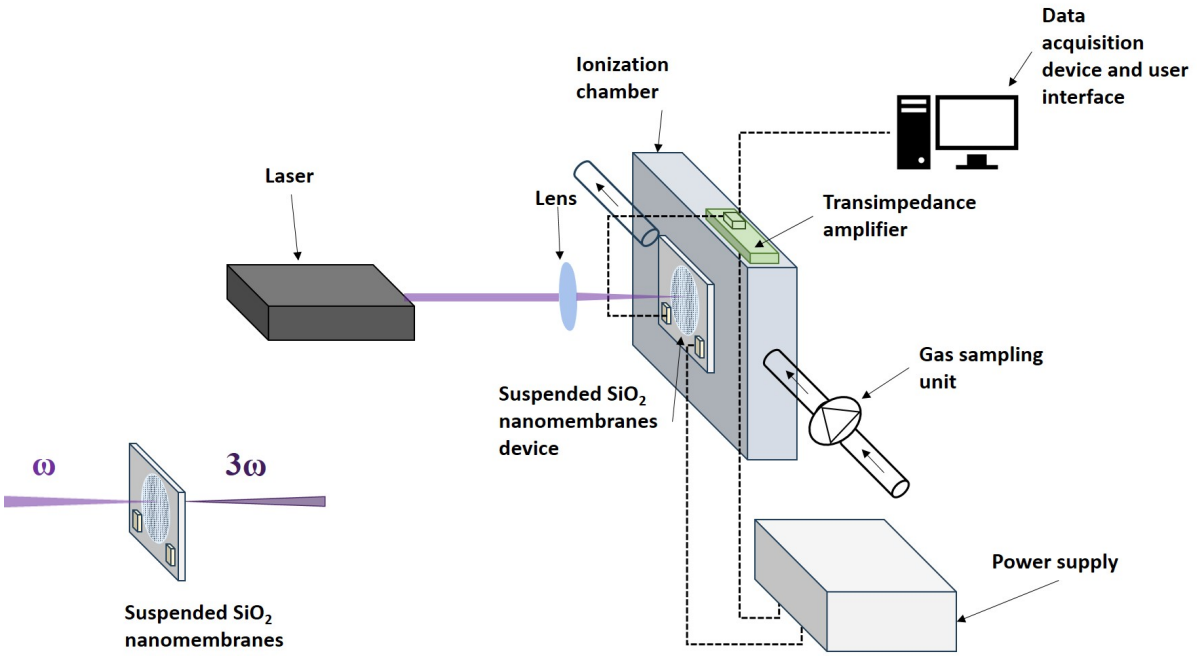


Figure 3.1: Schematic of the experimental setup. A femtosecond Yb laser pumps an OPA, the output is frequency-doubled in a BBO crystal, and the resulting near-UV beam is focused onto a suspended SiO_2 nanomembrane integrated within an ionization chamber. THG in the membrane generates VUV light that ionizes VOC molecules delivered by the gas sampling unit. The photo-ionization current is collected on AlSiCu electrodes patterned on the membrane device, amplified by an in-house transimpedance amplifier, and acquired through a NI DAQ under LabVIEW control. Dashed lines correspond to electrical connections; arrows indicate gas flow. Adapted from [1].

device hosts an array of circular suspended SiO_2 nanomembranes with 1 mm diameter and 600 nm thickness, fabricated on an 8-inch silicon-wafer support. Two AlSiCu metal electrodes with wire-bonded contact pads are integrated into the same chip, enabling small-current collection. The array layout provides redundancy: if one membrane is damaged by laser exposure, the device can be translated relative to the fixed laser path to engage a neighbouring membrane without realignment of the beam line.

3.2.2 Device fabrication

The fabrication flow is summarized in Fig. 3.2. The full fabrication run was carried out by H. Mahmoodnia together with the Micro and Nanofabrication Facility team at INL. It is documented here because the same process flow can be reused with the SiO_2 layer replaced by a high-index dielectric such as TiO_2, providing a guide to the nanostructured membranes proposed in Chapters 5–7. Starting from a double-side-polished 725 µm-thick (100) silicon wafer, a 3 µm SiO_2 hard-mask is deposited on the back side by plasma-enhanced chemical vapor deposition (PECVD; step 1). A first lithography step (step 2) patterns the membrane apertures, and the hard mask is opened by inductively coupled plasma reactive-ion etching (ICP–RIE) with fluorine chemistry (step 3). After resist strip, the process continues on the front side: a 2 µm sacrificial polyimide layer is spin-coated and cured at 250 °C for 14 h (step 4), and a 600 nm SiO_2 film is then deposited by PECVD to define the final membrane thickness (step 5). A bi-layer of lift-off resist (LOR-5B, 0.5 µm) and UV photoresist (AZ1505, 1.035 µm) is patterned by a second lithography step (step 6) to outline the metal pads, and a 250 nm AlSiCu layer is sputtered (step 7) and patterned by lift-off (step 8). A photoresist layer protects the front side, and a deep RIE process etches through the silicon from the back side until the polyimide layer acts as the etch stop (step 9). The free-standing 600 nm-thick SiO_2 membranes are finally released by an isotropic plasma-ashing step that simultaneously removes the back-side polyimide and the front-side protection (step 10).

The free-standing membranes exhibit compressive residual stress that produces a buckled surface profile, with the buckling amplitude increasing with the membrane diameter-to-thickness ratio. For the chosen 600 nm thickness, the 1 mm diameter was found to be a practical compromise between mechanical robustness and the wafer area consumed by the supporting silicon frame. The membranes are highly transmissive across the visible range, with optical transmittance and reflectance being indistinguishable between single-membrane and array measurements, so that linear-optical characterization performed on an isolated membrane is representative of the full device.

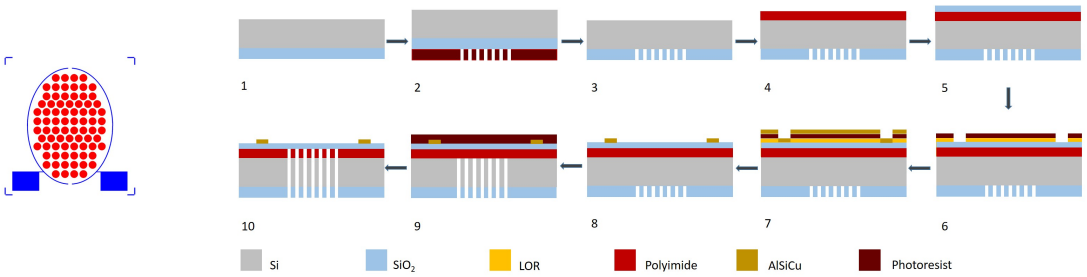


Figure 3.2: Schematic of the device cross-section and process flow for the suspended SiO_2 nanomembranes: (1) PECVD deposition of 3 µm SiO_2 hard mask on the back side; (2) photoresist patterning by DWL lithography; (3) ICP–RIE opening of the hard mask; (4) spin-coat and cure of a sacrificial polyimide layer; (5) PECVD deposition of the 600 nm membrane SiO_2; (6) LOR/photoresist bi-layer and DWL lithography for metal-pad definition; (7) sputter deposition of 250 nm AlSiCu; (8) lift-off; (9) deep RIE silicon etch through the back side to the polyimide etch-stop; (10) plasma-ashing release of the free-standing membrane. Adapted from [1].

3.2.3 Ionization chamber and gas handling

The ionization chamber, designed and built in-house, is an anodized-aluminium block of $80 \times 80 \times 30$ mm outer dimensions enclosing a 2427.3 mm^3 airtight volume with a gas inlet and outlet. The nanomembrane device sits inside the chamber, and the focused laser beam enters through a CaF_2 window at normal incidence onto a single membrane of the array (focused spot diameter < 100 µm). The positive terminal of a high-voltage power supply (Stanford Research Systems PS-350/5000V–25W) is connected to the chamber wall, biasing the chamber relative to the AlSiCu electrodes on the membrane device, which feed the transimpedance amplifier. A bias voltage V_{bias} of 300 V is applied across the chamber/electrode pair to drive photo-generated carriers toward the collection electrodes; this voltage was selected experimentally as a trade-off between signal collection efficiency and the laser-induced multiphoton background discussed in Section 3.4.3.

The gas-sampling subsystem provides the chamber with a controlled mixture of nitrogen carrier and saturated VOC vapor. A side-arm flask contains the liquid VOC at room temperature; a nitrogen flow regulated by mass flow controllers (MFCs; Aalborg Instruments) is split into a carrier line and a dilution line, the carrier line is bubbled through the headspace above the liquid to entrain the saturated vapor, and the diluted mixture is delivered to the chamber. Total flow rate is held at 200 mL/min throughout the experiments. Concentration ranges are calculated from the Antoine equation for the saturated vapor pressure and the dilution ratio (see Appendix D for the test-rig description that preceded this implementation).

The gas lines use Swagelok compression fittings throughout; I selected the 1/4 in OD tubing to match the chamber ports and the MFC fittings, built and leak-tested the gas

panel, integrated the mass flow controllers, and routed the gas lines in the laboratory in coordination with the facility management team. The layout is shown in Fig. 3.3.

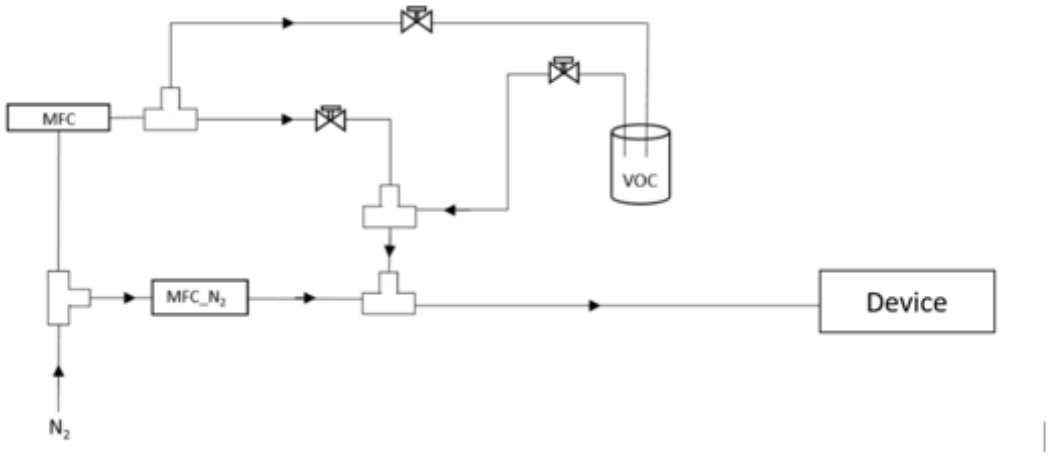


Figure 3.3: Air-sampling system for VOC generation. A nitrogen flow is split at a tee into a carrier line and a dilution line, each metered by a mass flow controller (MFC, MFC_N_2; Aalborg). The carrier line is bubbled through the headspace of the liquid VOC to entrain its saturated vapor, recombined with the dilution line at a second tee, and delivered to the device. The valves select the active line; the delivered concentration follows from the Antoine saturated-vapor pressure and the dilution ratio. Adapted from [1].

3.2.4 Photo-current acquisition chain

The photo-ionization current generated in the chamber is in the picoampere-to-nanoampere range and must be acquired against a substantial laser-induced background (Section 3.4.3). The front-end is a switched-integrator transimpedance amplifier built around the Burr-Brown / Texas Instruments IVC102U: input current is integrated onto a feedback capacitor over a programmable integration window, the resulting voltage step is sampled, and the capacitor is reset (Fig. 3.4). Low-leakage analogue switches (S1, S2) and a fully shielded enclosure minimize drift between integration cycles. The amplifier was assembled in-house on a custom printed circuit board; characterization with a controlled current injection (described in Appendix D) confirmed linear response over the relevant pA–nA range.

The integrator timing, the chamber bias supply, and the MFC set-points are controlled by a National Instruments data-acquisition module (cDAQ chassis with NI-9401 digital I/O and USB-6001 analogue I/O) running a LabVIEW 2023 state machine written for this work. The four-step photoionization protocol, namely (i) N_2 purge, laser OFF; (ii) laser ON, N_2 only; (iii) VOC vapor ON, laser ON; (iv) VOC OFF, laser ON, is enforced from this state machine, and the integrator output is logged synchronously with the protocol step labels for post-processing.

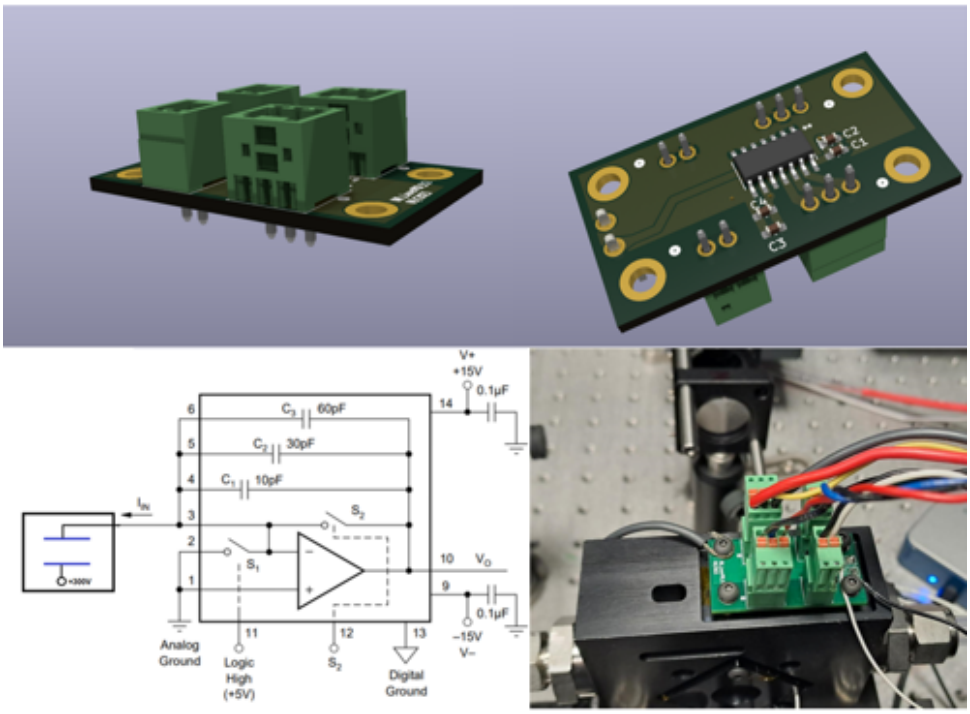


Figure 3.4: Switched-integrator transimpedance front-end. Top: the custom PCB (top and bottom faces). Bottom left: the circuit around the IVC102 integrator, with selectable feedback capacitors ($C_1 = 10\,\text{pF}$, $C_2 = 30\,\text{pF}$, $C_3 = 60\,\text{pF}$), the integrate/reset switches S_1 and S_2, and the $\pm 15\,\text{V}$ supply with $0.1\,\mu\text{F}$ decoupling; the input current I_{IN} is integrated onto the selected capacitor and read out as V_O. Bottom right: the board mounted on the ionization chamber. The front-end was validated against an external current source (Keithley 2400) over the pA–nA range. Adapted from [1].

3.3 Third-harmonic generation in suspended SiO_2 nanomembranes

The efficiency of the VUV generation step in Fig. 3.1 is governed by the dependence of the third-harmonic intensity on the membrane thickness and on the dispersion of SiO_2. This section establishes that dependence analytically (Section 3.3.1) and verifies it with FDTD simulations (Section 3.3.2); the corresponding experimental THG spectra are reported in Section 3.3.3.

3.3.1 Analytical model

For a thin slab of nonlinear material with thickness l, the transmitted third-harmonic electric field can be written as [129]

$$E_{3t}^{3\omega} = E_{2b}^{3\omega} \cdot A \cdot e^{i\vec{k}_{2t}^{3\omega}\cdot\vec{l}} \cdot e^{i(\vec{k}_{2b}^{3\omega}-\vec{k}_{2t}^{3\omega})\cdot\vec{l}}, \tag{3.1}$$

where $E_{2b}^{3\omega}$ is the bound-wave electric field, A is a transmission factor, and $\vec{k}_{2b}^{3\omega}, \vec{k}_{2t}^{3\omega}$ are the bound- and transmitted-wave wavevectors at 3ω. The transmitted third-harmonic intensity is

$$I^{3\omega} \propto \left|E_{3t}^{3\omega}\right|^2 \propto \left|e^{i\psi^{\omega}} - e^{i\psi^{3\omega}}\right|^2. \tag{3.2}$$

Following [13], the intensity can be re-expressed in terms of the phase mismatch $\Delta\psi = \psi^{\omega} - \psi^{3\omega}$ as

$$I^{3\omega} \propto \left|1 - e^{i\Delta\psi}\right|^2, \qquad \Delta\psi = \frac{6\pi l}{\lambda_{\omega}}\left(n^{\omega} - n^{3\omega}\right), \tag{3.3}$$

where λ_{ω} is the pump wavelength and $n^{\omega}, n^{3\omega}$ are the refractive indices of the medium at the fundamental and third-harmonic frequencies, respectively.

The intensity is therefore a periodic function of l at fixed λ_{ω}, with maxima located at thicknesses $l = 2m\,L_c$ (with $m \in \mathbb{N}$), where the coherence length is

$$L_c = \frac{\lambda_{\omega}}{6\left|n^{\omega} - n^{3\omega}\right|}. \tag{3.4}$$

Fig. 3.5 shows the resulting $I^{3\omega}(l, \lambda_{\omega})$ map computed from Eq. (3.3) over the OPA+SHG range 340–440 nm, together with the corresponding $L_c(\lambda_{\omega})$. The expected sinusoidal

modulation in l is recovered, and superimposed on it is a slow decay at large l that the simple lossless expression does not capture. This decay can be attributed to (i) longer optical paths inside the membrane, which increase scattering and diffraction losses, and (ii) the imaginary part of the refractive index at 3ω, which becomes non-negligible as the third harmonic approaches the SiO_2 bandgap edge. In the analytical curve we therefore retain the complex form $n = \eta + i\kappa$ [130]; setting $\kappa \to 0$ recovers the lossless result of Konishi *et al.* [13].

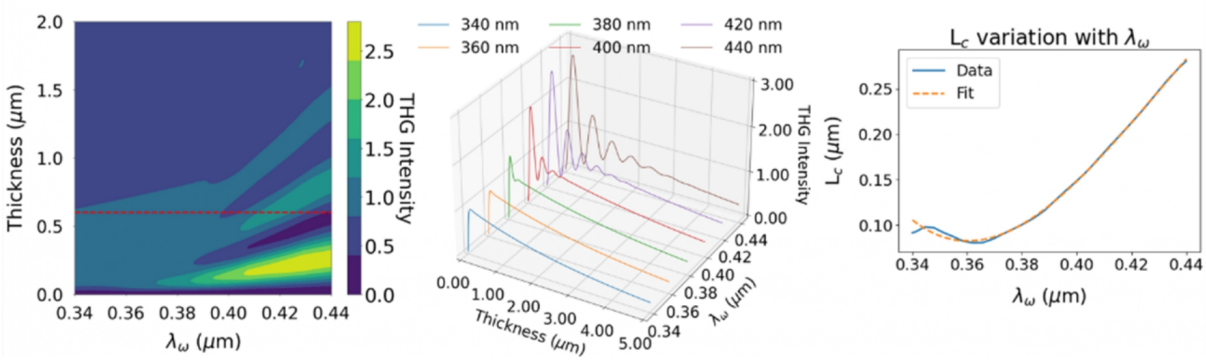


Figure 3.5: Analytical THG intensity from Eqs. (3.3) and (3.4). **(Left)** $I^{3\omega}$ as a function of membrane thickness l and pump wavelength λ_ω. **(Centre)** 3-D view of the same data, with one trace per simulated pump wavelength in 340–440 nm. **(Right)** Coherence length L_c as a function of pump wavelength (data points) and Cauchy-style fit (dashed line). The red dashed line in the left panel marks the 600 nm operating thickness used in this work.

The dashed red line at $l = 600$ nm in Fig. 3.5 (Left) corresponds to the membrane thickness used in this work and selects the THG intensity pattern realised experimentally.

3.3.2 FDTD verification

To verify the analytical model and to enable future extensions to non-planar nanostructures, the same calculation was repeated in a finite-difference time-domain (FDTD) framework (Lumerical FDTD [131]). The simulation domain (Fig. 3.6, inset) consists of a planar SiO_2 slab illuminated at normal incidence by a quasi-monochromatic plane-wave source; field and refractive-index monitors record $E^{3\omega}$ in transmission. The pump electric-field amplitude must be set to a high value ($\sim 10^9$ V/m) for the third-order process to produce a numerically resolvable response. Six pump wavelengths spanning 340–440 nm were simulated, and the resulting $I^{3\omega}(l)$ traces are compared with the analytical curves of Section 3.3.1 in Fig. 3.6.

The simulated and analytical curves agree closely across the longer-wavelength end of the OPA+SHG range. Small deviations appear for $\lambda_\omega = 340$ nm and 360 nm at large l, which we attribute to the Lumerical fit to the tabulated SiO_2 refractive index from Marcos *et al.* [130]: the software fits a single dispersive model to (η, κ) over the whole

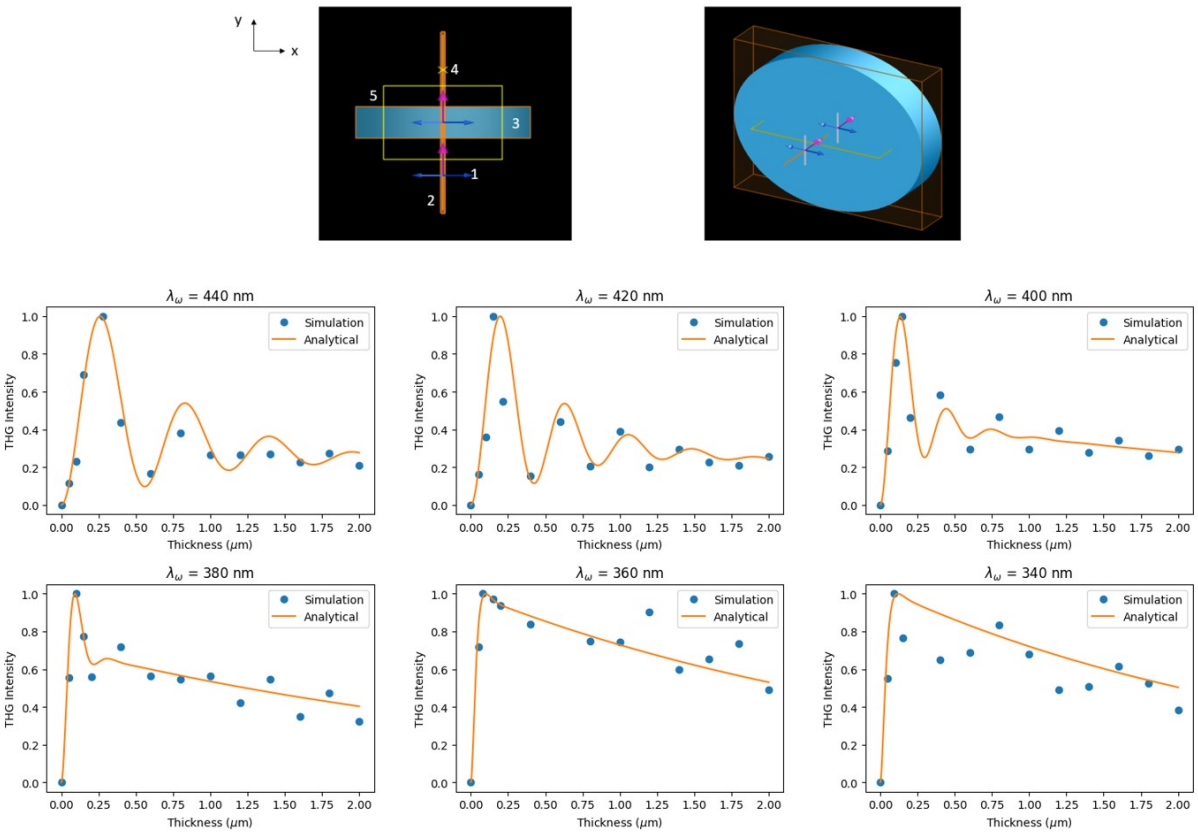


Figure 3.6: FDTD verification of the THG intensity model. **(Top)** Simulation domain: (1) light source, (2) FDTD region, (3) planar nanomembrane, (4) field monitor, (5) refractive-index monitor. **(Bottom)** $I^{3\omega}$ versus membrane thickness at six pump wavelengths in 340–440 nm. Solid orange curves: analytical expression Eq. (3.3); scattered blue points: FDTD simulation. Adapted from [1].

wavelength range, and at the short-wavelength edge the fit residuals on κ are amplified by the longer phase-accumulation path. Over the range relevant for our experiments ($l = 600\,\text{nm}$, $\lambda_\omega \in 340 - -440\,\text{nm}$), the FDTD prediction is consistent with the analytical model within $\sim 10\,\%$.

3.3.3 Experimental THG verification

Direct measurement of the VUV third harmonic (120–180 nm) is constrained by the strong absorption of air and the limited transparency of common optics. To verify THG performance on the fabricated membranes, we therefore measured the conjugate process at near-UV wavelengths: pumping with the OPA fundamental in the 1190–1380 nm range produces a THG signal at $\lambda_\omega/3$, i.e. in the 397–460 nm range, that can be detected with conventional silicon-based instrumentation. The measurement set-up (Fig. 3.7 a,c) uses a 10× objective to focus the laser onto a single membrane, a CMOS camera (uEye UI-1228LE-M; Fig. 3.7 b) to confirm spatial alignment with the membrane centre, and a fibre-coupled spectrometer (Ocean Optics USB 2000+) to record the spectrally resolved THG output, with a dichroic filter blocking the fundamental beam.

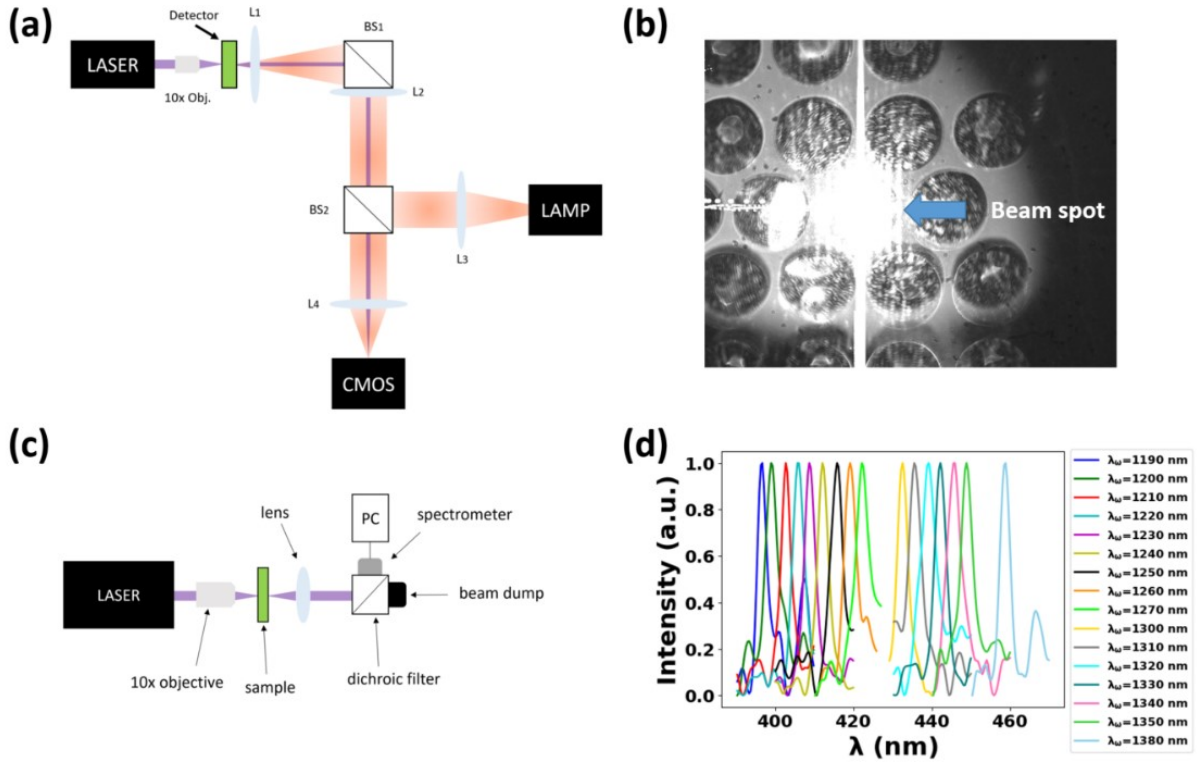


Figure 3.7: Near-UV THG verification on the 600 nm-thick SiO_2 nanomembranes. **(a)** Alignment setup. **(b)** CMOS image of the focused laser spot centred on a single suspended membrane. **(c)** Optical layout used for spectrally resolved THG detection. **(d)** THG emission spectra recorded for pump wavelengths from 1190 nm to 1380 nm in 10 nm steps. Adapted from [1].

Fig. 3.7(d) shows the recorded THG spectra. The third-harmonic peaks track $\lambda_\omega/3$ as

expected, with a measured spectral bandwidth of ~ 3.7 nm FWHM, smaller than the ~ 10 nm bandwidth of the OPA fundamental in this range. Four pump wavelengths (1280, 1290, 1360, 1370 nm) are omitted because of a spurious background generated by an OPA stage transition that overlaps the THG peak. No significant fluctuations in peak intensity were observed across the recorded set, confirming run-to-run stability of the THG process at the membrane.

This near-UV verification, combined with the analytical/FDTD agreement of Section 3.3.2, establishes that the same membrane geometry should deliver THG into the deep-UV / VUV when pumped with the BBO+SHG output at 340–440 nm, producing third-harmonic radiation down to 120 nm. Section 3.4 confirms this through VOC photoionization at 120 nm, which constitutes indirect experimental evidence of VUV generation at the membrane.

3.4 Proof-of-concept VOC photoionization

The platform of Section 3.2 was validated on two representative VOCs: ethanolamine and eucalyptol. Both compounds are low-toxicity and have saturated vapor pressures (SVPs) sufficient for measurable photoionization at room temperature (0.404 mmHg and 1.90 mmHg at 25 °C, respectively). Ethanolamine has a reported ionization potential of 8.96 eV and is the simplest abundant phospholipid head-group with established interest in astrochemistry and life-precursor studies [132, 133]; eucalyptol (1,8-cineole) is a monoterpene oxide widely present in essential oils and consumer products [134, 135, 136, 137, 138], with an ionization potential not previously reported in the literature.

The chamber bias was held at $V_{\text{bias}} = 300$ V, the total gas flow rate at 200 mL/min, and the integrator window at 200 ms, throughout. The four-step acquisition protocol (Section 3.2.4) was repeated three times for each measurement. Concentrations cited below are corrected for the laser-induced background characterized in Section 3.4.3.

3.4.1 Ethanolamine

Photoionization of ethanolamine was investigated with the BBO+SHG output set to $\lambda_\omega = 390$ nm, so that the expected third harmonic at $\lambda_{3\omega} = 130$ nm (9.5 eV) lies just above the ionization threshold (8.96 eV; 138.4 nm[1]). Fig. 3.8(d) shows a representative current trace at 178 ppm ethanolamine in nitrogen: a clear photoionization step appears as soon as the VOC vapor is admitted to the chamber, and disappears when the vapor is shut off. The laser-only background (blue trace) is flat across the same time window. The chemical

[1]The published article [1] quotes ≈138.7 nm; the value 138.4 nm given here is the direct conversion of 8.96 eV via $\lambda = hc/E$.

structure of ethanolamine is shown in Fig. 3.8(b).

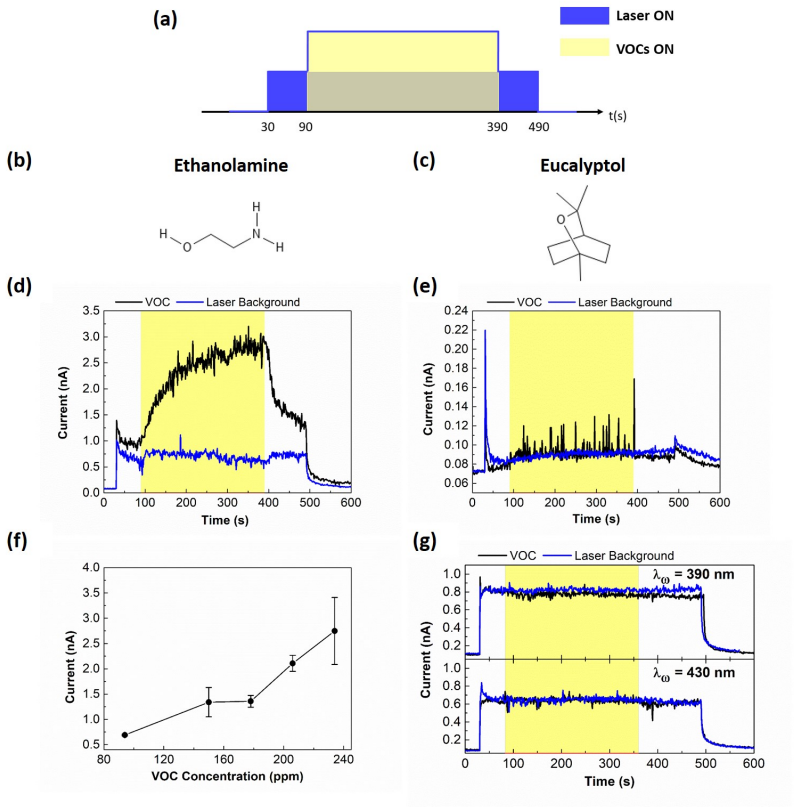


Figure 3.8: Photoionization signals in the in-house ionization chamber. **(a)** Four-step acquisition protocol: laser OFF / laser ON / VOC ON / VOC OFF. **(b,c)** Chemical structures of ethanolamine and eucalyptol. **(d,e)** Current variation for the laser background (blue) and with VOC present (black): **(d)** 178 ppm ethanolamine at $\lambda_\omega = 390\,\text{nm}$ ($\lambda_{3\omega} = 130\,\text{nm}$, 9.5 eV); **(e)** 711 ppm eucalyptol at $\lambda_\omega = 360\,\text{nm}$ ($\lambda_{3\omega} = 120\,\text{nm}$, 10.3 eV). **(f)** Concentration linearity for ethanolamine over 94–234 ppm. **(g)** Eucalyptol response at $\lambda_\omega = 390\,\text{nm}$ (no photoionization) and $\lambda_\omega = 430\,\text{nm}$ (no photoionization), bracketing the eucalyptol ionization threshold. Adapted from [1].

Fig. 3.8(f) reports the photoionization current as a function of ethanolamine concentration over 94–234 ppm, prepared by diluting the SVP-saturated flow with nitrogen at fixed total flow rate. The response is monotonic and approximately linear across the range, as expected for a single-photon threshold process below saturation of the chamber. This linearity is the basis of the quantitative SIPS calibration developed in Chapter 4.

3.4.2 Eucalyptol and bounding the ionization potential

For eucalyptol the OPA+SHG output was set to $\lambda_\omega = 360\,\text{nm}$, corresponding to $\lambda_{3\omega} = 120\,\text{nm}$ ($E_{ph} = 10.3\,\text{eV}$). A clear photoionization step is again observed (Fig. 3.8(e), 711 ppm in nitrogen), confirming that the membrane is generating VUV light at 120 nm, a wavelength at which most bulk frequency-conversion crystals are opaque. The eucalyptol

step is roughly an order of magnitude smaller than the ethanolamine step at comparable concentration, consistent with its higher ionization threshold and different photoionization cross-section.

To bracket the unknown eucalyptol ionization potential, the same sample was illuminated at $\lambda_\omega = 390\,\text{nm}$ ($E_{3\omega} = 9.5\,\text{eV}$) and $\lambda_\omega = 430\,\text{nm}$ ($E_{3\omega} = 8.7\,\text{eV}$); in both cases the photoionization signal collapses to the laser background (Fig. 3.8(g)). Combined with the positive signal at 360 nm (10.3 eV), this places the eucalyptol ionization potential in the interval 9.5–10.3 eV. To our knowledge, this is the first experimental bound on the eucalyptol IP.

3.4.3 Laser-induced background and the multiphoton regime

When the laser is switched on in nitrogen-only flow, a small steady current is observed even in the absence of VOCs: $\sim$1.3 nA at $\lambda_\omega = 390\,\text{nm}$ and $\sim$0.14 nA at $\lambda_\omega = 360\,\text{nm}$. We attribute this to multiphoton excitation of carriers in the SiO_2 membrane itself: the bandgap of our PECVD SiO_2 (determined by XPS-based onset of inelastic losses to be $\sim 8.8\,\text{eV}$) is much larger than the per-photon energy of the pump (3.2 eV at 390 nm), but the intensity at the focus is high enough to drive multiphoton absorption. The Keldysh parameter

$$\gamma = \frac{2\pi c\sqrt{mE_g}}{\lambda|e|E}, \tag{3.5}$$

where c is the speed of light, m the electron mass, $|e|$ the electron charge, λ the pump wavelength, E_g the bandgap, and E the peak electric-field amplitude at the focus, evaluates to $\gamma \approx 46$ at $\lambda = 400\,\text{nm}$. Values $\gamma \gg 1$ correspond to the multiphoton regime, in agreement with this interpretation.

Three control experiments support this assignment (full details in Supplement 1 of [1]). First, the laser-induced current scales nonlinearly with input power at fixed wavelength, consistent with multiphoton order $n \geq 2$ at the relevant photon energy. Second, the same current scales monotonically with V_{bias} from 0 V to 300 V, confirming that the signal is carried by photo-generated charges drifting in the chamber field rather than by drift-independent dark currents in the amplifier. Third, the laser-induced current correlates strongly with the laser output power over the full OPA+SHG range 320–460 nm, and an additional reabsorption contribution becomes visible below 422 nm as the per-photon energy approaches the bandgap edge.

The practical consequence for VOC photoionization is two-fold: (i) the multiphoton background must be characterized at each pump wavelength and subtracted from the VOC signal, as done implicitly through the four-step protocol of Section 3.2.4; and (ii)

further sensitivity improvement requires either lower V_{bias} (which reduces both background and VOC signal, but with a more favourable ratio in some regimes), reduced electrode spacing within the chamber (which preserves the driving field at lower bias), or a material with a larger bandgap and/or smaller multiphoton cross-section in the same wavelength range.

3.5 Conclusion of the chapter

This chapter demonstrated a table-top, wavelength-tunable VUV source based on third-harmonic generation in suspended SiO_2 nanomembranes, coupled to a custom photoionization chamber. The platform delivers quasi-monochromatic VUV light nominally tunable across 120–180 nm, with generation demonstrated experimentally at 130 nm and at 120 nm (10.3 eV), well above the 8.8 eV bandgap of the SiO_2 membrane and within the ionization-energy range of most VOCs. The third-harmonic dependence on membrane thickness and pump wavelength is well captured by a simple analytical model and verified by FDTD simulations, and direct optical measurement at near-UV pump wavelengths confirms the expected $\lambda_\omega/3$ scaling of the THG output. Linear photoionization of ethanolamine and eucalyptol was observed at SVP-derived concentrations, and a switching experiment between three pump wavelengths placed the previously unreported eucalyptol ionization potential in the interval 9.5–10.3 eV.

Two outcomes motivate the remainder of the thesis. First, the linearity of the photoionization signal in concentration combined with the wavelength tunability of the source motivates the SIPS protocol of Chapter 4, in which switching the pump wavelength across molecular ionization thresholds yields molecule-resolved detection in complex mixtures. Second, the THG efficiency at the membrane, currently $\sim 10^{-8}$, sets the sensitivity of the present platform; reaching the ppb regime relevant for occupational exposure thresholds requires field-enhancing nanostructured media. The TiO_2 thin-film and metasurface chapters that constitute the second half of the thesis (Chapters 5–7) develop this enhancement strategy. Both the slow-light and the anapole-resonance regimes aim to amplify the third-harmonic response of a high-index dielectric at the same wavelength range explored here.

4 Real-Time Molecular Specific BTEX Detection at TLV-TWA Concentration Levels

4.1 Introduction

Sample ionization potential spectroscopy (SIPS) is the application for which the tunable source of Chapter 3 was developed. This chapter shows that scanning the photon energy across molecular ionization thresholds suffices to distinguish individual VOCs in a flowing mixture at occupationally relevant concentrations. A conventional lamp-based photoionization detector cannot do this: its fixed-energy emission ionizes every compound whose ionization potential lies below the lamp energy and reports only their summed current, so the BTEX components, whose ionization potentials all fall below the standard 10.6 eV krypton lamp, are indistinguishable. Replacing the lamp with the wavelength-tunable VUV source removes that limit in principle, and the rest of this chapter develops the acquisition and analysis protocol that turns the scan into a quantitative, molecule-resolved measurement.

Together, Chapters 3 and 4 form the complete experimental sensing work of the thesis, and motivate the nanophotonic enhancement strategies pursued in the later chapters. The chapter is laid out as follows. Section 4.1 builds the SIPS principle from the linear photoionization response established in Chapter 3. Section 4.2 describes the second-generation platform, which differs from the Chapter 3 setup in chamber volume, membrane thickness, and front-end integration. Sections 4.3 to 4.5 cover the acquisition protocol, the BTEX results and the reconstructed spectrograph, and the per-compound sensitivity and detection limits, before Section 4.6 closes the chapter.

This chapter is based on the peer-reviewed article *Real-time molecular specific BTEX detection at TLV-TWA concentration levels* (R. Vilarinho, H. Mahmoodnia, C. Maibohm, D. Poeta, C. Calaza, S. Costa, A. Gómez-Varela, M. López-García, A. Miranda, P. De Beule), *Environmental Technology & Innovation* **42** (2026) 104895, DOI: 10.1016/j.eti.2026.104895 [2], on which I am a co-author. It applies the tunable VUV photoionization system directly to the SIPS protocol, demonstrating molecular-specific, real-time detection of individual BTEX compounds within a BTEX/N_2 mixture at occupationally relevant concentrations. My contribution covers the earlier photoionization-detection lamp project that established the experimental lineage reused here (see Section D.1.1), together with research discussion and the writing and editing of the manuscript across its revisions. The principal experimental work, the instrument modifications, and the BTEX data acquisition were carried out by the co-authors, as detailed in the article's author-contributions statement.

4.1.1 Limits of fixed-photon-energy photoionization detection

A conventional lamp-based photoionization detector (PID) uses a single fixed-energy VUV emission and reports the total ionization current generated by all compounds in the sampled gas whose ionization energy lies below the lamp photon energy [9, 37]. As reviewed in Chapter 2 (Section 2.2), this configuration cannot identify which compound contributes to the signal: the response is an aggregate measurement weighted by compound-specific photoionization cross sections, and BTEX components (benzene, toluene, ethylbenzene, and the xylene isomers), which have ionization potentials 8.44–9.24 eV *all* below the standard 10.6 eV Kr lamp, are indistinguishable by lamp-based PID.

Replacing the lamp with the wavelength-tunable VUV source of Chapter 3 removes this limit in principle. Fig. 4.1 contrasts the two architectures: a traditional lamp PID (top branch) versus a SIPS PID (bottom branch), in which the photon energy is scanned across the molecular ionization thresholds of the compounds of interest. The remainder of this chapter develops the data-analysis protocol that makes this scan quantitative.

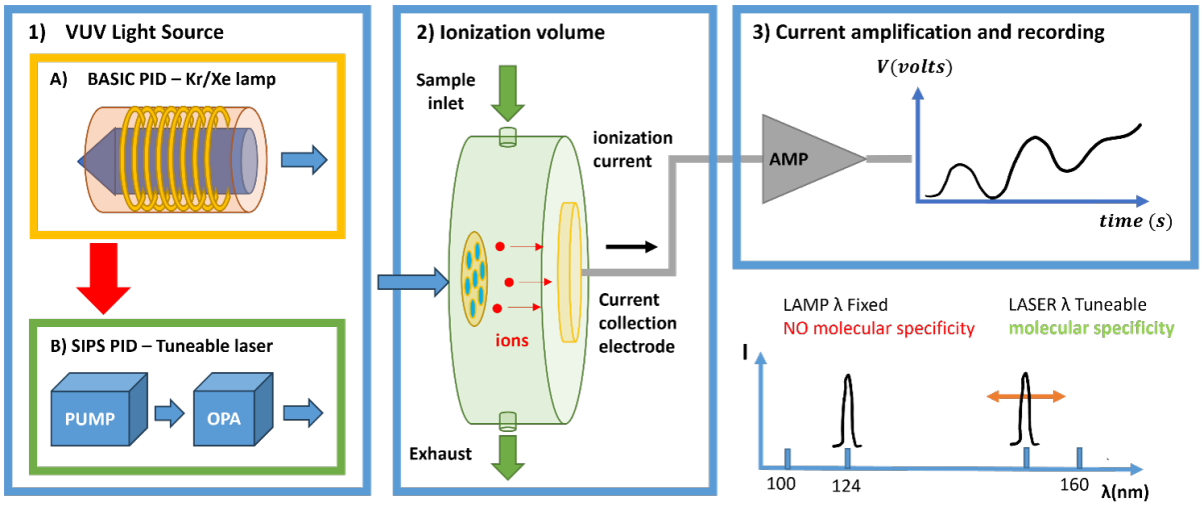


Figure 4.1: Traditional lamp-based PID compared with the SIPS PID. The traditional architecture uses a fixed-energy Kr/Xe lamp and reports the total photoionization current: compounds with $E_{IP} < h\nu$ all contribute, with no molecular selectivity. The SIPS architecture replaces the lamp with the femtosecond-laser + OPA system of Chapter 3, producing quasi-monochromatic VUV that can be tuned across molecular ionization thresholds, enabling molecular-specific detection. Adapted from [2].

4.1.2 Threshold scanning and the SIPS derivative

For a gas mixture containing N compounds with ionization potentials $\{E_{IP,i}\}_{i=1\ldots N}$ and concentrations $\{C_i\}$, the total photoionization current I at photon energy E_{ph} takes the form of a sum of Heaviside-weighted contributions:

$$I(E_{ph}) \;=\; \sum_{i=1}^{N} r_i \, C_i \, \Theta(E_{ph} - E_{IP,i}) \,, \tag{4.1}$$

where r_i is a per-compound photoionization response factor (cross-section, electron-collection efficiency, etc.) and Θ is the step function. As E_{ph} is increased, each successive compound becomes ionizable in turn, producing a monotonically rising staircase. Fig. 4.2 illustrates this idealised behaviour: the tunable VUV source (Fig. 4.2a) produces a monotonically rising current as a function of photon energy (Fig. 4.2b), and the derivative

$$\frac{\partial I}{\partial E_{ph}} \;=\; \sum_{i=1}^{N} r_i \, C_i \, \delta(E_{ph} - E_{IP,i}) \tag{4.2}$$

peaks at each ionization potential, with peak amplitude proportional to $r_i C_i$ (Fig. 4.2c). Equation (4.2) is the operational definition of the SIPS measurement: the derivative of the photoionization current with respect to photon energy is a molecular spectrograph of the gas mixture, with each peak located at an ionization potential and weighted by concentration.

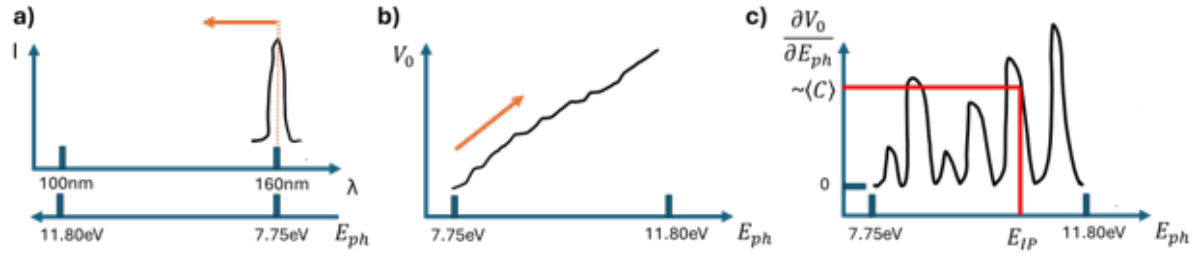


Figure 4.2: SIPS principle. **(a)** The tunable VUV source delivers quasi-monochromatic photons that are scanned (decreasing λ, equivalently increasing E_{ph}) across the VUV range 160–105 nm, covering most VOC ionization potentials (7.75–11.80 eV). **(b)** For a multi-component mixture, the photoionization current rises monotonically with E_{ph}, each compound contributing as E_{ph} crosses its $E_{IP,i}$. **(c)** The derivative $\partial I/\partial E_{ph}$ produces peaks at each $E_{IP,i}$, with amplitude proportional to $r_i C_i$, providing a molecular spectrograph of the mixture. Adapted from [2].

4.1.3 Two-wavelength switching as a finite-difference approximation

Continuous wavelength scanning is impractical with the present OPA+SHG chain because mode hops and OPA stage transitions introduce intensity jumps that contaminate $\partial I/\partial E_{ph}$. Instead, for each compound of interest we approximate the derivative at its known ionization potential $E_{IP,i}$ by a two-point finite difference,

$$\left.\frac{\partial I}{\partial E_{ph}}\right|_{E_{IP,i}} \;\approx\; \frac{I(\lambda_{\mathrm{VUV-H},i}) - I(\lambda_{\mathrm{VUV-L},i})}{E_{ph}(\lambda_{\mathrm{VUV-H},i}) - E_{ph}(\lambda_{\mathrm{VUV-L},i})}, \tag{4.3}$$

where $\lambda_{\mathrm{VUV-H},i}$ and $\lambda_{\mathrm{VUV-L},i}$ are two VUV wavelengths bracketing $\lambda_{IP,i} = hc/E_{IP,i}$ from above and below, respectively. In practice $|\lambda_{\mathrm{VUV-H},i} - \lambda_{\mathrm{VUV-L},i}|$ is set to a few nanometres around the ionization threshold of the compound of interest. The BTEX ionization potentials span 8.44–9.24 eV, with the closest separation between ethylbenzene (8.77 eV) and toluene (8.83 eV) corresponding to a wavelength separation of 1.0 nm at the corresponding VUV wavelength. As discussed in the published article [2], the required spectral separation between $\lambda_{\mathrm{VUV-H}}$, λ_{IP}, and $\lambda_{\mathrm{VUV-L}}$ must take into account the spectral bandwidth of the laser, which is highly dependent on the laser pulse duration.

4.2 Second-generation experimental system

The system used here is a second-generation integration of the source built in Chapter 3, redesigned for lower SIPS background. The ionization chamber, the suspended membranes and the read-out electronics all changed.

The new in-house chamber is an anodized-aluminium block of $100 \times 38 \times 18$ mm outer dimensions enclosing a 307 mm^3 airtight volume, almost 8 times smaller than the chamber of Chapter 3. At the total flow rate of 200 mL/min the nominal residence time is below 0.1 s, whereas the observed response to a change in gas composition takes seconds (Section 4.4.1). Mixing and the dead volume of the gas lines therefore set the response time, not the chamber volume.

The suspended membranes are thinner. The PECVD recipe of Chapter 3 (Section 3.2.2) was modified to give 300 nm-thick SiO_2 membranes of reduced diameter (0.5 mm). The analytical THG model of Section 3.3.1 predicts that halving the thickness moves the working point closer to the first coherence length maximum at the OPA+SHG wavelengths used here, and therefore raises the per-pulse VUV generation efficiency.

The read-out is now integrated. The dielectric membrane device, the bias electrodes and the transimpedance front-end sit on a single printed circuit board, shown in Fig. 4.3. This removes the long cable run between the chamber electrodes and the IVC102U amplifier, which contributed to the laser induced background of Chapter 3. The chamber bias is reduced to $V_{\mathrm{bias}} = 50$ V across a ~ 5 mm electrode separation, giving a field of ~ 100 V/cm, within the range used in ion mobility spectrometry [139].

The optical chain upstream of the membrane is unchanged from Chapter 3: Yb laser (Carbide), OPA (Orpheus, 720–880 nm), SHG in BBO (Lyra, 360–440 nm), with THG in the SiO_2 membrane producing the VUV at 120–180 nm. The transimpedance integration window is 200 ms, and the data-acquisition path is the same NI DAQ + LabVIEW state machine described in Section 3.2.4.

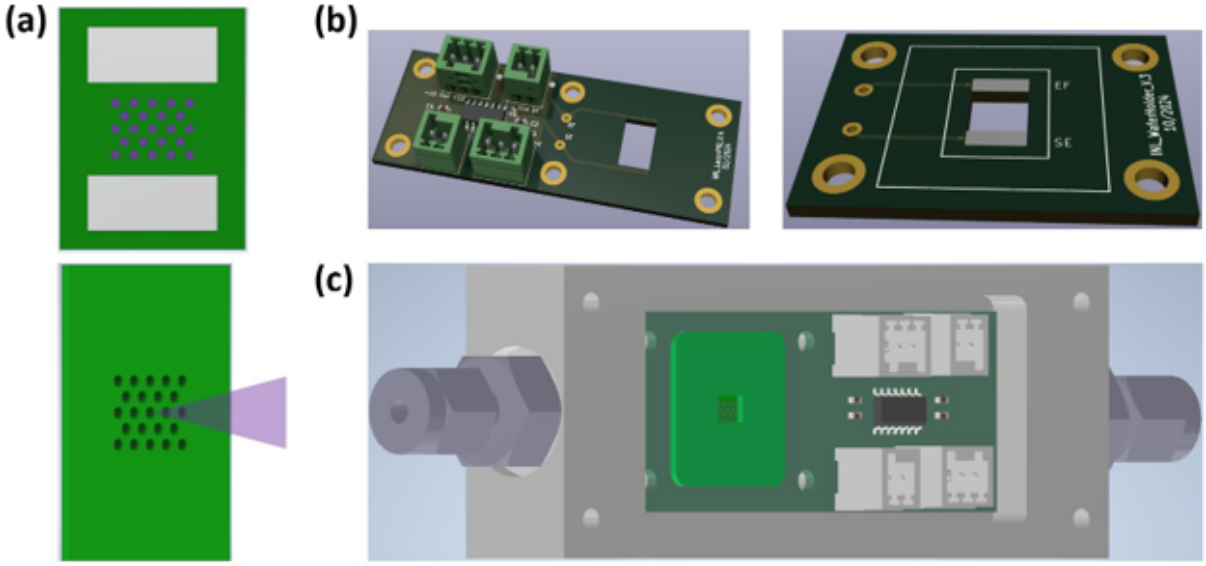


Figure 4.3: Sensor integration. (a) Membrane device: SiO_2 suspended nanomembranes (purple), metal electrodes (grey), and the silicon substrate (green); the focused laser is aligned to the centre of one nanomembrane. (b) Transimpedance board (left) and the small carrier PCB that contacts the membrane with silver paste and is soldered to the transimpedance board (right). (c) Assembly of the ionization chamber, electronics, and membrane device. Adapted from [2].

For the BTEX measurements, the gas mixture was supplied from a calibrated cylinder purchased from Linde Portugal, containing benzene (0.5 ppm), toluene (50 ppm), ethylbenzene (20 ppm), *para*-xylene (50 ppm), and *ortho*-xylene (50 ppm) in N_2 (Fig. 4.4).

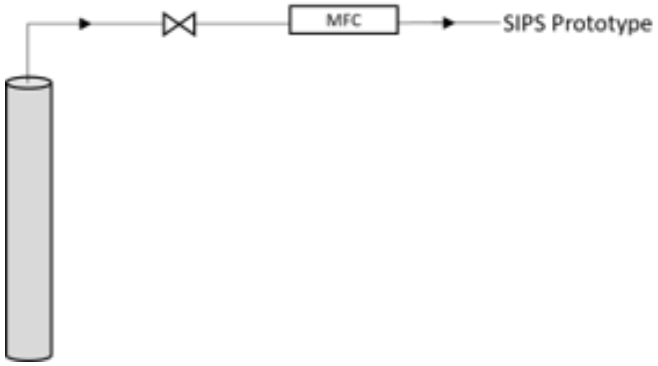


Figure 4.4: Gas delivery for the BTEX measurements: the BTEX mixture is supplied from a certified gas cylinder, regulated through a shut-off valve and a mass flow controller (MFC), and delivered to the SIPS prototype. Adapted from [2].

4.3 SIPS acquisition protocol

The two-point finite difference of Eq. (4.3) is realised by alternating the OPA wavelength between $\lambda_{\mathrm{VUV-H},i}$ and $\lambda_{\mathrm{VUV-L},i}$ for each compound of interest while a constant BTEX mixture flows through the chamber. Fig. 4.5 shows the timing diagram. After a 30 s N_2 purge with the laser off, the laser is switched on at the first pump wavelength and

held for 90 s (180 s for benzene, see below) while only N_2 flows; this establishes the pre-exposure baseline under laser illumination. The BTEX mixture is then admitted to the chamber, and the wavelength is alternated between $\lambda_{\mathrm{VUV-H},i}$ and $\lambda_{\mathrm{VUV-L},i}$ *twice*, giving two independent estimates ΔI_1 and ΔI_2 of the SIPS step at $E_{IP,i}$. The laser is switched off at $t = 570$ s (1140 s for benzene).

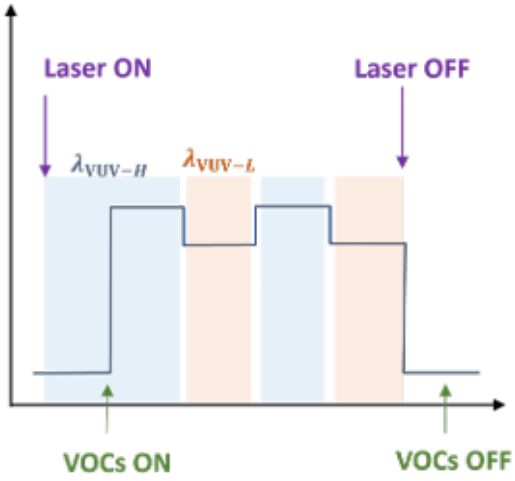


Figure 4.5: Acquisition timing for one SIPS measurement. The laser is switched on after a N_2 purge; the BTEX mixture is admitted (VOCs ON); the OPA wavelength is alternated twice between $\lambda_{\mathrm{VUV-H}}$ (blue band) and $\lambda_{\mathrm{VUV-L}}$ (orange band) bracketing the ionization potential of the target compound; the laser is switched off (Laser OFF) at the end of the sequence. The benzene measurement uses doubled hold times because of the lower target concentration. Adapted from [2].

The longer hold time for benzene reflects its lower target concentration (0.5 ppm versus 50 ppm for toluene, ortho-xylene, and para-xylene, and 20 ppm for ethylbenzene), which in turn reflects benzene's classification as a human carcinogen and the correspondingly stringent occupational threshold. The concentrations were chosen with respect to the time-weighted-average occupational exposure limits (OEL-TWA) defined for the European Union: for toluene and the xylene isomers the experimental concentration is at the OEL-TWA, and for benzene and ethylbenzene it is below.

4.4 Results

A calibrated BTEX mixture in nitrogen (benzene 0.5 ppm, toluene 50 ppm, ethylbenzene 20 ppm, ortho-xylene 50 ppm, para-xylene 50 ppm; Linde Portugal) was loaded into the chamber, and five separate SIPS acquisitions were performed using the wavelength pairs listed in Table 4.1.

Table 4.1: BTEX target compounds: ionization potentials, corresponding VUV wavelengths, the two-point switching wavelengths used in this work, and the mixture concentrations. Ionization potentials taken from [4].

VOC	E_{IP} (eV)	λ_{IP} (nm)	$\lambda_{\mathrm{VUV-H}}$ (nm)	$\lambda_{\mathrm{VUV-L}}$ (nm)	C (ppm)
Benzene	9.24	134.2	133.3	135.0	0.5
Toluene	8.83	140.4	139.3	140.7	50
Ethylbenzene	8.77	141.4	140.3	141.7	20
o-Xylene	8.56	144.9	144.0	145.7	50
p-Xylene	8.44	146.9	144.3	147.3	50

4.4.1 BTEX SIPS traces

Fig. 4.6 shows the five SIPS current traces, one panel per compound. In every panel, a well defined step in current is observed when the OPA wavelength is switched between $\lambda_{\mathrm{VUV-H}}$ and $\lambda_{\mathrm{VUV-L}}$, with the higher current at the shorter wavelength as expected from Eq. (4.3).

The 40 nA baseline is not laser induced. It is present before the laser is switched on, and switching the laser on leaves it unchanged. In Chapter 3, by contrast, switching the laser on added a background that we attributed to electron hole pair formation in the 600 nm membrane. No such background appears with the platform of Section 4.2. The response to a wavelength switch settles within a few seconds, set by the 200 ms integration window of the transimpedance front end and by the OPA wavelength-shift time [2].

The two SIPS steps differ by a fixed factor, $\Delta I_2/\Delta I_1 \approx 1.2$, for all five compounds (Table 4.2). We attribute the $\sim 20\,\%$ rise to slow adsorption of BTEX on the chamber walls: the walls take up BTEX during the first wavelength cycle and saturate, so more of it stays in the gas phase during the second. The signal returns to baseline when the laser is switched off, so the current is laser driven rather than set by fluctuations in the BTEX flux.

4.4.2 Reconstructed spectrograph and per-compound calibration

The discrete set of SIPS steps in Fig. 4.6 can be reconstructed into the spectrograph and derivative representations of Fig. 4.2. Fig. 4.7(a) plots the cumulative photoionization current as a function of photon energy from 8.4 eV to 9.4 eV, obtained by ordering the BTEX compounds by their ionization potential and summing each ΔI step in turn. The current rises monotonically as expected from Eq. (4.1). Fig. 4.7(b) shows the corresponding derivative, presented as a bar at each ionization potential with height $\delta I/\delta E_{ph}$. Each bar is labelled with the corresponding BTEX compound and its concentration in the mixture, and the bar heights summarize the SIPS sensitivities reported in Table 4.2.

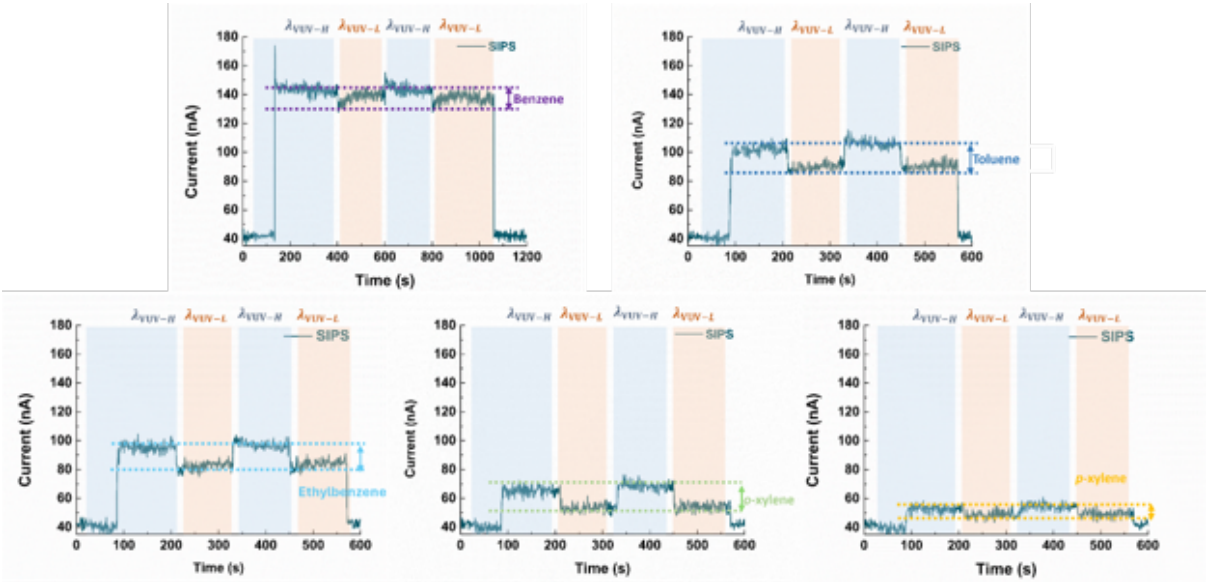


Figure 4.6: SIPS current traces for the five BTEX compounds at the wavelength pairs listed in Table 4.1. Blue bands: $\lambda_{\mathrm{VUV-H}}$; orange bands: $\lambda_{\mathrm{VUV-L}}$. Top row: (a) benzene at 0.5 ppm, (b) toluene at 50 ppm. Bottom row: (c) ethylbenzene at 20 ppm, (d) o-xylene at 50 ppm, (e) p-xylene at 50 ppm. In each panel a clear SIPS step is observed when the wavelength crosses the corresponding ionization potential. Adapted from [2].

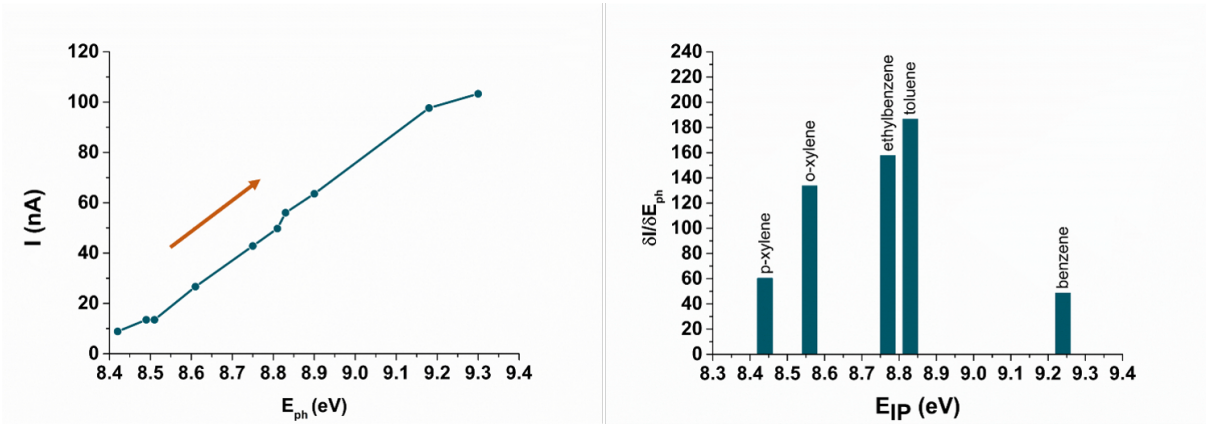


Figure 4.7: Reconstructed sample ionization potential spectrograph for the BTEX mixture. **(a)** Cumulative photoionization current as a function of photon energy, obtained from the SIPS steps of Fig. 4.6. **(b)** Derivative $\delta I/\delta E_{ph}$ at each BTEX ionization potential; bar colours indicate the contributing compound, and labels above each bar give the mixture concentration. Adapted from [2].

Table 4.2: SIPS measurement summary for the five BTEX compounds. $\Delta I_1, \Delta I_2$: first and second SIPS steps. SIPS sensitivity S_{BTEX}: SIPS step normalised by mixture concentration. Limit of detection $\mathrm{LOD}_{\mathrm{VOC}}$ defined as in Eq. (4.4).

VOC	ΔI_1 (nA)	ΔI_2 (nA)	$\Delta I_2/\Delta I_1$	S_{BTEX} (nA/ppm)	$\mathrm{LOD}_{\mathrm{VOC}}$ (ppm)
Benzene	5.15	6.12	1.19	10.29–12.25	0.26
Toluene	11.95	15.73	1.31	0.24–0.31	12.22
Ethylbenzene	12.77	13.44	1.05	0.64–0.67	5.13
o-Xylene	11.91	14.42	1.21	0.24–0.29	12.23
p-Xylene	4.17	5.09	1.22	0.08–0.10	25.95

4.5 Discussion and limitations

The five SIPS sensitivities $S_{\mathrm{BTEX}} = \Delta I/C$ in Table 4.2 span almost two orders of magnitude, from $\sim$0.1 nA/ppm for p-xylene to $\sim$12 nA/ppm for benzene. As reported in the published article [2], the sensitivity of the SIPS sensor, defined by the current response relative to the VOC concentration, is compound specific, just as for traditional PID sensors. Per-compound calibration with known concentrations, as performed here through the calibrated BTEX mixture, is therefore required for quantitative use.

The limit of detection for each compound is defined by comparing the SIPS step to the current fluctuations $\Delta I_{\mathrm{noise}}$ observed under stable operating conditions:

$$\mathrm{LOD}_{\mathrm{VOC}} \;=\; \frac{\Delta I_{\mathrm{noise}}}{\Delta I_{\mathrm{VOC}}}\, C_{\mathrm{VOC}}. \tag{4.4}$$

This definition corresponds to a signal-to-noise ratio of unity. Applying the more conservative 3σ criterion commonly used for analytical detection limits would raise the benzene limit of detection from 0.26 ppm to $\approx$ 0.78 ppm, which remains below the European Union OEL-TWA of 1 ppm but with a substantially reduced margin.

The LOD column of Table 4.2 confirms the cross-section ordering: the LOD for benzene is below 1 ppm, while the LOD for p-xylene is 26 ppm. The benzene LOD of 0.26 ppm is below the European Union OEL-TWA of 1 ppm, demonstrating that SIPS detection of benzene at occupationally-relevant concentrations is achievable with the present platform. For toluene and the xylenes, the LOD is above the respective OEL-TWA at the present sensitivity, and further improvement is required for compliance-grade monitoring at the legal limit; this is the practical motivation for the nanophotonic field-enhancement strategies pursued in the remainder of the thesis.

4.6 Conclusion of the chapter

This chapter demonstrated sample ionization potential spectroscopy (SIPS) as a route to molecular-specific, real-time detection of BTEX compounds in mixture, using the tunable VUV source of Chapter 3. The SIPS measurement is a two-point finite-difference approximation to the derivative of the photoionization current with respect to photon energy, and its peaks at the ionization potentials of the five BTEX compounds were clearly resolved in a single calibrated mixture. The benzene limit of detection of 0.26 ppm is below the European Union OEL-TWA, showing that the technique is viable for occupational monitoring of the most toxic BTEX component without GC–MS infrastructure. Improving sensitivity for the other BTEX compounds toward their respective OEL-TWAs requires more efficient VUV generation, which motivates the nanophotonic enhancement strategies developed in the remaining chapters: a high-index dielectric (TiO_2) thin-film in Chapter 5, the slow-light in Chapter 6, and the anapole-metasurface design in Chapter 7.

5 Material Study of ALD-grown TiO_2 Thin Films for UV Nanophotonics

5.1 Publication context and author contributions

This chapter presents the material characterization that forms the supplementary material of the manuscript introduced in Chapter 6; the manuscript has been submitted to *Optical Materials Express*, with me as lead author. I designed the ALD growth study and ran the spectroscopic ellipsometry and the UV–Vis–NIR measurements myself. The depositions were carried out together with Andrea Gouvêa, Carlos Calaza and Hélder Fonseca. The XRD was measured by Elvira Paz, the XPS, REELS, and UPS by Oleksandr Bondarchuk, and the AFM by Adelaide Miranda. In each case the collaborators operated the instrument and provided the raw images or spectra, while I processed and interpreted all of the data and wrote the manuscript.

5.2 Scope of the chapter

The aim is to establish how the ALD oxidant sets the optical constants on which the energy-velocity and anapole analyses depend. Growth is described first (Section 5.3), including the Al_2O_3 calibration used to fix the O_3 flow rate. Optical characterization follows in Section 5.4: ellipsometric modelling and depolarization, the angle-dependent UV–Vis–NIR reflectance of the thick O_3 film, and the matching transfer-matrix and FDTD simulations. Section 5.5 covers surface and structure (AFM roughness and the XRD evidence for partial anatase), and Section 5.6 the chemistry and electronic structure (XPS stoichiometry, and REELS and UPS for the bandgap and Ti^{3+} defects).

5.3 Thin-film growth

5.3.1 ALD process and oxidant selection

To investigate the effect of oxidizing agents on the properties of atomic layer deposited (ALD) TiO_2 films, we utilized an ALD system (Beneq TFS200) operated in a cleanroom environment. We chose tetrakis(dimethylamido)titanium (TDMAT) as the Ti precursor for its compatibility with both thermal and plasma enhanced ALD. Three oxidizing agents were compared: deionized water (H_2O), ozone (O_3), and hydrogen peroxide (H_2O_2). All depositions were carried out on silicon substrates at a fixed temperature of 200 °C and a working chamber pressure of 0.2 Torr.

Each ALD cycle consisted of a TDMAT pulse, followed by a nitrogen purge, then an oxidant pulse, and a final purge step. For each oxidant (H_2O, O_3, H_2O_2), we performed 1250 ALD cycles to grow TiO_2 films. In addition, an O_3 film of 2135 cycles was deposited (targeting near 100 nm).

As shown in Fig. 5.1, the deposited TiO_2 films prepared using different oxidants exhibit different colors. Even for the same number of ALD cycles (1250), the samples exhibit clearly different visible colors, with the H_2O-grown film appearing golden, the O_3-grown film dark brown, and the H_2O_2-grown film dark blue/purple. The thicker O_3-grown film deposited using 2135 cycles appears green. These color differences qualitatively indicate that the choice of oxidant affects the optical response of the deposited films, likely through changes in film thickness and/or refractive index under otherwise identical deposition conditions.

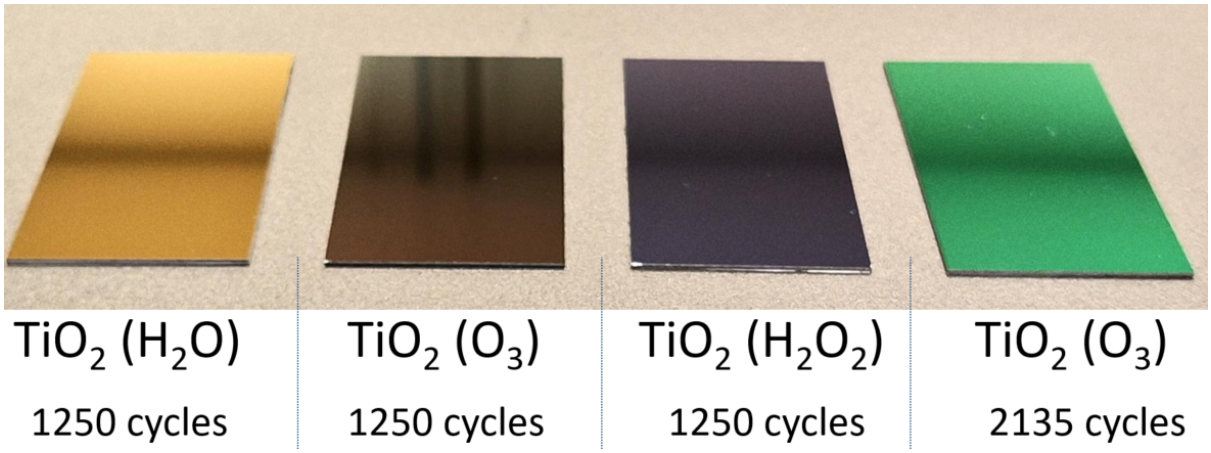


Figure 5.1: Photographs of ALD-grown TiO_2 coatings deposited using different oxidants and cycle numbers: H_2O (1250 cycles), O_3 (1250 cycles), H_2O_2 (1250 cycles), and O_3 (2135 cycles), from left to right.

To ensure reliable growth and account for our chamber's larger volume (which can affect growth per cycle relative to prior studies [86]), we first performed calibration runs using Al_2O_3. Reflectometry data (Hyperion NanoCalc) showed that varying the O_3 flow rate from 12–40 L/min on Si wafers nearly doubled the Al_2O_3 thickness for a given number of cycles, demonstrating a strong oxidant-flow dependence. (Details of this calibration are provided in Section 5.3.2 and Table 5.1). We then applied the optimized O_3 flow conditions to TiO_2 deposition.

Due to hardware limitations, we did not use an O_2-plasma step. Thus, comparisons with plasma-enhanced ALD recipes (e.g., Vazquez-Arce *et al.*, [86]) were not performed, and we focused only on differences between thermal oxidants.

5.3.2 Growth calibration and thickness control

Table 5.1 and the associated reflectometry / ellipsometry measurements were used to calibrate the ALD process. For TiO_2, initial depositions (e.g., 20 nm) showed a high refractive index and negligible absorption ($k \approx 0$), consistent with a transparent TiO_2 film. These were well fitted by a Tauc–Lorentz model, typical for amorphous dielectrics [140].

However, thicker TiO_2 films (e.g., 100 nm) exhibited spectral features inconsistent with the same model, suggesting increased absorption. A Tauc–Lorentz fit underestimated the true thickness, indicating a deviation from purely amorphous behavior. When adjusted to include absorption tails, the optical model better matched the measured spectra, consistent with observations of "black TiO_2" reported in the literature [86].

To prevent such changes, we adjusted ALD recipes and conducted control experiments. These included Al_2O_3 reference layers (with TMA and O_3) to confirm proper ozone flow during deposition, followed by a series of TiO_2 films using different oxidants (H_2O, O_3, and H_2O_2) and precursor combinations. These comparisons were used to understand and avoid deposition conditions that promote excess absorption. We then adopted the Cauchy model described in Section 5.4 to fit the optical constants of the thicker TiO_2 films.

Table 5.1: Thickness and uniformity of Al_2O_3 films deposited at different O_3 flow rates (725 μm Si substrates).

Wafer	O_3 Flow (L/min)	t_{mean} (nm)	t_{std} (nm)	t_{min} (nm)	t_{max} (nm)	Non-uniformity (%)
(a)	40	96.7	1.3	94.6	99.8	2.69
(b)	12	49.9	0.7	48.5	52.0	3.51
(c)	40 (readjusted)	89.7	0.5	88.5	90.4	1.06
(d)	45	93.8	0.9	92.2	95.4	1.71

5.4 Optical characterization

5.4.1 Ellipsometric modelling and extraction of optical constants

SE determines TiO_2 film thickness and refractive index by fitting Ψ (λ) and Δ (λ) spectra between 370–1000 nm using a Cauchy dispersion model appropriate for transparent dielectrics. Ellipsometry (J.A. Woollam M-2000 SE) was modeled with a Cauchy + Urbach absorption function and a thin surface-roughness layer (Table 5.2). Depolarization analysis shows a strong near-UV increase for the thick O_3 film, indicating a small but non-negligible absorption; including a tiny k improves the fit (Fig. 5.2). This supports using an absorptive model for n, k in the 370–400 nm range.

The optical constants of the TiO_2 film were modeled using a Cauchy dispersion model combined with Urbach absorption to account for sub-bandgap absorption. A surface roughness layer was included. Parameters were extracted using CompleteEASE software

(J.A. Woollam Co.) from spectroscopic ellipsometry measurements over 370–1000 nm. The mean squared error (MSE) obtained for this sample (O_3-grown TiO_2 2135 cycles) is higher than that of thinner films, as the increased thickness amplifies sensitivity to optical interference effects and fitting errors.

Table 5.2: Ellipsometry fit parameters for the O_3-grown TiO_2 thin film (2135 cycles) using the Cauchy + Urbach absorption model.

Parameter	**Value**
Model	Cauchy + Urbach absorption
Film thickness (nm)	100.9 ± 0.05
Surface roughness (nm)	1.66 ± 0.04
Cauchy A	2.421 ± 0.001
Cauchy B (μm^2)	0.0074 ± 0.00009
Cauchy C (μm^4)	0.0106 ± 0.0002
Urbach amplitude	0.0124
Urbach exponent	1.86 ± 0.15
Absorption edge (nm)	400
Mean squared error (MSE)	7.39

5.4.2 Depolarization and near-UV absorption

To validate the optical model used for the different deposited films, we analyzed the depolarization response of the films [141]. Depolarization in spectroscopic ellipsometry arises from factors such as surface roughness, inhomogeneities, or optical absorption. A strong depolarization signal near 370 nm was observed for the O_3-grown TiO_2 film, indicating absorption in the near-UV that is incompatible with the previously fitted model with $k = 0$. We therefore introduced a small extinction coefficient (k) in the model, which significantly decreased the mean squared error (MSE) of the fit. By contrast, the H_2O_2-grown film showed low and stable depolarization across the measured spectral range, consistent with lower absorption and reduced sensitivity to the extinction term. These findings indicate that strongly oxidizing ALD processes such as O_3 and thicker films lead to increased absorption, even in relatively thin TiO_2 layers.

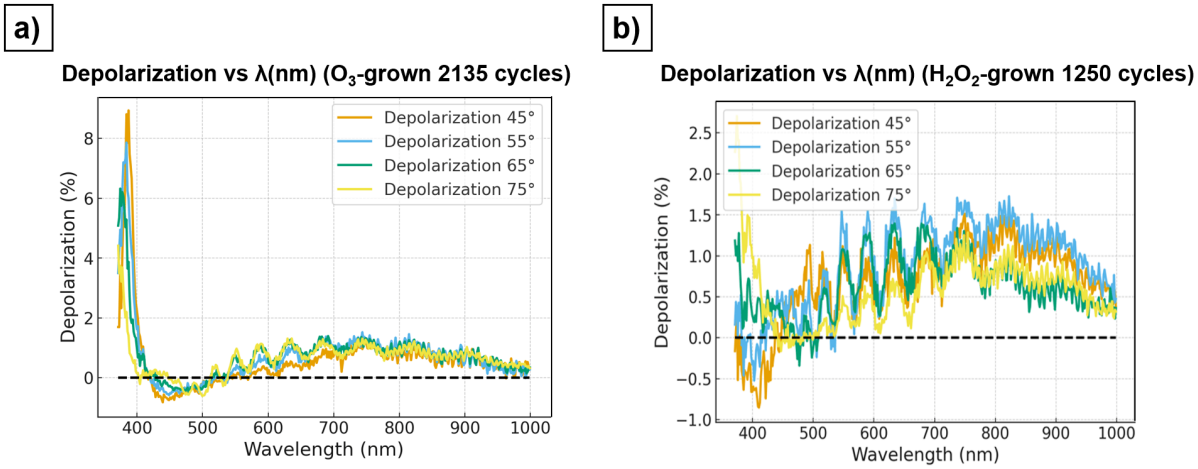


Figure 5.2: Depolarization percentage as a function of wavelength at different incident angles for TiO_2 films grown with **a)** O_3-grown (2135 ALD cycles) and **b)** H_2O_2-grown (1250 cycles).

5.4.3 UV–Vis–NIR reflectance

Optical reflectance was evaluated through UV-Vis-NIR using (Lambda 950, PerkinElmer). Specular reflectance was measured from 300–800 nm using a variable-angle relative reflectance accessory. Prior to each measurement, the system was calibrated using a mirror at 45° incidence to establish a baseline, ensuring accurate comparison across all wavelengths. The investigated samples can be described as an asymmetric Fabry–Pérot-type thin-film stack consisting of air/TiO_2/Si [142]. In the UV-A spectral range considered here, the Si substrate is strongly absorbing and can therefore be treated as optically semi-infinite, such that the relevant optical response is determined by the air/TiO_2/Si stack. This stack determines the interference conditions and the field distribution inside the TiO_2 layer, and is therefore also relevant to the THG response discussed below.

Fig. 5.3 **a)** compares the specular reflectance of TiO_2 films grown with different oxidants (H_2O, H_2O_2, O_3) at 45° incidence. The thicker O_3-grown film shows a pronounced dip at $\approx$ 380 nm. Similar dips in TiO_2 film reflectance spectra have been reported in the literature [143]. In Fig. 5.3 **b)**, we examine only this O_3-grown $\approx$ 98 nm film at different incidence angles (30°, 45°, 60°) and find that the dip remains at the same wavelength across all angles. We believe that the dip is not due to angular dependent interference but is an intrinsic optical feature of the film, consistent with enhanced light trapping near the band edge.

Fig. 5.3 also compares O_3 films at 59 nm and 98 nm thickness. Only the thicker 98 nm film shows the pronounced dip near 380 nm, while both films remain flat beyond 700 nm (250–1400 nm sweep). The sharp feature at 830 nm is a lamp-switch artifact of the PerkinElmer Lambda instrument.

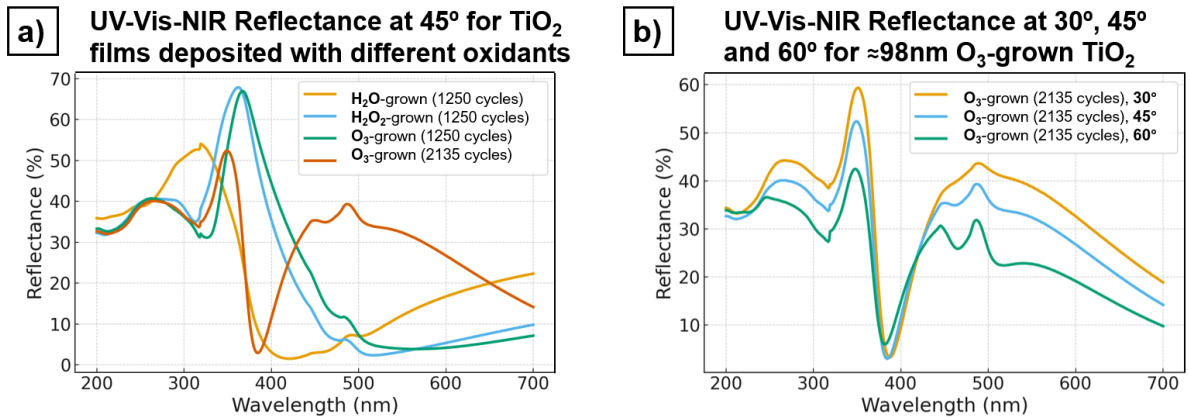


Figure 5.3: UV–Vis–NIR specular reflectance measurement of ALD-grown TiO_2 films. **a)** Comparison of films deposited with different oxidants, all measured at a 45° incidence angle. The O_3-grown film (2135 cycles, orange) shows reduced reflectance across the visible range compared to the O_3-grown film (1250 cycles, green), H_2O_2-grown film (1250 cycles, blue), and H_2O-grown film (1250 cycles, yellow). **b)** Angle-dependent reflectance of the O_3-grown TiO_2 film (2135 cycles, $\approx$ 98 nm thick) measured at 30° (yellow), 45° (blue), and 60° (green).

5.5 Surface and structural characterization

5.5.1 AFM morphology and roughness

AFM topography scans were performed in intermittent contact (tapping) mode in air (JPK Nanowizard 3 AFM, Bruker Nano GmbH, Berlin, Germany). Silicon nitride cantilevers with pyrex-nitride tips (PPP-NCHAu, Nanosensors) were used; they had a nominal resonance frequency of 330 kHz in air and a spring constant of 29 N/m. The AFM instrument is mounted on a noise vibration platform (i4, Accurion GmbH, Göttingen, Germany) and the entire set-up is installed on an ultra-low noise floor constructed to comply with the NIST-A vibration specifications [144]. The system has a demonstrated noise floor of $\approx$ 75 pm [145]. All investigations were carried out with out-of-the-box cantilevers. Images with a scan size of ($2\,\mu m \times 2\,\mu m$) were acquired in three different areas for each sample. Image analysis and determination of the roughness parameters were performed using the JPK Data Processing software (version 6.1.116).

Fig. 5.4 shows representative $2 \times 2\,\mu m$ AFM topography scans for TiO_2 films grown with **a)** H_2O, **b)** H_2O_2, and **c)** O_3, each deposited with 1250 cycles, and **d)** O_3 with 2135 cycles. The surface roughness is notably higher in the thick O_3-grown film (d), which exhibits larger grains and surface features compared to the thinner films.

AFM measurements show that the surface morphology depends strongly on the choice of oxidant. H_2O-grown films (1250 cycles) show relatively large granular features despite their smaller thickness, while O_3-grown films develop larger grains and defect-like features

once the thickness reaches ≈ 98 nm (2135 cycles), with more pronounced surface structures overall. Such morphological changes can enhance light scattering and contribute to increased optical losses, but they do not necessarily reduce the intrinsic refractive index of the material; indeed, our ellipsometry showed that the thicker O_3 film has the highest refractive index. The formation of crystalline anatase phases, confirmed by XRD (Section 5.5.2, Fig. 5.5), generally leads to a higher refractive index due to increased packing density and order within the film [3].

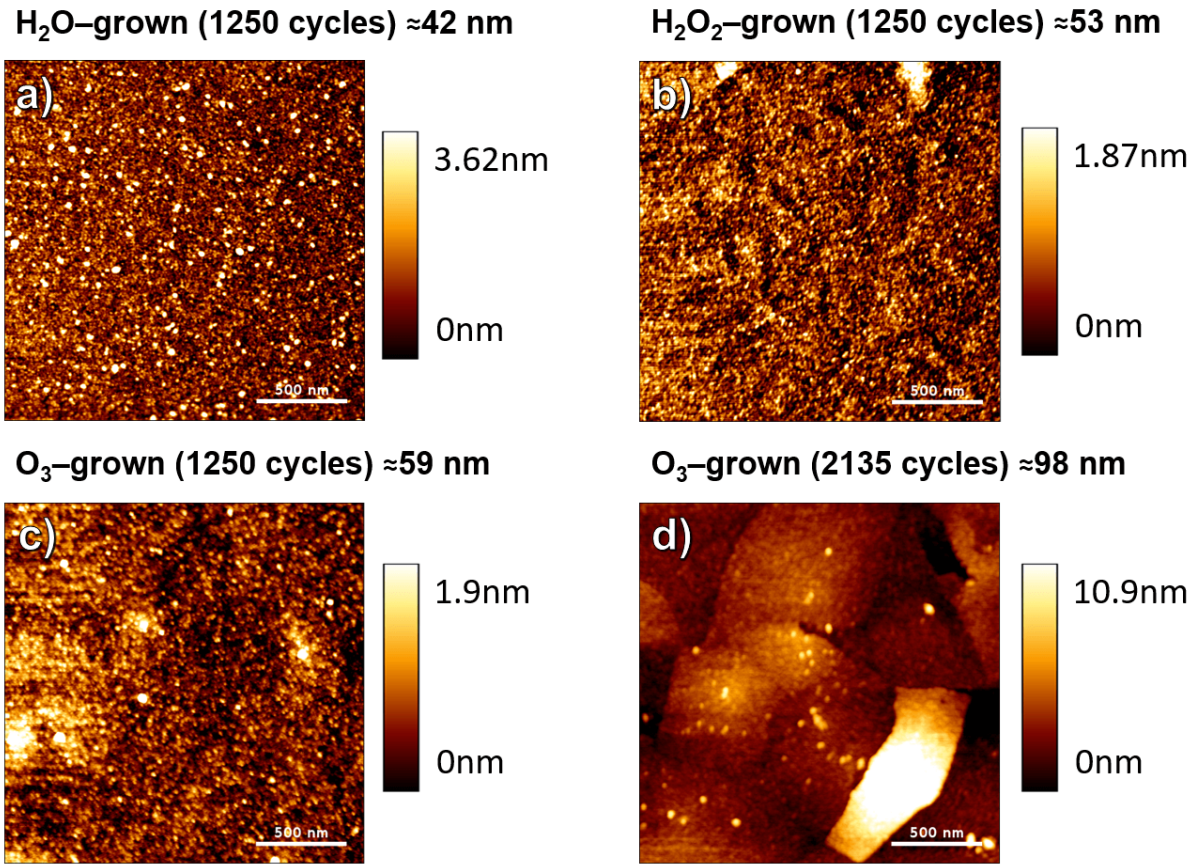


Figure 5.4: AFM representative topography images of TiO_2 films deposited via ALD using different oxidizing agents. **a)** H_2O-grown (1250 cycles, thickness ≈ 42 nm), **b)** H_2O_2-grown (1250 cycles, thickness ≈ 53 nm), **c)** O_3-grown (1250 cycles, thickness ≈ 59 nm), and **d)** O_3-grown (2135 cycles, thickness ≈ 98 nm). The color bars indicate height variations across each scan.

Table 5.3 presents the corresponding values for R_a, R_q, and R_t, which quantify the arithmetic mean roughness, root-mean-square roughness, and total height variation, respectively. Among the samples with 1250 cycles, the film grown with H_2O_2 exhibits the lowest roughness. In contrast, the thicker O_3-grown film (2135 cycles) shows an increase in roughness, consistent with the AFM scans in Fig. 5.4. These trends confirm that both precursor chemistry and thickness strongly influence the surface morphology of TiO_2 ALD films.

Table 5.3: Roughness parameters for TiO_2 films deposited by ALD using different oxidizing agents and cycle counts. Values are mean ± standard deviation from three measurements over a $2 \times 2\,\mu$m area.

Oxidant	**Cycles**	R_a (nm)	R_q (nm)	R_t (nm)
H_2O	1250	0.527 ± 0.025	0.790 ± 0.046	10 ± 2
H_2O_2	1250	0.342 ± 0.017	0.449 ± 0.024	7 ± 1
O_3	1250	0.298 ± 0.003	0.416 ± 0.016	8.0 ± 1.0
O_3	2135	1.341 ± 0.351	1.878 ± 0.554	18.2 ± 2.5

Note that surface roughness can affect ellipsometry measurements if not properly accounted for, since light interacts not only with the dense film but also with surface voids or topographical variations. Huang *et al.* [146] demonstrated that the accuracy of effective medium approximation (EMA) models in ellipsometry degrades when the surface roughness exceeds $\approx \lambda/10$; in such cases, the "effective" optical constants measured can differ from the true bulk values. Accurate refractive index extraction therefore requires an optical model that incorporates surface roughness or effective medium approximations. While any ellipsometric model is an approximation, the reliability of our extracted $n(\lambda)$ is improved by accounting for surface roughness (e.g., via an EMA layer) and by cross-validating with AFM and XRD data.

5.5.2 XRD and partial anatase crystallization

XRD measurements were carried out using a PANalytical X'Pert PRO diffractometer in Bragg–Brentano geometry, with a Cu anode X-ray source ($K\alpha_1 = 1.540598$ Å, $K\alpha_2 = 1.544426$ Å, intensity ratio = 0.5). Scans were performed in continuous mode over a 2θ range of 10° to 80° with a step size of 0.0098°, using a fixed divergence slit (0.76 mm). The generator was operated at 45 kV and 40 mA. A fine calibration offset of $-0.0193°$ 2θ was applied. No monochromator was used during measurement.

Fig. 5.5 shows XRD patterns of the TiO_2 films. A clear anatase (101) diffraction peak appears at $2\theta \approx 25°$ for the films grown with O_3, indicating that these ALD films are not purely amorphous but have partially crystallized into anatase nanocrystals. We therefore refer to our films as mixed-phase TiO_2, consisting of anatase crystallites embedded in an amorphous matrix. This mixed-phase structure may influence index contrast and absorption and could have implications for optical resonances in these films, although a detailed analysis is beyond the scope of this work.

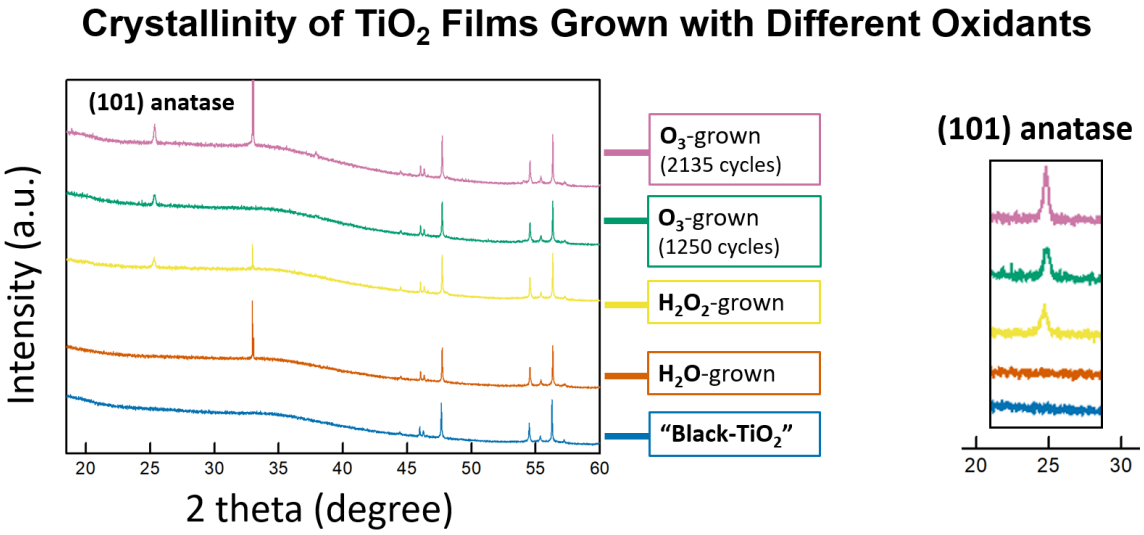


Figure 5.5: (**Left**) XRD patterns of TiO_2 thin films grown using different oxidizing agents via ALD. Samples include O_3-grown films with 2135 cycles (pink) and 1250 cycles (green), H_2O_2-grown (yellow), H_2O-grown (orange), and black-TiO_2 (blue). (**Right**) Magnified view of the (101) anatase reflection near 25°, highlighting differences in peak intensity and crystallinity among the samples.

5.6 Chemical and electronic characterization

5.6.1 XPS stoichiometry and defect states

Elemental composition and oxidation states were assessed via XPS using a Thermo Fisher ESCALAB 250Xi instrument with a Mg Kα source. High-resolution spectra were acquired using the standard high-resolution settings of the facility; the acquisition parameters were set by the instrument operator. Deconvoluting these spectra enabled quantification of Ti^{3+} content and assessment of oxygen stoichiometry. Analyzed samples included TiO_2 films grown with H_2O, H_2O_2 and O_3 oxidants. Films grown using H_2O as the oxidant resulted in visibly darker films, referred to as *black* TiO_2 [86].

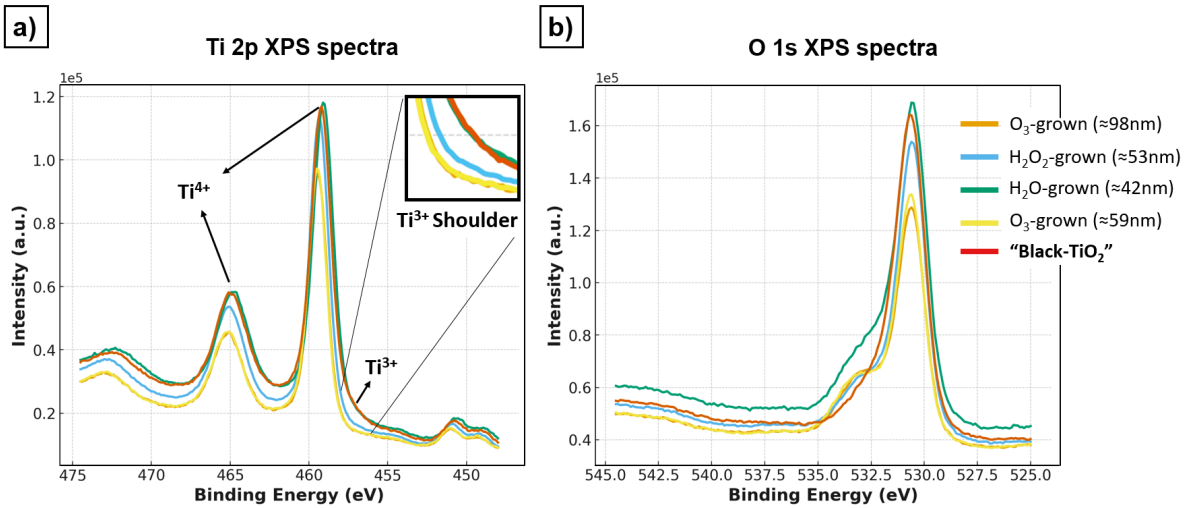


Figure 5.6: High-resolution XPS spectra of TiO_2 films grown with different oxidants. **a)** Ti *2p* region for O_3-grown (≈ 98 nm, orange), O_3-grown (≈ 59 nm, yellow), H_2O_2-grown (≈ 53 nm, blue), H_2O-grown (≈ 42 nm, green), and black TiO_2 (red) films. **b)** O *1s* spectra of the same films reveal differences in the oxygen bonding states.

We used X-ray photoelectron spectroscopy (XPS), reflected electron energy loss spectroscopy (REELS), and ultraviolet photoelectron spectroscopy (UPS) to characterize the chemical and electronic properties of ALD-grown TiO_2 films deposited using ozone (O_3), water (H_2O), or hydrogen peroxide (H_2O_2) as the oxidant. Five samples were analyzed: two O_3-grown films (2135 and 1250 cycles), two H_2O-grown films (one being the black-TiO_2 sample), and one H_2O_2-grown film.

Fig. 5.6 **a)** shows the Ti *2p* high-resolution XPS spectra for TiO_2 films grown with different oxidants. All samples exhibit the characteristic Ti *2p* doublet, with the $2p_{3/2}$ peak near 458–459 eV and the $2p_{1/2}$ peak at 5.72 eV higher binding energy. For O_3-grown films, the $2p_{3/2}$ peak appears at 459.5 eV, with no shoulder (fully Ti^{4+}). In contrast, H_2O-grown films exhibit a shoulder near 457.8 eV indicating Ti^{3+}. This is consistent with partial reduction in water-grown films as reported in the literature [86]. While no formal peak deconvolution (i.e., removal of instrumental broadening) was performed, the shoulder is highlighted in Fig. 5.6 **a)**.

Fig. 5.6 **b)** shows the O *1s* high-resolution XPS spectra of the same TiO_2 films. All samples exhibit a dominant peak near 530 eV (lattice oxygen), but H_2O-grown films also exhibited a higher energy shoulder (532 eV), attributed to surface hydroxyls or adsorbed oxygen species, consistent with literature reports for O *1s* in oxygen-deficient TiO_2. O_3-grown films, by contrast, show more symmetric O *1s* peaks at 529.7 eV.

The derived XPS stoichiometry shows that the O_3-grown films are nearly stoichiometric (O/Ti ≈ 2.00), while the H_2O-grown films are slightly oxygen-deficient (O/Ti ≈ 1.92–1.95).

Table 5.4: XPS parameters for TiO_2 samples. Binding energies (BE) for Ti *2p₃/₂* and O *1s* peaks are reported.

Oxidant, cycles	Ti *2p₃/₂* BE (eV)	O *1s* BE (eV)	O/Ti ratio
(H_2O, Black)	459.2 (shoulder at 457.8)	529.9	1.92
(H_2O, 1250)	459.1 (shoulder at 457.8)	529.9	1.95
(H_2O_2, 1250)	459.3	530.6	1.60
(O_3, 1250)	459.5	529.7	1.99
(O_3, 2135)	459.4	529.7	2.00

The H_2O_2-grown film also shows moderate oxygen deficiency, with an O/Ti ratio of ≈ 1.60 (Ti = 20.57%, O = 32.95%). This oxygen deficiency correlates with the Ti^{3+} presence and may explain the subtle differences in optical properties [86]. A summary of key XPS parameters is provided in Table 5.4.

High-resolution XPS peak-fit parameters for the black TiO_2 film are summarized in Table 5.5, parameters for the O_3-grown film are listed in Table 5.6. Although minor signals from C, N, and S were detected in the spectra, these are attributed to surface contamination or residual precursor species from the ALD. As such, they are excluded from the tables in order to focus on the intrinsic TiO_2 film chemistry.

Table 5.5: High-resolution XPS peak-fit parameters for the black H_2O-grown TiO_2 film. Binding energies (BE) and FWHM are in eV, atomic percentages are derived from area analysis.

Peak	BE (eV)	FWHM (eV)	Atomic %
Ti $2p_{3/2}$ (Ti^{4+})	459.2	1.53	14.2
O $1s$ (lattice O^{2-})	530.5	1.41	25.9
O $1s$ (–OH / vacancies)	531.1	1.87	18.3
O $1s$ (adsorbed organics)	532.9	1.86	4.3

Table 5.6: High-resolution XPS peak parameters for O_3-grown TiO_2 film. Binding energies (BE) and FWHM are in eV; atomic percentages are derived from area analysis.

Peak	BE (eV)	FWHM (eV)	Atomic %
Ti $2p_{3/2}$ (Ti^{4+})	459.4	1.38	17.09
O $1s$ (lattice O^{2-})	530.6	1.49	27.17
O $1s$ (adsorbed organics)	532.8	2.56	11.50

5.6.2 REELS and UPS analysis of bandgap and defect features

The reflected electron energy loss spectra (REELS) were acquired using a 1 keV primary electron beam. Bandgap energies were estimated by linear extrapolation of the low-loss onset. Spectra were recorded in constant analyzer energy (CAE) mode with a 0.1 eV energy step size. The extracted bandgaps are summarized in Table 5.7. The REELS-derived bandgaps exceed the optical values. This behavior has been reported in the

literature, where electron spectroscopies such as REELS yield larger bandgap estimates due to the additional kinetic energy required to overcome excitation thresholds and satisfy momentum conservation, especially in indirect or disordered materials [147, 148]. We find a bandgap of 3.58 eV for the Black TiO_2 sample, 3.55 eV for H_2O-grown TiO_2, 3.53 eV for H_2O_2-grown TiO_2, 3.57 eV for O_3-grown TiO_2 (1250 cycles), and 3.45 eV for O_3-grown TiO_2 (2135 cycles).

Table 5.7: Bandgap energies estimated from REELS spectra of TiO_2 films.

Sample (oxidant, cycles)	**Bandgap** (eV)
Black TiO_2	3.58
TiO_2 + H_2O (1250 cycles)	3.55
TiO_2 + H_2O_2 (1250 cycles)	3.53
TiO_2 + O_3 (1250 cycles)	3.57
TiO_2 + O_3 (2135 cycles)	3.45

Ultraviolet photoelectron spectroscopy (UPS) measurements were performed using He I excitation (21.2 eV) to probe the valence band edge of the same TiO_2 samples. All films show the broad O *2p*–Ti 3d valence band region between approximately 12 and 18 eV binding energy. Significant variation is observed near the Fermi level. The H_2O-grown films, particularly the Black TiO_2, show higher intensity close to 0 eV binding energy, indicating the presence of occupied mid-gap Ti^{3+} states. In contrast, the O_3-grown samples show a sharp and well-defined valence band edge with no evidence of sub-gap emission. These results are consistent with the XPS data and support the interpretation that oxygen-deficient, reduced films develop Ti^{3+} defect states, while the fully oxidized O_3-grown TiO_2 remains close to stoichiometric and shows no measurable mid gap signature within our detection limits. Mid gap peaks in UPS are attributed in the literature to oxygen-vacancy-induced Ti^{3+} states [149], in agreement with studies of reduced TiO_2 surfaces showing electronic modifications due to such vacancies [150].

For completeness, we note that the bandgap values were obtained by linear fitting of the low-loss edge in REELS data, whereas the UPS spectra were analyzed qualitatively for the valence band onset and presence of defect features; no quantitative cutoff energies were available from the UPS datasets.

5.7 Conclusions and implications for Chapter 6

In summary, the ALD oxidant chemistry impacts the optical, structural, and electronic properties of the TiO_2 thin films relevant for UV nanophotonics. Ellipsometric modelling established the wavelength dependent optical constants and revealed differences in near-UV absorption behaviour between the films, while UV–Vis–NIR reflectance measurements and corresponding simulations clarified the interference response associated with thickness

and incidence angles. AFM and XRD further showed that roughness and partial anatase crystallization depend strongly on both oxidant and film thickness. Then, XPS, REELS, and UPS identified clear differences in stoichiometry, Ti^{3+} defect signatures, and bandgap features across the different films. These results establish the material foundation for the analysis in Chapter 6, where the measured optical constants, absorption behaviour, crystallinity, roughness, and defect chemistry are used to interpret the energy velocity calculations and the observed third-harmonic response.

6 Energy Velocity and Third-Harmonic Generation in UV TiO_2 Thin-Films

6.1 Publication context and author contributions

This chapter presents the energy velocity and third harmonic study that forms the main text of the manuscript whose supplementary material is given in Chapter 5; the manuscript has been submitted to *Optical Materials Express*, with me as lead author. I carried out the energy velocity analysis and implemented the Brillouin energy density evaluation in Fourier-domain FDTD. The third harmonic measurements were performed together with Christian Maibohm on the nonlinear microscopy setup. In each case the collaborators operated the instrument and provided the raw spectra, while I processed and interpreted all of the data and wrote the manuscript.

6.2 Introduction

The capacity to reduce the velocity of light in matter much below the speed of light in vacuum has long been recognized as a powerful tool in nanophotonics [151]. Such light is referred to as slow light and has been exploited in atomic vapors, photonic crystals, and metamaterials to amplify nonlinear response, improve sensing performance, and manipulate information transfer [152, 153, 154]. Slow-light significantly enhances light-matter interactions and can sometimes lead to the enhancement of nonlinear optical phenomena [155]. Slow-light has proved harder to achieve in compact, solid-state platforms operating at UV wavelengths. Harmonic generation depends on absorption, dispersion and refraction together, a dependence analyzed in detail by Dachraoui et al. for high-order harmonic generation in gases [156]. We address the same question here for third-harmonic generation in a solid-state UV platform.

Several characteristic velocities of light can be defined in dispersive photonic media, including the phase velocity v_p, the group velocity v_g valid for linear, lossless band structures, and the energy velocity v_e describing the speed of energy transport defined as the ratio of the time-averaged Poynting flux to the time-averaged energy density. These velocities are respectively described by

$$v_p = \frac{\omega}{k} \quad ; \quad v_g = \frac{\partial \omega}{\partial k} \quad ; \quad v_e = \frac{\langle S \rangle}{\langle u \rangle}, \tag{6.1}$$

whereby ω is the angular frequency and k the angular wavenumber of the light wave under consideration. $\langle \mathbf{S} \rangle$ is the cycle-averaged Poynting vector and $\langle u \rangle$ is the stored electromagnetic energy density [104]. The group velocity becomes ill-defined when absorption is present, exceeding the speed of light in vacuum or even becoming complex [157, 102, 103]. In such cases, v_g no longer represents the transport of optical energy and can lead to misleading physical interpretations. TiO_2 offers a high refractive index in the UV-A, making it a suitable dielectric for nanophotonic structures; its finite absorption in this range is precisely why the group velocity becomes unreliable and the energy velocity is the relevant metric.

Intense optical excitation of TiO_2 strongly perturbs its permittivity. Nonlinearities such as the electronic Kerr effect (intensity-dependent refractive index), multi-photon absorption, and free-carrier generation can induce large, ultrafast changes in both the real and imaginary components of the permittivity $\varepsilon(\omega)$. For example, rutile TiO_2 exhibits strong nonlinear absorption near 800 nm, with reported two-photon and three-photon absorption coefficients on the order of 1×10^{-3} cm/GW and 1×10^{-6} cm^3/GW^2 [158]. Such absorption creates free carriers and associated losses, while the Kerr nonlinearity adds a dynamic index shift. These nonlinear processes invalidate a simple linear dispersion relation $k(\omega)$, as the material's band structure dynamically evolves with the optical pump intensity. Consequently, using a small-signal group velocity derived from the unperturbed TiO_2 band structure to represent actual energy flow is generally unreliable. In contrast, the energy velocity v_e inherently tracks how much energy is actually carried by the field. In principle, v_e can be computed from a full electromagnetic model. For instance, finite-difference time-domain (FDTD) simulations can include Kerr and carrier effects by directly evolving Maxwell's equations in a dispersive nonlinear medium [159]. The key point is that v_e includes the effects of absorption and dispersion, by weighting the Poynting flux with the local energy density (including any stored or dissipated energy). This provides an accurate measure of energy propagation speed, even in the presence of material losses and nonlinearities. Crucially, v_e respects causality: it never exceeds the speed of light in vacuum. By contrast, v_g or related group-delay velocities can formally become superluminal in anomalous dispersion or gain-assisted media, but this does not correspond to superluminal information or energy transfer. Frias *et al.* [160] demonstrated this explicitly in a dispersive slab, showing that under strong dispersion conditions ($\varepsilon < 1$), the group velocity can exceed c, yet the calculated energy velocity remains below c. Only at a resonance (where the slab is effectively transparent) did v_g and v_e coincide, and only then did the energy velocity approach the vacuum speed of light [160]. This is consistent with general principles of causality and the front velocity limit. Therefore, in slow-light nanostructures that exhibit strong nonlinearities and absorption, $v_e = \langle S \rangle / \langle u \rangle$ is the correct metric for quantifying pulse propagation speed.

Titanium dioxide is an abundant material that exists in multiple crystalline phases (rutile and anatase) as well as in an amorphous form. The material is well known for its high refractive index and photocatalytic activity. In photocatalysis, TiO_2 is used to degrade pollutants under UV illumination [161, 162, 163], and strategies such as introducing oxygen vacancies or dopants have been employed to extend its activity into the visible range [164, 165]. Beyond photocatalysis, TiO_2 has attracted growing interest in nanophotonics due to its compatibility with scalable deposition techniques and its optical properties [166, 167, 168, 169, 170]. TiO_2 thin films and metasurfaces have been extensively investigated for both linear and nonlinear optical applications, including waveguides, resonators, and harmonic generation [171, 108]. Semmlinger *et al.* demonstrated a TiO_2 metasurface generating third-harmonic light down to the vacuum UV [16]. Amorphous TiO_2 thin films can contain small, randomly oriented anatase crystallites that are optically anisotropic [172], but are nonetheless well described by an isotropic dispersion model [172]. This approach is consistent with Mueller-matrix ellipsometry results for sol-gel derived TiO_2, whereby no measurable birefringence was observed, even after partial crystallization to anatase [172]. Likewise, studies report that both amorphous and partially crystalline TiO_2 films obtained through Atomic Layer Deposition (ALD) are best modelled with isotropic dispersion functions such as the Tauc-Lorentz model, without the need for anisotropic terms [173, 174].

Suppression of far-field scattering and strong field confinement leads to an increased third-harmonic-generation efficiency of a material. This can be achieved through the creation of anapoles or so-called dark modes. Anapole excitation has been demonstrated in various dielectric nanostructures (Si, Ge, etc.) across visible and near-IR wavelengths [14, 19, 20, 175, 15, 176, 177, 178, 21, 179, 180, 181], and even at microwave scales using high-permittivity resonators [182, 183]. Anapoles in the UV-A region (350–380 nm) have been theoretically predicted for high-index amorphous TiO_2 nano-rectangular cuboids (190 nm length, 160 nm width, and 100 nm thickness) deposited atop a fused silica substrate [108]. Experimental demonstration of anapole modes in the UV-A region however remains a significant challenge due to TiO_2 material properties and nanostructuring tolerances. Obtaining TiO_2 films with a thickness of 100 nm, high refractive index, and low extinction coefficient is challenging, as illustrated by the formation of oxygen-deficient "black-TiO_2" when using H_2O as oxidant (see Section 5.6). Furthermore, nanosized cuboids require lithography processes creating perfect edges placing very stringent engineering requirements.

Even though nano patterning and the subsequent generation of anapoles can lead to strongly increased THG efficiency, unstructured sputtered nanocrystalline TiO_2 thin films with a thickness of 180 nm have also been demonstrated for the efficient generation

of the third-harmonic of 800 nm [184]. It remains an open question how TiO_2 thin films can be employed to efficiently generate VUV light originating from UV-A pump wavelengths through THG. In order to investigate this, we (i) quantify slow-light behaviour in weakly absorbing TiO_2 films through the energy velocity as derived from ellipsometry data combined with FDTD simulations; (ii) show how O_3-grown ALD at 200 °C tunes stoichiometry, crystallinity, and optical constant to realize $v_e \approx 0.18\ c$ in the UV-A; and (iii) experimentally connect this reduced energy velocity to enhanced nonlinear optics, such as THG from the same films. All experiments were done at ambient temperature and pressure conditions.

6.3 Methods

6.3.1 Energy velocity

The energy velocity analysis is presented for a 100 nm TiO_2 slab, corresponding to the thicker O_3-grown film (2135 ALD cycles, ≈ 98 nm), which is the target thickness for UV-A anapole excitation in our prior work and the film for which a well fitted optical model is available from ellipsometry. For the thinner films (42–59 nm), the ellipsometric extraction of the complex refractive index, particularly the extinction coefficient, was insufficiently constrained to support a reliable energy velocity calculation. The thinner films (1250 cycles of O_3, H_2O, and H_2O_2) are therefore characterized experimentally through their THG response rather than through simulation.

Analytical approximation

Assuming a homogeneous, isotropic, linearly absorbing dielectric, we determine v_g using the group index n_g defined by [185]

$$v_g = \frac{c}{n_g} \quad ; \quad n_g = n - \lambda \frac{dn}{d\lambda} \tag{6.2}$$

We obtain $dn/d\lambda$ from ellipsometry measurements as discussed below. For homogeneous, isotropic, non-magnetic media that are effectively lossless (or only weakly absorbing) and for propagating plane-wave pulses, the energy velocity is approximated by the group velocity, $v_e \approx v_g$. We use $v_e \approx c/n_g$ as an analytical estimate, considering the limit of negligible absorption and moderate dispersion.

FDTD Simulations

For the rigorous accounting of dispersion and finite absorption in the determination of v_e, the Brillouin energy density valid for dispersive media was determined through

finite-difference time-domain (FDTD) simulations.[186]

$$\langle u \rangle = \frac{1}{4}\left[\frac{d(\omega\varepsilon'(\omega))}{d\omega}|\mathbf{E}|^2 + \frac{d(\omega\mu'(\omega))}{d\omega}|\mathbf{H}|^2\right] \tag{6.3}$$

Philbin [102] derived the electromagnetic energy momentum tensor from an action principle, showing that Brillouin's expression is exact in dispersive, lossless media. Luan [187] obtained the same result via power conservation arguments. For absorptive materials, Nunes et al. [188] and Welters et al. [103] extended the framework, demonstrating that the Brillouin energy density corresponds to the portion of energy that can be returned to the field, while the dissipated part contributes to irreversible heating. In practice, the Brillouin form is often adopted as the operational definition of energy density in simulations, with the understanding that it corresponds to the recoverable energy. For nonmagnetic TiO_2 ($\mu' = \mu_0$), the second term reduces to $\mu_0|\mathbf{H}|^2/4$. This expression ensures that the stored energy remains positive and causal even in the presence of strong material dispersion and loss.

The FDTD simulations were performed using the open-source package MEEP (MIT Electromagnetic Equation Propagation) 1.31.0 [189] on Ubuntu 24.04.3 LTS (GNU/Linux 6.6.87.2-microsoft-standard-WSL2), installed and managed via the *Micromamba* environment (conda-forge channel). MEEP solves Maxwell's equations in the time domain on a Yee grid. Because each wavelength is analyzed independently at a single frequency ω_0, the field distribution inside the slab is determined entirely by the value of the complex permittivity at that frequency. The simulated medium was therefore specified so as to reproduce the measured $\varepsilon(\omega_0)$ exactly, using a non-dispersive real permittivity together with an ohmic conductivity as given in Eqs. (6.4)–(6.6). The frequency derivative that enters the Brillouin energy density, $\mathrm{d}(\omega\varepsilon')/\mathrm{d}\omega$, cannot be obtained from a medium defined at a single frequency and was instead evaluated from the measured dispersion, using the ellipsometric $n(\lambda)$ and $k(\lambda)$ tabulated in Table 6.1. Fields and energy density are thus both consistent with the measured optical constants: the former through $\varepsilon(\omega_0)$, the latter through the measured dispersion.

The simulations were carried out in a two-dimensional (TE_z-polarized) configuration. The computational domain consisted of a 100 nm-thick TiO_2 film surrounded by air regions of 1 μm thickness above and below (representing semi-infinite space), with a total cell size of 4 μm × 4.1 μm and perfectly matched layers (PML) of 1 μm thickness on all boundaries. The simulations used a spatial resolution of 20 pixels per 100 nm, with the time step set by the automatic Courant condition ($\Delta t \approx 0.5\,\Delta x/c$) and a total simulation time of 200 fs to ensure full field decay.

The complex field distributions at the frequency of interest were recorded using MEEP's Fourier domain monitors, while flux planes provide the net transmitted power. The dispersive permittivity model supplied both $\varepsilon'(\omega)$ and its derivative $d(\omega\varepsilon')/d\omega$, allowing evaluation of the Brillouin energy density voxel-by-voxel. Integration over the simulation domain yields the stored energy, and the ratio of transmitted power to stored energy provides the energy velocity. In the lossless limit, this definition reduces to the usual group velocity, i.e., it coincides with the speed of energy transport for propagating waves. In the presence of absorption, however, the group velocity can become superluminal ($v > c/n$) or even complex and therefore cannot be interpreted as the velocity of energy transport, whereas $v_e = \langle S \rangle / \langle u \rangle$ remains well defined.

In all simulations, v_e is computed as the ratio of the Fourier-domain Poynting flux to the Brillouin energy density (Eq. 6.3), ensuring causality in dispersive, weakly absorptive media. At wavelengths $\lambda \in \{380\,\mathrm{nm}, 400\,\mathrm{nm}, 800\,\mathrm{nm}\}$, we enforced the target complex index $n(\lambda) + i\,k(\lambda)$ (Table 6.1) at $\omega_0 = 2\pi c/\lambda$ by setting the real and imaginary parts of the permittivity and the ohmic conductivity as in Eqs. (6.4)-(6.6):

$$\varepsilon' = n^2 - k^2, \tag{6.4}$$

$$\varepsilon'' = 2nk, \tag{6.5}$$

$$\sigma = \omega_0\,\varepsilon'', \tag{6.6}$$

so that Meep's linear ohmic medium realizes $\varepsilon_{\mathrm{eff}}(\omega_0) = \varepsilon' + i\,\sigma/\omega_0$. We have chosen the wavelengths of 380–400 nm to highlight the impact of absorption onset of TiO_2 moving into the ultraviolet. Additionally, we have added 800 nm as it is a frequently utilized laser wavelength for SHG and THG studies due to the availability of high-power ultrafast laser sources in this wavelength region. For the energy-density calculation, dispersion was taken from the tabulated $n(\lambda)$ and $k(\lambda)$ in Table 6.1. The volume-averaged v_e was computed over the non-PML (perfectly matched layer) interior of the domain; the reduced $|v_e|$ visible near the lateral edges of the maps corresponds to the absorbing PML and is excluded from the average. As a validity check, we note that for a plane wave in a homogeneous, weakly absorbing dispersive medium the Brillouin form reduces analytically to $v_e = c/n_g$, since $\mathrm{d}(\omega\varepsilon_r)/\mathrm{d}\omega = 2nn_g - n^2$ and the magnetic term contributes n^2.

The optical constants for SiO_2 and Si used in the simulations were taken from the literature [190, 191].

Table 6.1: Complex refractive index $n + i\,k$ at selected wavelengths (interpolated from the provided datasets).

λ (nm)	n (TiO_2)	k (TiO_2)	n (Si)	k (Si)	n (SiO_2)	k (SiO_2)
380	2.9786	0.0192	6.4734	0.9665	1.4778	0.0028
400	2.8797	0.0141	5.5695	0.3870	1.4755	0.0027
800	2.4586	0.000 79	3.6923	0.0064	1.4607	0.0013

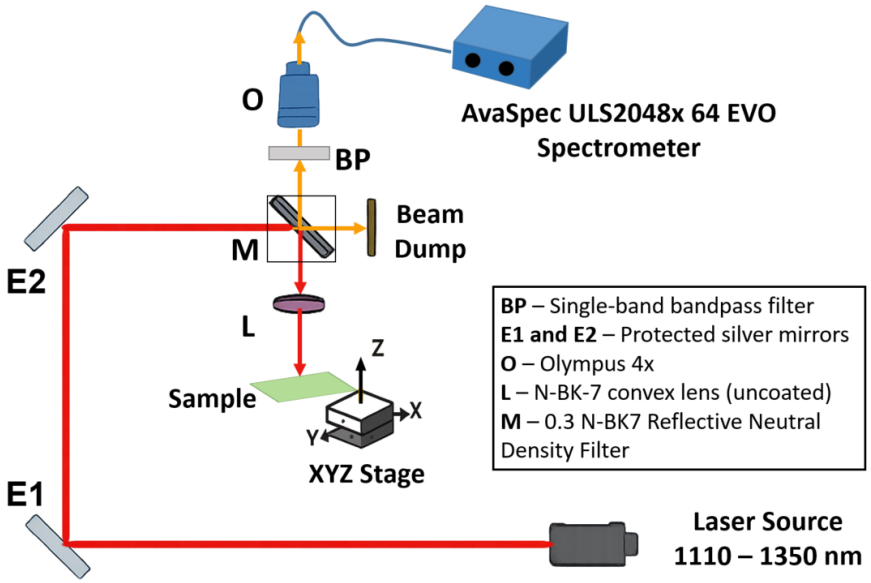


Figure 6.1: Experimental setup for third-harmonic measurements. A tunable infrared source (1110–1350 nm) is directed by protected-silver mirrors ($E1$,$E2$) onto a reflective neutral density filter (M),then focused by an uncoated N-BK7 lens (L) onto the sample on an XYZ stage. The backscattered third-harmonic is collected by (L), spectrally filtered by a bandpass filter (BP), coupled through an Olympus 4× objective (O), and recorded by an AvaSpec ULS2048×64 EVO spectrometer.

6.3.2 Third-harmonic generation

THG measurements were performed in reflection using the optical arrangement shown in Fig. 6.1. Femtosecond pulses from a Carbide CB5 (Light Conversion, 1030 nm, 60 KHz, 6 W, 100 uJ) laser pumping an Orpheus optical parametric amplifier (Light Conversion) provided the used tunable infrared radiation in the 1110–1350 nm range. The beam was steered by protected silver mirrors ($E1$, $E2$, Thorlabs, PF10-03-P01) onto a reflective neutral density filter (M, Thorlabs, ND03B), which attenuated the pump and directed it through an uncoated plano convex N-BK7 lens (L, Thorlabs, LA1252) onto the sample. The sample was mounted on an XYZ translation stage to position the focus on the TiO_2 and Si samples. The THG signal was collected in EPI configuration by the same lens (L). In the detection path the residual pump was suppressed with a single-band bandpass filter (BP, Semrock, FF01-439/154-25, 439+/-77 nm) tuned to the third-harmonic wavelength, and the filtered signal was coupled through an Olympus 4× objective (O, Olympus, RMS4X, WD 18.5 mm, NA 0.1) into a fiber coupled spectrometer (AvaSpec ULS2048×64 EVO). The beam transmitted through the reflective neutral-density filter was blocked in a beam dump. All experiments were conducted in air at a temperature of 20 °C.

6.4 Results

Spectroscopic ellipsometry was performed on the four TiO_2 thin film ALD samples detailed in Chapter 5. Detailed ellipsometric modelling, fit parameters, and depolarization analysis are provided in Table 5.2 and Fig. 5.2. The ellipsometry derived thickness confirmed the ALD growth trends: ≈ 41.99 nm for H_2O-grown film, ≈ 58.57 nm for O_3-grown film, and ≈ 52.87 nm for H_2O_2-grown film (all at 1250 cycles), and ≈ 98 nm for the thicker O_3 film (2135 cycles). The stronger the oxidant, the greater the film thickness achieved for the same number of ALD cycles, as its higher reactivity promotes faster growth and results in a denser film with a higher refractive index. Based on the ellipsometry data, the energy velocity analysis is presented in Table 6.2 for the ≈ 98 nm TiO_2 slab, corresponding to the O_3-grown film in 2135 ALD cycles. This film thickness corresponds to the target thickness required for UV-A anapole excitation [108]. Results indicate that the energy velocity is reduced to $\approx 0.20\ c$ at 380 nm and $\approx 0.21\ c$ at 400 nm, while assuming a value of $\approx 0.39\ c$ at 800 nm. This significant slowdown of electromagnetic energy transport enhances the effective interaction time within the TiO_2 films, which is beneficial for THG processes [192]. Fig. 6.6 displays the optical constant dispersion data extracted from spectroscopic ellipsometry of the O_3-grown TiO_2 film, reporting a high refractive index of approximately $n \approx 3.0$ at 380 nm and a very small extinction coefficient k. This indicates that the film is both optically dense and highly transparent in the UV-A, which is advantageous for enhancing nonlinear interactions without significant absorption losses.

Table 6.2: Optical constants of the ≈ 98 nm TiO_2 film grown using O_3 (≈ 200 °C), the numerical derivative of the refractive index (computed from spectroscopic ellipsometry), the resulting group index, and the estimated energy velocity. The ratio v_e/c highlights how much the energy velocity is reduced relative to c in vacuum.

λ **(nm)**	$\approx n$	$\approx k$	$\mathrm{d}n/\mathrm{d}\lambda$ (1/nm)	**Group index** (n_g)	**Energy velocity** v_e **(m/s)**	v_e/c
380	2.975	0.019	-0.0056	5.13	$\mathbf{5.85 \times 10^7}$	$0.20c$
400	2.881	0.014	-0.0044	4.66	$\mathbf{6.44 \times 10^7}$	$0.21c$
800	2.459	0.00079	-0.0001	2.59	$\mathbf{1.16 \times 10^8}$	$0.39c$

FDTD simulations have been performed for 100 nm thick slabs of TiO_2, Si, and SiO_2 embedded in air, illuminated at normal incidence with transverse-electric (TE_z) polarization, as depicted in Fig. 6.2 **a)**. 2D domain spanning 4 μm in-plane with 1 μm of air above and below the film and 1 μm perfectly matched layers (PMLs) on all boundaries have been considered. The spatial resolution was 20 pixels per 100 nm, corresponding to a grid spacing of 5 nm. A Gaussian-pulse plane wave centered at the analysis wavelength λ_0 is launched from the lower air region, and the energy velocity is evaluated at the single analysis frequency $f_0 = c/\lambda_0$ using the Fourier-domain monitors. Each wavelength $\lambda_0 \in \{380\,\mathrm{nm}, 400\,\mathrm{nm}, 800\,\mathrm{nm}\}$ is simulated independently, so the source is narrowband at f_0 rather than broadband.

Fig. 6.2 **b)** indicates v_e/c computed via $v_e = \langle \mathrm{S} \rangle / \langle u \rangle$ using the cycle-averaged Poynting vector $S = \frac{1}{2}\mathrm{Re}\{\mathrm{E} \times \mathrm{H}^*\}$ and the Brillouin energy density $u = \frac{1}{4}[\partial_\omega(\omega\varepsilon')|E|^2 + \partial_\omega(\omega\mu')|H|^2]$. In air, one observes $v_e \approx c$, while dispersion and absorption reduce v_e well below c inside the TiO_2 slab, consistent with the definition and with Meep's frequency-domain flux formalism (Poynting flux of the Fourier-transformed fields) and Poynting's theorem. Dispersion and the small but finite absorption reduce the local energy velocity well below c, with interior regions at $\lambda_0 = 380$ nm showing $v_e/c \simeq 0.2$ (see also Fig. 6.4 for the 800 nm TiO_2 map and Fig. 6.5 for the air/SiO_2 maps). Volume averaged values extracted from the FDTD simulations yield $v_e/c = 0.185$ at 380 nm, 0.201 at 400 nm, and 0.328 at 800 nm, in good agreement with the analytical estimates $v_e/c \approx 0.20$, 0.21 and 0.39 reported in Table 6.2. v_e is systematically lower than the corresponding group velocity ($v_g/c = 0.195$, 0.216, 0.386), so $v_e/v_g \approx 0.95$, 0.93 and 0.85, respectively, indicating that the actual energy transport is 5–15 % slower than what group velocity alone would suggest. This confirms that, for TiO_2 films, using v_g alone would slightly overestimate the actual energy transport speed, whereas $v_e = \langle S \rangle / \langle u \rangle$ correctly captures the reduced energy flow velocity.

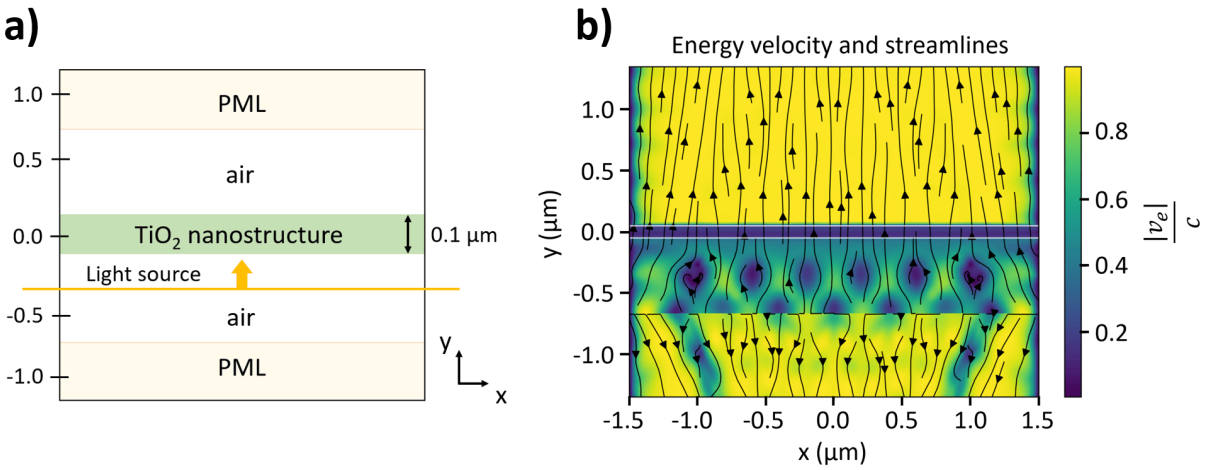


Figure 6.2: **a)** Simulation layout for the 2D FDTD model (Meep): a 100 nm TiO_2 film is centered at $y = 0$, surrounded by air and terminated by PMLs. A TE_z line source in the lower air region launches a normally-incident plane wave in the $+y$ direction. **b)** Energy velocity map and streamlines at $\lambda_0 = 380$ nm. Colors show $|v_e|/c$ computed as $v_e = \langle \mathrm{S} \rangle / \langle u \rangle$. Streamlines indicate energy flowing upward from the source through the slab into the upper half space. The reduced $|v_e|/c$ near the left and right edges of the map corresponds to the 1 µm absorbing PML at the lateral boundaries and is not physically significant.

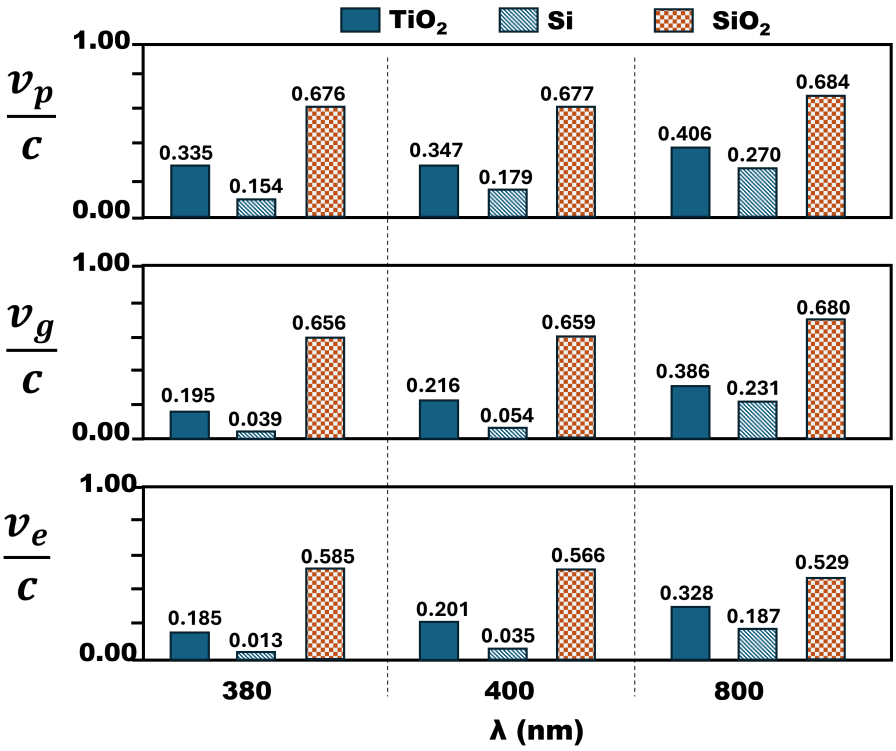


Figure 6.3: Normalized velocities for 100 nm slab under TE_z excitation at normal incidence, evaluated at $\lambda = 380$ nm, 400 nm and 800 nm. The three panels show (top) phase velocity v_p/c, (middle) group velocity v_g/c, and (bottom) energy velocity v_e/c. The horizontal axis is wavelength λ (nm) and the vertical axis is the corresponding normalized velocity. Bar colour and hatching indicate the material: solid blue = TiO_2, hatched blue = Si, and orange cross = SiO_2. Numeric labels give the values of v/c.

Fig. 6.3 summarizes the normalized phase, group, and energy velocities, v_p/c, v_g/c, and v_e/c, for TiO_2, Si, and SiO_2 at 380 nm, 400 nm and 800 nm (same 100 nm slab, TE_z, normal incidence). Two trends stand out: (i) for TiO_2 and Si, $v_e < v_g$ at all wavelengths, because even small k and/or steep $n(\lambda)$ reduce the recoverable electromagnetic energy and slow energy flow; thus, using v_g alone overestimates transport speed in these materials; (ii) for SiO_2, with very small k and comparatively weak dispersion in this band, v_e and v_g track more closely, so v_g is often an adequate descriptor for transport in low-loss silica films. Figs. 6.4 and 6.5 show v_e/c maps and streamlines across UV-NIR wavelengths for a 100 nm slab. As dispersion and absorption increases (towards the UV or in higher-index media), the spatial distribution of v_e becomes highly non-uniform and v_e falls below c, supporting that v_g can overestimate the actual energy transport speed. Control maps in air and in a 100 nm SiO_2 slab (Fig. 6.5) show: air gives $v_e/c \approx 1$ while low-loss SiO_2 shows $v_e/c \approx 0.585$ and $v_e \approx v_g$, emphasizing that light velocity within SiO_2 is equivalent to v_g [193], while TiO_2 or Si require v_e to accurately describe light velocity within.

6.4.1 Complementary simulations

The simulated maps (Figs. 6.4 and 6.5) illustrate how dispersion and absorption in TiO_2 influence the energy flow at different wavelengths and in different media.

At 800 nm (Fig. 6.4), the color map shows $|v_e|$ only partially reduced within the slab, indicating efficient energy transport. In contrast, at 380 nm, near the TiO_2 band edge, the interior of the film exhibits $|v_e|$ as low as $\approx 0.2\ c$, with dense energy streamlines. This slowdown at 380 nm (also seen in Fig. 6.2 **b)**) shows that strong material dispersion together with absorption produces the pronounced slow-light effect. As the wavelength approaches the TiO_2 absorption edge, the energy velocity drops significantly, while in the transparency range v_e approaches the normal light speed in the medium.

We note that the volume average of $|\mathbf{S}|$ and u is taken over a domain that includes the slab together with a thin region of surrounding air and the immediately adjacent interface cells. This is visible in the extracted v_e at $\lambda_0 = 800$ nm, where TiO_2 is essentially transparent ($k \approx 7 \times 10^{-4}$) and weakly dispersive, so the volume-averaged v_e would be expected to approach c/n_g. The residual deviation reflects this choice of integration window rather than additional physics. The same effect is visible in the SiO_2 control discussed below.

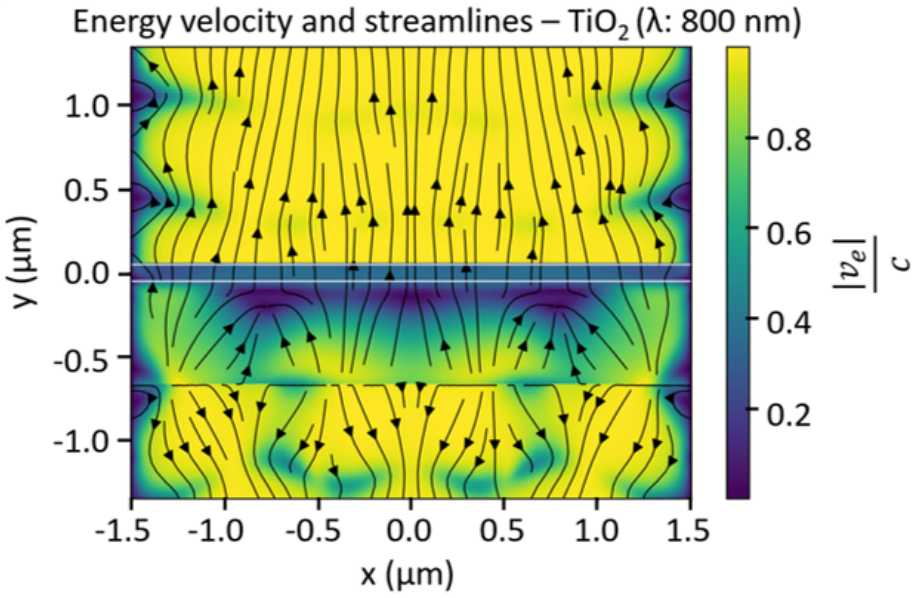


Figure 6.4: **a)** Energy velocity map and streamlines $\lambda_0 = 800$ nm. Colors show $|v_e|/c$ computed as $v_e = \langle \mathrm{S} \rangle / \langle u \rangle$. Streamlines indicate energy flowing upward from the source through the slab into the upper half space.

For air (Fig. 6.5 **a)**) the $|v_e|/c$ map is essentially uniform and ≈ 1 throughout the domain, with straight, upward streamlines, confirming that in a homogeneous, lossless medium the energy velocity approaches c.

For the SiO_2 slab (Fig. 6.5 **b)**) at $\lambda_0 = 380\,\text{nm}$, $n \approx 1.478$ and $k \approx 2.8\times10^{-3}$ (Table 6.1), so the medium is essentially transparent and weakly dispersive. The corresponding velocities are $v_p/c \approx 0.676$, $v_g/c \approx 0.656$, and $v_e/c \approx 0.585$. The phase and group velocities differ by only $\approx 3\%$, while the volume-averaged v_e lies ≈ 10–15% below them. As discussed above, this offset is consistent with the volume-averaging window rather than with intrinsic dispersion or loss, since at $\lambda_0 = 380$ nm SiO_2 is essentially transparent. The standard description of light propagation in low-loss silica in terms of v_g therefore remains adequate, in contrast with higher-index or absorbing dielectrics such as TiO_2 or Si, where v_e is the more appropriate metric.

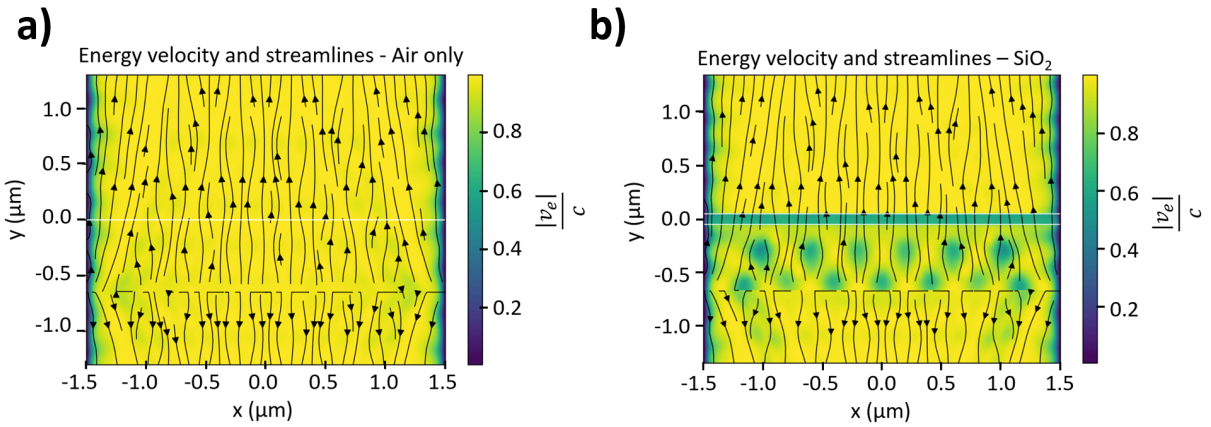


Figure 6.5: Energy velocity maps at $\lambda_0 = 380\,\text{nm}$. Maps show $|v_e|/c$ (color) and energy-flow streamlines ($\langle \mathbf{S} \rangle$) computed from Fourier-domain fields; $v_e = \langle \mathbf{S} \rangle / \langle u \rangle$ with Brillouin energy density u. **a)** Air (reference): uniform $|v_e|/c \approx 1$. **b)** SiO_2 slab (100 nm thick) in air.

6.4.2 Third-harmonic response

Fig. 6.7 shows third-harmonic spectra from bare Si and TiO_2/Si samples grown with different oxidants in the UV-A wavelength region. A narrow peak is observed at the expected $\lambda_{3\omega}$, confirming third harmonic frequency conversion. The peak amplitude depends strongly on film chemistry and thin film thickness. The H_2O_2-grown TiO_2 film and the thicker O_3-grown TiO_2 film (2135 cycles, $\approx 98\,\text{nm}$) exhibit the strongest THG signals at $\lambda_{3\omega} = 380\,\text{nm}$, whereas the H_2O-grown film yields the weakest response. The THG response at $\lambda_{3\omega} = 400\,\text{nm}$ differs significantly, whereby the H_2O_2-grown film remains highest followed by the thinner O_3-grown film (1250 cycles). Here, the thick O_3 film and bare Si provided intermediate signal strength. To enable a direct comparison between the two excitation settings, the third-harmonic spectra in Fig. 6.7 were normalized by the cube of the measured pump power at each excitation wavelength, $P^3_{\text{pump}}(\lambda)$, following the cubic power dependence of THG. The fundamental wavelengths for panels **a)** and **b)** were 1140 nm and 1200 nm, respectively. This normalization therefore introduces only a minor correction and does not affect the relative ordering between samples.

Ellipsometry modelling of the H_2O_2-grown TiO_2 thin film is complicated by a non-negligible absorption that is difficult to parametrize reliably, unlike the O_3 film, for which the Cauchy + Urbach model fits cleanly. Hence, one cannot analytically extract the energy velocity for this film with the same confidence as for the O_3-grown TiO_2 thin film. We attribute its strong THG response qualitatively to the combined effect of its intermediate refractive index, its film thickness ($\approx 53\,\mathrm{nm}$), and its defect chemistry as detailed below, while noting that resolving this quantitatively requires a more complete optical model. In reflection geometry, the measured response originates from an asymmetric Fabry–Pérot-type air/TiO_2/Si stack, in which the TiO_2 layer and the absorbing Si substrate determine the interference and field-enhancement conditions. Since measurements are performed on TiO_2/Si in reflection geometry, the detected third-harmonic should be interpreted as the response of the full TiO_2/Si stack: the Si substrate can contribute directly, and the TiO_2 layer modifies both the pump field distribution and the absorption conditions for the generated harmonic.

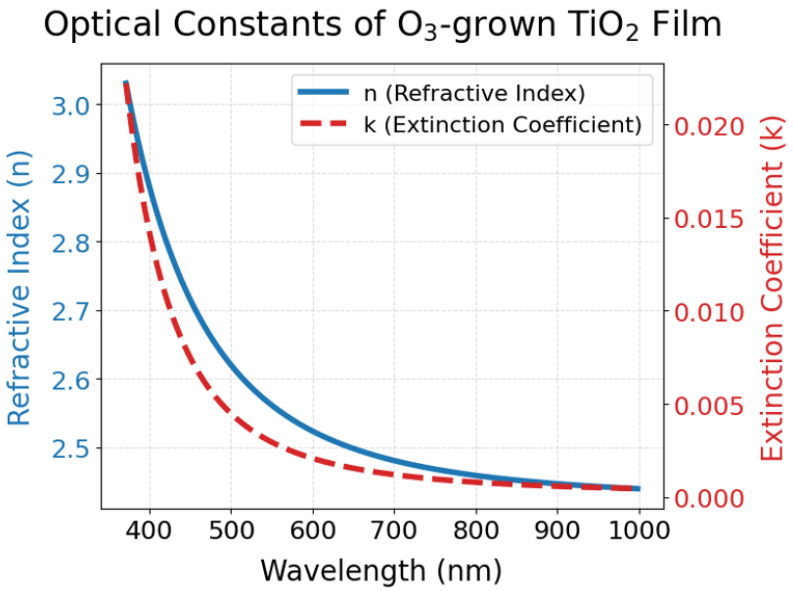


Figure 6.6: Fitted ellipsometry data for the O_3-grown TiO_2 film. **(a)** Real (n) (in blue) and imaginary (k) (in red) parts of the refractive index.

Table 6.3 summarizes material properties most relevant for the interpretation of the third-harmonic response derived from SE, AFM, XRD and XPS. The H_2O-grown film exhibits the clearest defect signature, while the O_3-grown films are closest to stoichiometric TiO_2. The H_2O_2-grown film shows intermediate material properties but yields the strongest measured third-harmonic signal, indicating that the observed response is not determined only by stoichiometry, but also by the combined effects of optical dispersion, defect chemistry, and field distribution in the TiO_2/Si structure. Interestingly, we observe the anatase (101) peak in O_3-grown TiO_2 films deposited at only 200°C, even without any

post-deposition annealing. We attributed this partial crystallization to the high oxidative reactivity of ozone. The O_3 process likely facilitates more complete ligand removal and Ti–O bond formation during deposition, providing enough local energy for nuclei of anatase to form. These crystallites seem to nucleate at the surface (where atomic mobility and reaction likelihood are highest) while the bulk of the film remains amorphous, resulting in a hybrid phase. A similar phenomenon was reported by Jin *et al.*, [194], who observed anatase formation in ALD TiO_2 at low temperatures using O_3. Chapter 5 provides an in-depth study of the different oxidants.

Table 6.3: Material characterization results for the TiO_2 films used in the third-harmonic-generation study. Film thickness values are approximate.

Oxidant	ALD cycles	Film thickness (nm)	R_q (nm)	O/Ti	Ti^{3+} presence	REELS bandgap (eV)	THG relevance
H_2O	1250	42	0.79	1.95	Yes	3.558	weakest response
H_2O_2	1250	53	0.45	1.60	moderate	3.531	strongest response
O_3	1250	59	0.42	1.99	No	3.571	strong response
O_3	2135	98	1.88	2.00	No	3.452	low-energy velocity regime

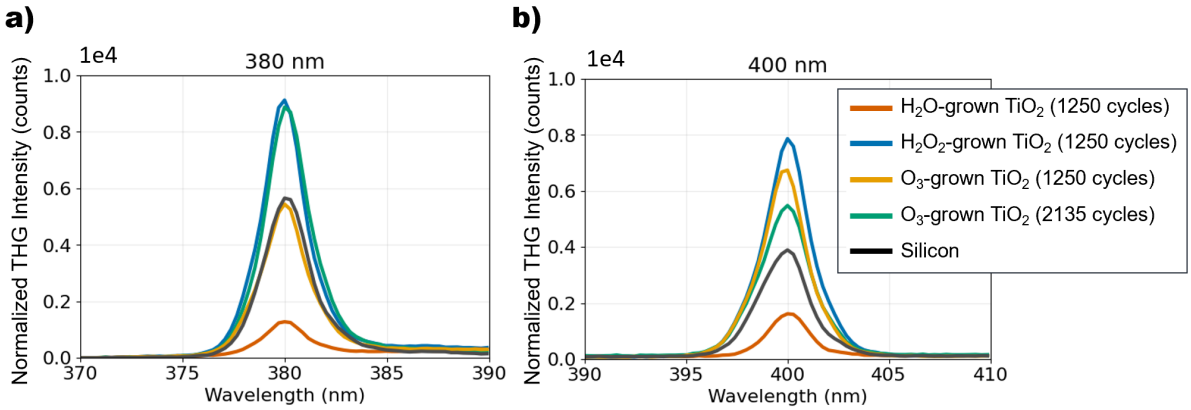


Figure 6.7: Reflected third-harmonic spectra recorded from bare silicon and from TiO_2 coated samples at $\lambda_{3\omega} = 380$ nm and 400 nm. **a)** Spectra centered at 380 nm. **b)** Spectra centered at 400 nm. The comparison includes TiO_2 films grown with H_2O and H_2O_2 at 1250 cycles and with O_3 at 1250 and 2135 cycles.

6.5 Discussion

The materials data show that the ALD oxidant controls stoichiometry, roughness, and (partial) crystallinity, which jointly determine the optical constants and therefore the

energy-transport properties of the films. O_3-grown TiO_2 is closest to stoichiometric TiO_2 (no clear Ti^{3+} signature) and exhibits a high refractive index with only small extinction in the UV-A, while H_2O-grown films show Ti^{3+} related defect signatures. The thicker O_3 film additionally shows increased roughness and optical features near the band edge (Fig. 5.3), consistent with a change in the internal field distribution and light trapping in the UV-A.

Our energy-velocity analysis captures this near-band-edge regime directly. Using ellipsometry-derived dispersion together with Fourier-domain FDTD simulations, we find that the energy velocity in O_3-grown TiO_2 can drop to $v_e/c \approx 0.18$–0.21 around $\lambda = 380\,\text{nm}$–$400\,\text{nm}$, whereas it is substantially higher at longer wavelengths. In this regime, v_e provides a more rigorous measure of electromagnetic energy transport than the group velocity, because it remains meaningful even when absorption and strong dispersion are present.

The reflected THG measurements (Fig. 6.7) show that the detected third-harmonic depends sensitively on the film chemistry and thickness. The H_2O_2-grown film provides the strongest signal at $\lambda_{3\omega} = 400\,\text{nm}$, while the thicker O_3-grown film becomes competitive at $\lambda_{3\omega} = 380\,\text{nm}$. This wavelength-dependent ranking indicates that the measured signal is shaped by multiple factors beyond stoichiometry alone, including (i) the linear optical response of the TiO_2/Si stack (interference and field enhancement at both ω and 3ω), (ii) absorption and scattering losses, and (iii) the nonlinear contribution of the Si substrate.

The non-negligible third-harmonic from bare Si in Fig. 6.7 reinforces that the reflection geometry probes the full stack rather than the film in isolation. In this configuration, the TiO_2 layer can enhance or suppress the detected third-harmonic by modifying the pump field distribution in Si and by changing the extraction conditions for the generated harmonic. Therefore, while reduced v_e near the UV band edge is a useful design indicator for enhanced light-matter interaction in TiO_2, a direct, quantitative link between v_e and an intrinsic film $\chi^{(3)}$ cannot be established from reflection on Si spectra alone. For rutile TiO_2 at $1\,\mu\text{m}$, where the photon energy lies well below the band edge, the non-resonant bound electronic value is $\chi^{(3)} \approx 2.6 \times 10^{-21}\ \text{m}^2/\text{V}^2$ [195]. Thin-film THG and Z-scan measurements closer to the edge give values one to two orders of magnitude higher [196, 197]. A single $\chi^{(3)}$ therefore cannot be assumed across the UV-A.

A decisive next step is to suppress the substrate contribution and quantify the film nonlinearity on an optically transparent substrate (e.g., fused silica), using transmission and reflection geometries with wavelength dependent normalization by P^3_{pump} and calibrated detection. Combined with the energy-velocity framework, such measurements would enable a thickness and dispersion optimization strategy for UV-range nonlinear conversion in

ALD-grown TiO_2 (both in simple slabs and in patterned resonators).

6.6 Conclusion

We established energy velocity $v_e = \langle S \rangle / \langle u \rangle$ as a suitable metric to describe optical transport in the UV-A wavelength range for TiO_2, where strong dispersion and finite absorption render the well-known group velocity unreliable for estimation of material's optical nonlinearity. O_3-grown thermal ALD at 200 °C yields high-index TiO_2 thin films with small extinction in the UV-A and enables an energy-transport slowdown to $v_e/c \approx 0.18$–0.21 around $\lambda = 380\,\text{nm}$–$400\,\text{nm}$, confirmed by ellipsometry and FDTD. Experimentally, backscattered THG spectra from TiO_2/Si stacks exhibit intensities that depend strongly on oxidant chemistry and thickness. The H_2O_2-grown film produces the strongest signal at 400 nm, while the thicker O_3-grown film becomes stronger at 380 nm. The third harmonic signal from bare Si indicates that reflection measurements report a coupled film-substrate response, emphasizing the need for substrate suppressed geometries to extract intrinsic film nonlinear coefficients. In summary these results indicate that ALD process chemistry can be utilized to engineer both the UV-A energy transport regime and the observed nonlinear response of TiO_2 nanostructures. Because v_e follows directly from the ellipsometric $n(\lambda)$ and $k(\lambda)$, the analytical estimate carries essentially no computational cost, and the FDTD maps are lightweight two-dimensional single-frequency calculations rather than demanding simulations. The approach is therefore practical for screening film thickness, oxidant chemistry, and operating wavelength against a target slow-light band prior to fabrication. Because v_e is computed directly from the simulated fields, the same FDTD analysis can be applied to patterned geometries, including the UV-A anapole supporting TiO_2 cuboids for which this film thickness was targeted [108], although interpreting a volume averaged v_e requires more care in strongly resonant structures. Its advantage over the group velocity is that it remains causal and bounded by c in the absorbing, strongly dispersive UV-A band, where v_g can exceed c or become complex. The main limitation is that v_e describes linear energy transport, so it identifies favorable conditions for light-matter interaction but does not by itself yield an intrinsic $\chi^{(3)}$, and its accuracy near the band edge depends on a well constrained extinction coefficient. Coupling the energy velocity analysis with fully nonlinear dispersive FDTD that includes Kerr and free-carrier terms, together with substrate suppressed measurements on a transparent substrate, would extend the present design indicator into a quantitative thickness and dispersion optimization for UV harmonic generation.

7 Design and Simulation of a TiO_2 Anapole Metasurface

7.1 Publication context and author contributions

This chapter is based on our article on ultraviolet anapole metasurfaces (Poeta, Vilarinho, López-García, De Beule, *Optics Letters* **49**(24), 7198–7201, 2024) [108], for which I am the lead author. Rafael Vilarinho found and set up the MENP multipole-decomposition package, and together we reproduced results from the package authors' reference papers to confirm that our implementation was correct. From that point I carried out the work on my own: designed the TiO_2 nanostructures, ran the electromagnetic simulations, performed the multipole decomposition that identified the UV-A anapoles, and wrote the manuscript. Co-authors contributed to the discussion of the results.

7.2 Introduction

We report the excitation of optical anapole states at ultraviolet wavelengths. Numerical simulations indicate that TiO_2 nano-rectangles with varying length-to-width ratios can support such modes within the 350–380 nm range. We further propose a two-dimensional periodic arrangement of these nano-rectangles deposited atop a fused silica substrate [108]. Understanding and manipulating optical anapole states in the ultraviolet spectrum is crucial for advancing next-generation photonic devices and enhancing nonlinear optical processes, such as generation of highly energetic vacuum ultraviolet light through third harmonic generation.

Light scattering of nano-structures is fundamental to how we perceive colors and shapes and essentially determines our visual environment [74, Ch. 1, pp. 1–11]. It namely results in radiation that contributes to the visibility of objects. Scattering depends on both the geometrical properties of the nano-structure and the wavelength and polarization properties of the illuminating beam. One intriguing question that has been addressed is the identification of the geometrical properties of a nano-structure that reduce the overall scattered light intensity to a minimum, thereby creating strong confined electric and magnetic fields inside the nano-structure [198], a quest culminating in the creation of a three-dimensional invisibility cloak [199].

The concept of an anapole mode was translated to the nanophotonics field, characterized by their ability to confine electromagnetic fields inside and near the nano-structure [106], largely eliminating far-field radiation. This confinement makes the object quasi-invisible at the resonant wavelength of the incident light. This minimizes scattering and absorption,

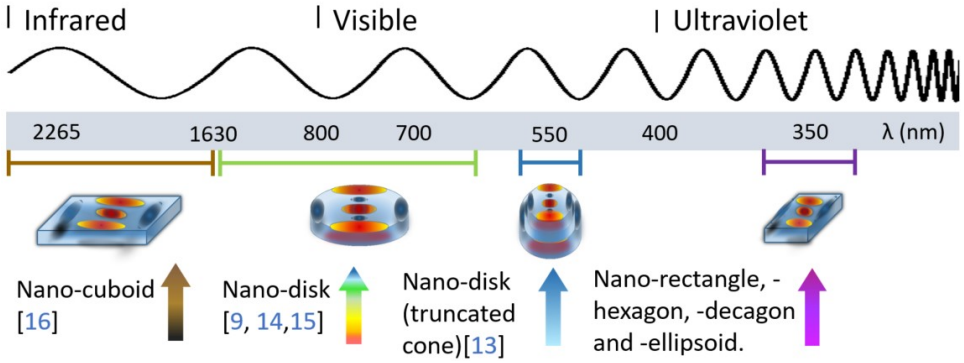


Figure 7.1: Illustration of some geometries that are proven of exciting anapoles for wavelengths between 550–2265 nm. Our goal is to excite anapole states in the UV regime.

thereby reducing visibility in the far-field.

The family of toroidal moments are retrieved as next-order terms from the expansion of the Cartesian multipoles [105]. For instance [200], we have that the total electric dipole moment $\mathbf{D}$ is given by Eq. (7.1):

$$\mathbf{D} = \mathbf{p} + \frac{i\left(\frac{2\pi}{\lambda_0}\right)}{c}\mathbf{T} \tag{7.1}$$

, where $\mathbf{p}$ is the electric dipole moment, $\mathbf{T}$ is the toroidal dipole moment, λ_0 is the wavelength in vacuum, and c is the speed of light. The condition $\mathbf{D} = 0$, with $|\mathbf{p}| \neq 0$ and $|\mathbf{T}| \neq 0$ is necessary for exciting the electric anapole state [201]. In a real scenario, the total electric dipole moment approaches a minimum value or remains close to zero, indicating proximity to the anapole state, highlighting the significance of electromagnetic phenomena. Exciting anapole states in the UV spectrum has potential applications in various fields such as photonics, sensing technologies, and nonlinear optics [15, 75, 202, 18]. As a result, it has been increasingly studied for its role in enhancing light-matter interactions and its potential to minimize energy losses in these applications. This distinction highlights that anapole states minimize electromagnetic field scattering, rather than eliminate it.

Fig. 7.1 illustrates various geometries that have been shown to excite anapoles for wavelengths between 550–2265 nm [16, 203, 15, 204, 180]. These geometries utilize nano-structures specifically designed to support the formation of toroidal dipole moments, achieving non-radiating resonances by inducing particular electromagnetic field distributions. By leveraging nano-structures with varying aspect ratios, we aim to excite anapole states using ultraviolet (UV) light.

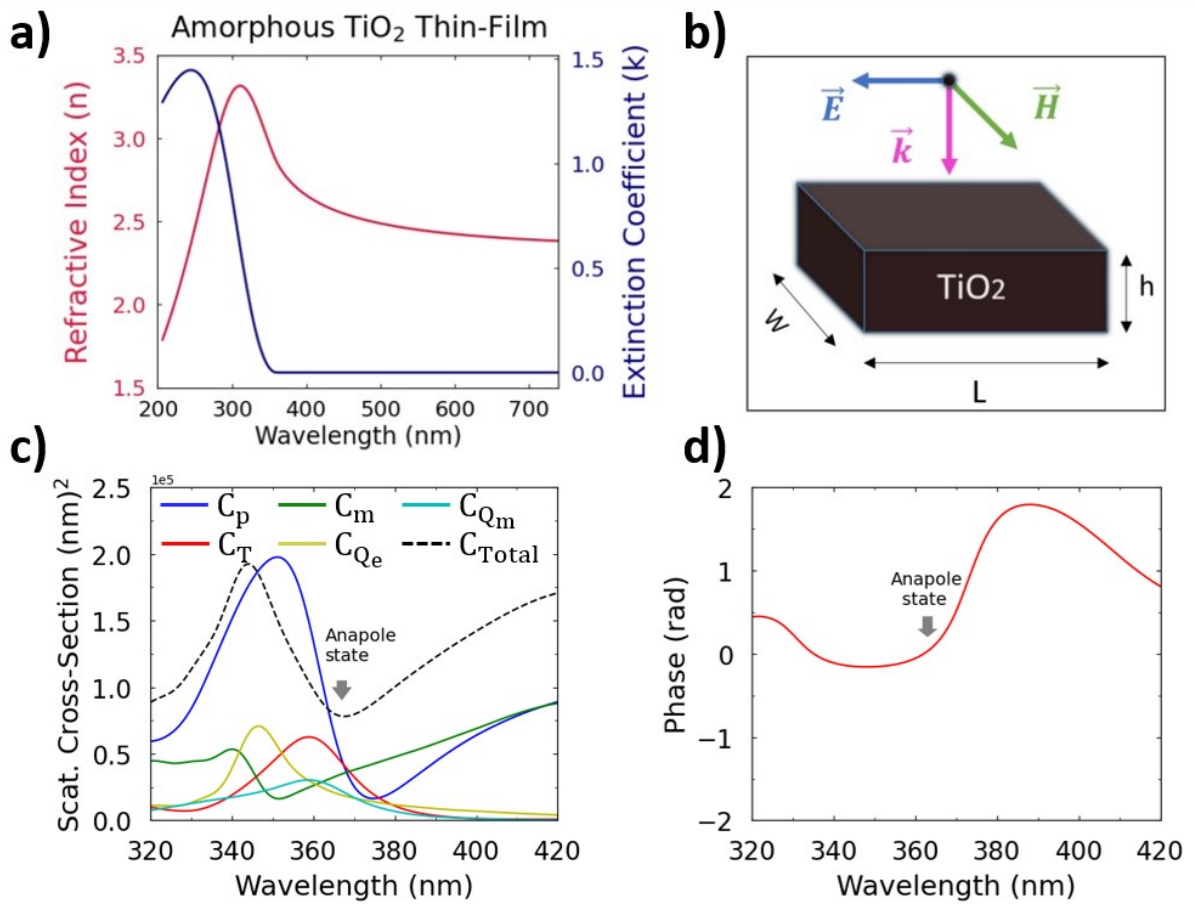


Figure 7.2: **a)** Dispersion profile of the amorphous TiO_2 refractive index (n) and extinction coefficient (k). **b)** Schematic of a free-standing TiO_2 nano-rectangle with dimensions: length L = 190 nm, width w = 160 nm, and height h = 100 nm, showing light propagation direction ($\vec{k}$) and polarization direction ($\vec{E}$) directions. **c)** Contributions of various multipole moments: C_p (blue), C_T (red), C_m (green), C_{Qe} (yellow), C_{Qm} (cyan blue) and total contribution C_{total} (black). **d)** Phase difference between the electric and toroidal dipoles (allusive to Eq.7.1).

7.3 In-depth analysis of the resonance profile

The resonance profiles exhibited between a Fano and an anapole resonance are similar, although we analyze some key distinctions in this section. The observed dip in the scattering cross-section (Fig. 7.2) is indicative of an anapole state rather than a Fano resonance. To distinguish between these two, we observe the dip in the total scattered field results from the destructive interference between the electric dipole and the toroidal dipole moment. While in the case of a Fano resonance there is also interference, it can result from a broad spectral continuum, we quote from [205]: *"Fano resonance structures occur when a discrete state and a continuum are coupled, creating two pathways from the ground state to the continuum that interfere".*

To complement this analysis we would like to highlight the electric field intensity map

(Fig. 7.3 a), which is significantly enhanced (reaching a value of 6) consistent with strong near-field localization with scattering remaining suppressed. Also, the magnetic field intensity map (Fig. 7.3 b) showing a value close to zero, indicative of minimal contribution to radiation. Those patterns, of the electric and magnetic field intensity maps, show that the resonance is trapped in the nano-rectangle with minimal radiation to the outside, indicative of the uniqueness of the anapole resonance.

7.4 Material selection and geometry definition

We identified that nano-structures with equal length-to-width are ineffective at exciting anapole states at short wavelengths, particularly within the UV spectrum below 400 nm. This limitation arises from their inability to achieve the necessary field configurations for destructive interference. In contrast, structures with varied aspect ratios and specific geometric modifications were able to support anapole states effectively. This difficulty stems from the limited availability of materials that possess the necessary optical properties at these wavelengths, which are strong refractive index and very low extinction coefficient [206, 207]. This constraint complicates the design and realization of anapole states in the UV region, thus demanding appropriate material selection to overcome the inherent limitations posed by the available material properties [208, 18].

Our investigation into nano-structures capable of supporting non-radiating anapole states in the UV regime involves a two-step approach [130]. First, we select a material with appropriate optical properties for UV operation. Second, we design nano-structures using this material to achieve anapole states at specific UV wavelengths. Amorphous TiO_2 was chosen for this study due to its excellent dielectric properties, specifically, its refractive index (n) and extinction coefficient (k) [74, Ch. 11, pp. 286–324], which are well-suited for operation at UV wavelengths below 400 nm [67]. As shown in Fig. 7.2 **a)** the dielectric properties of this material, according to Jolivet et al. (2023) [3], indicate that the material possesses a high refractive index with a low extinction coefficient over a specific range of wavelengths, suggesting that these optical properties are conducive to the excitation of anapoles for wavelengths higher than 350 nm, where the extinction coefficient (k) approaches zero. The selection of TiO_2 was influenced by the findings in M. Semmlinger et al. (2019) [16], where the authors demonstrate the effective utilization of TiO_2 metasurfaces for generating vacuum UV light. Their work highlights the harnessing of anapole resonances within TiO_2 nano-structures, optimizing conversion efficiency at short wavelengths. These characteristics establish TiO_2 as a compelling material for advancing applications in UV light generation [16].

Once we have defined the material of choice as a candidate to operate in the UV, it is necessary to determine which nanostructure geometry is more suitable for the anapole

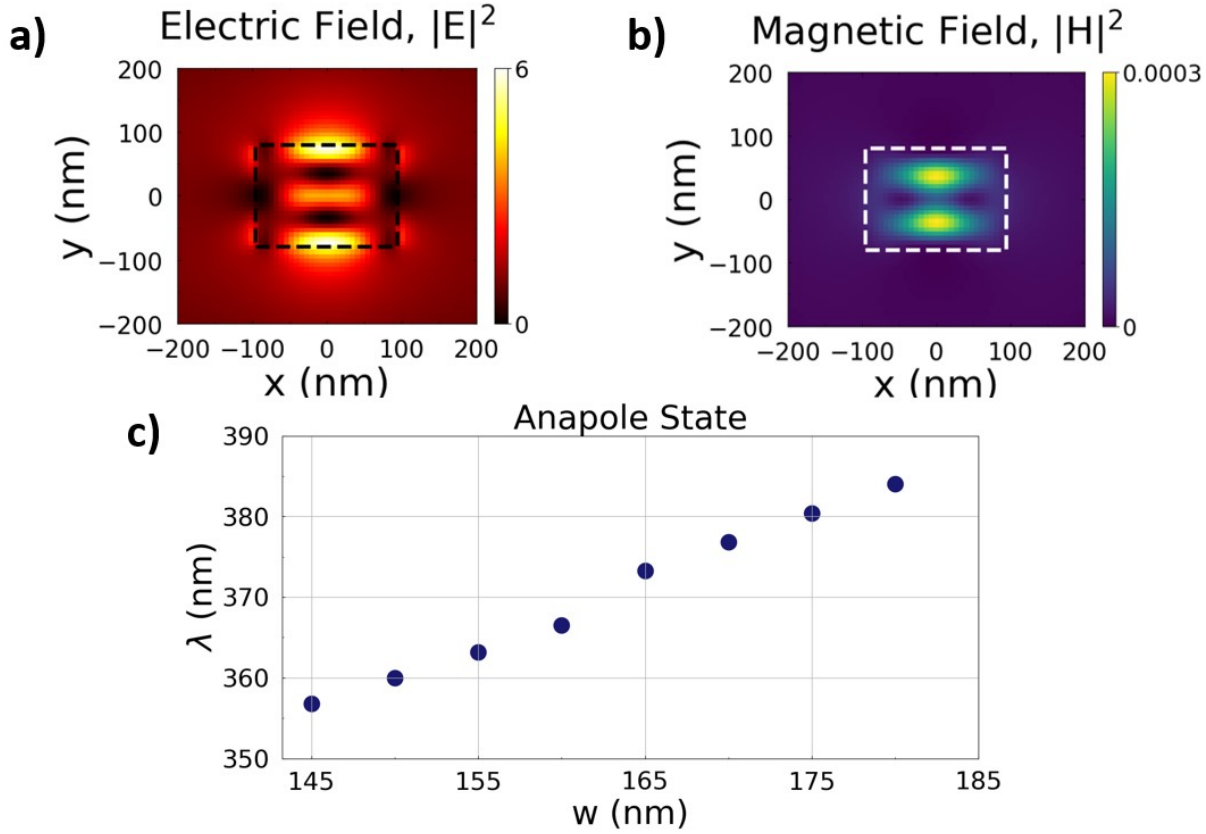


Figure 7.3: **a)** Electric and **b)** magnetic field intensities in the free-standing nano-rectangle structure. **c)** Shift of the anapole wavelength when varying the width (w) of the nano-rectangle. The electric and magnetic fields are normalized relative to the source.

excitation. The selected rectangular shape (Fig. 7.2**b)**) is free-standing rather than positioned on a substrate. In order to probe the anapole excitation, we consider a plane wave incident perpendicular to the rectangular shape, as indicated in Fig. 7.2**b)**. The incident electric field ($\vec{E}$) is aligned parallel to the length of the nano-rectangle, which is critical for the excitation of the anapole state due to the polarization-dependent response of the structure. The propagation direction of the incident light is represented by the wave vector ($\vec{k}$), and the induced magnetic field ($\vec{H}$) is depicted orthogonal to $\vec{E}$.

We investigated the scattering characteristics of different nano-structured metasurfaces using the Multipole Expansion Nanophotonics Package (MENP) tool developed by Hinamoto and Fujii [209]. This tool, adapted for our study, enabled us to decompose the scattering cross-section into partial contributions of multipoles in Cartesian coordinates up to the second order. Specifically, these multipoles include the electric dipole C_p, toroidal dipole C_T, magnetic dipole C_m, electric quadrupole C_{Qe}, magnetic quadrupole C_{Qm}, and total scattering cross-section C_{total}. This decomposition offers valuable insight into the complex light-matter interactions within the metasurface. The results from the multipolar decomposition are visually represented in Figs. 7.2 **c)** and **d)**, illustrating the

electric anapole condition achieved within the nano-rectangle. The plot demonstrates the interaction between multipole moments, with the electric anapole condition marked by a significant dip in scattering, a direct consequence of destructive interference between C_p and C_T [14]. This condition is characterized by the cancellation of the electric dipole **p** and toroidal dipole **T** moments. Notably, at approximately 366.5 nm, a pronounced minimum in the total scattering cross-section (C_{total}) indicates the occurrence of the electric anapole condition.

Subsequent simulations, using the finite-difference time-domain method (Lumerical FDTD), provided spatially resolved electric and magnetic field distributions (Figs. 7.3 **a)** and **b)** respectively), allowing us to examine the anapole resonance in detail. The electric field is confined within the structure, while the magnetic field distribution shows localized regions of intensity, characteristic of anapole states. This characteristic ensures that the resonance is an anapole, a non-radiating source, as it minimizes electromagnetic fields outside its excitation region [210]. Additionally, the spectral position of the anapole resonance within the specified UV wavelength range is demonstrated to shift by varying the width of the free-standing nano-rectangle. The results highlight the adaptability of the design, showing a direct correlation between the width of the resonator and its optical response. A similar relationship was observed by Y. Li *et. al* [211]. Notably, there is no data for a width of 190 nm (see Fig. 7.3 **c)**), as there is no anapole excitation for this size. This reinforces the need for specific aspect ratios of length and width to excite anapoles in the UV range.

7.5 Effect of the incident angle on the formation of the anapole state

The anapole state's spectral position remains constant for incident angles up to 25° (Fig. 7.4), suggesting robustness of the meta-atom, important for applications where light incidence is variable or non-perpendicular. Beyond this angle the scattering increases and a minimum in the total scattering cross-section is no longer observed. This stability ensures that the desired anapole resonance is preserved even in experimental scenarios where perfect alignment cannot be guaranteed. Temporal analysis (Fig. 7.5 and 7.6) reveals the anapole state undergoes distinct stages of formation and decay. Electric and magnetic field patterns evolve across different moments, with a brief radiative phase before the resonance stabilizes [212].

Fig. 7.4 illustrates the dependence of the anapole state on the incident angle (θ), showing the total scattering cross-section (C_{sca}) as a function of wavelength for incident angles ranging from 0° to 50°. The anapole state is characterized by a dip in the scattering spectrum, observed around 350-360 nm for normal incidence (0°).

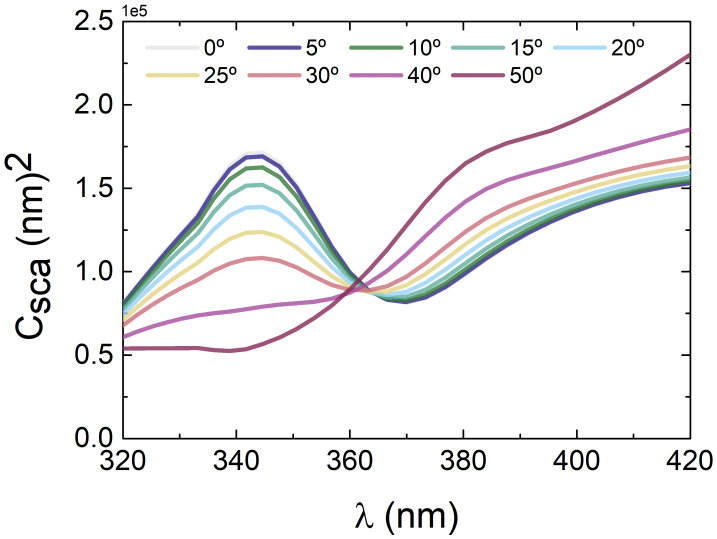


Figure 7.4: Simulated scattering cross-section for several angles of incidence θ.

As the incident angle increases, the position of the anapole state remains relatively stable up to approximately 25°. Beyond 25°, the dip in the C_{sca} deteriorates and the scattering cross-section increases, indicating that the scattering increases.

These findings reinforce the angular stability of the anapole state up to moderate incident angles, while also highlighting the limits of this stability.

7.6 Temporal evolution of the Anapole State in the UV Spectrum

Fig. 7.5 shows the temporal evolution from 8.36 to 10.26 fs of the electric (Time $|E|^2$) and magnetic (Time $|H|^2$) fields for four distinct instances. For each instance, three (or four) key moments are highlighted: the formation of the anapole pattern, the stabilization point where the anapole is fully formed and robust, and the final stage where the pattern begins to decay.

For each key moment in the four anapole instances, there is a temporal mismatch ($Time|H|^2 - Time|E|^2$) between the electric and magnetic fields. At 366 nm the optical period is 1.22 fs, so this delay corresponds to approximately one quarter of an optical cycle. Electric and magnetic fields in quadrature is precisely the signature of stored, reactive energy rather than propagating radiation, and the observed delay therefore provides an independent confirmation that the excitation is non-radiating, consistent with the suppressed far-field scattering obtained from the multipole decomposition. These time scales are critical when working in the femtosecond regime, where such delays can have noticeable impacts on the resonance behavior and field distributions. Our results are in agreement with the literature for other nanostructures [213]. In addition, the E-field

First Anapole Instance

Time|E|²	Time|H|²	|E|²/|E0|²
8.36 fs	8.63 fs	0.733
8.42 fs	8.68 fs	0.745
8.47 fs	8.73 fs	0.645

Second Anapole Instance

Time|E|²	Time|H|²	|E|²/|E0|²
8.89 fs	9.20 fs	0.543
8.94 fs	9.26 fs	0.678
9.00 fs	9.31 fs	0.716
9.05 fs	9.36 fs	0.649

Third Anapole Instance

Time|E|²	Time|H|²	|E|²/|E0|²
9.47 fs	9.78 fs	0.433
9.52 fs	9.84 fs	0.580
9.57 fs	9.89 fs	0.651
9.63 fs	9.94 fs	0.625

Fourth Anapole Instance

Time|E|²	Time|H|²	|E|²/|E0|²
10.15 fs	10.41 fs	0.554
10.20 fs	10.47 fs	0.564
10.26 fs	10.52 fs	0.495

Figure 7.5: Temporal evolution of the electric and magnetic fields during the excitation of the anapole state in the UV spectrum.

intensity ($|E|^2/|E_0|^2$) is consistent in the four anapole instances.

Following these results (analysis of the temporal evolution of the anapole state), we now investigate the radiated power during the transient state. Fig. 7.6 illustrates the calculated radiated power during the transient phase leading to the excitation of the anapole state. The plot shows an initial radiated power followed by a rapid decay, converging near-zero emission. This brief period of radiative loss can be attributed to transient effects as the system evolves toward a steady-state configuration.

Our results show a strong correlation between the temporal evolution of the electric and magnetic fields (Fig. 7.5) and the radiated power (Fig. 7.6). The anapole instances depicted in Fig. 7.5 coincide with the brief radiative bursts (with a maximum value of approximately 2.5×10^{-16} W), followed by rapid decay, observed in early stages of the

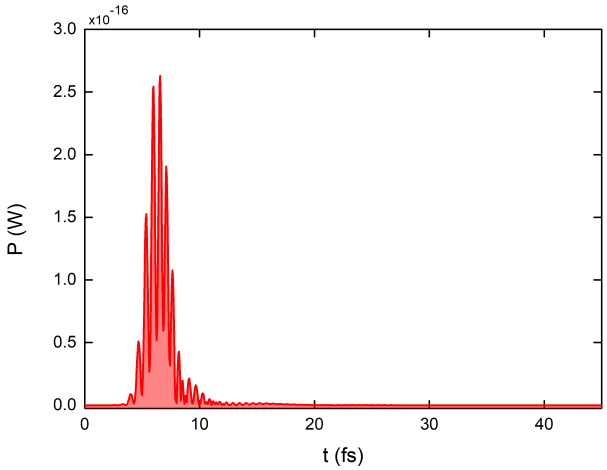


Figure 7.6: This plot illustrates the radiated power as a function of time during the transient phase of the anapole state.

transient simulations (Fig. 7.6). Such dynamics highlight the complex interplay between dipole moments and field confinement during anapole excitation.

7.7 Geometry variation and multipole analysis

Our focus shifted towards understanding how different geometric configurations could influence and potentially enhance the excitation of anapole states within this material framework [214, 215]. We compared the behavior of different free-standing nano-structures: nano-hexagon, nano-octagon, nano-decagon, and nano-ellipsoid of similar dimensions (L = 190 nm, w = 160 nm, and h = 100 nm), to understand how these geometric configurations affect the excitation of anapole states. Our results show that while the nano-rectangle and other structures can excite the anapole state, the nano-ellipsoid cannot, despite having similar dimensions.

Figs. 7.7 and 7.8 present the multipolar distributions and phase difference of the electric and toroidal dipoles, respectively, for different geometries: a) nano-hexagon, b) nano-octagon, c) nano-decagon, d) nano-ellipsoid.

Fig. 7.9 presents the anapole resonances for different nano-structures, including nano-hexagons, nano-octagons, nano-decagons, and nano-ellipsoids. The structures nano-hexagon **a)**, nano-octagon **b)** and nano-decagon **c)** all have dimensions L = 190 nm, w = 160 nm, h = 100 nm and support anapole states within the 350–380 nm range.

A nano-ellipsoid **d)** with similar dimensions does not sustain these states. However, increasing the dimensions to L = 200 nm, w = 180 nm, h = 130 nm enables the nano-ellipsoid to achieve anapole resonance.

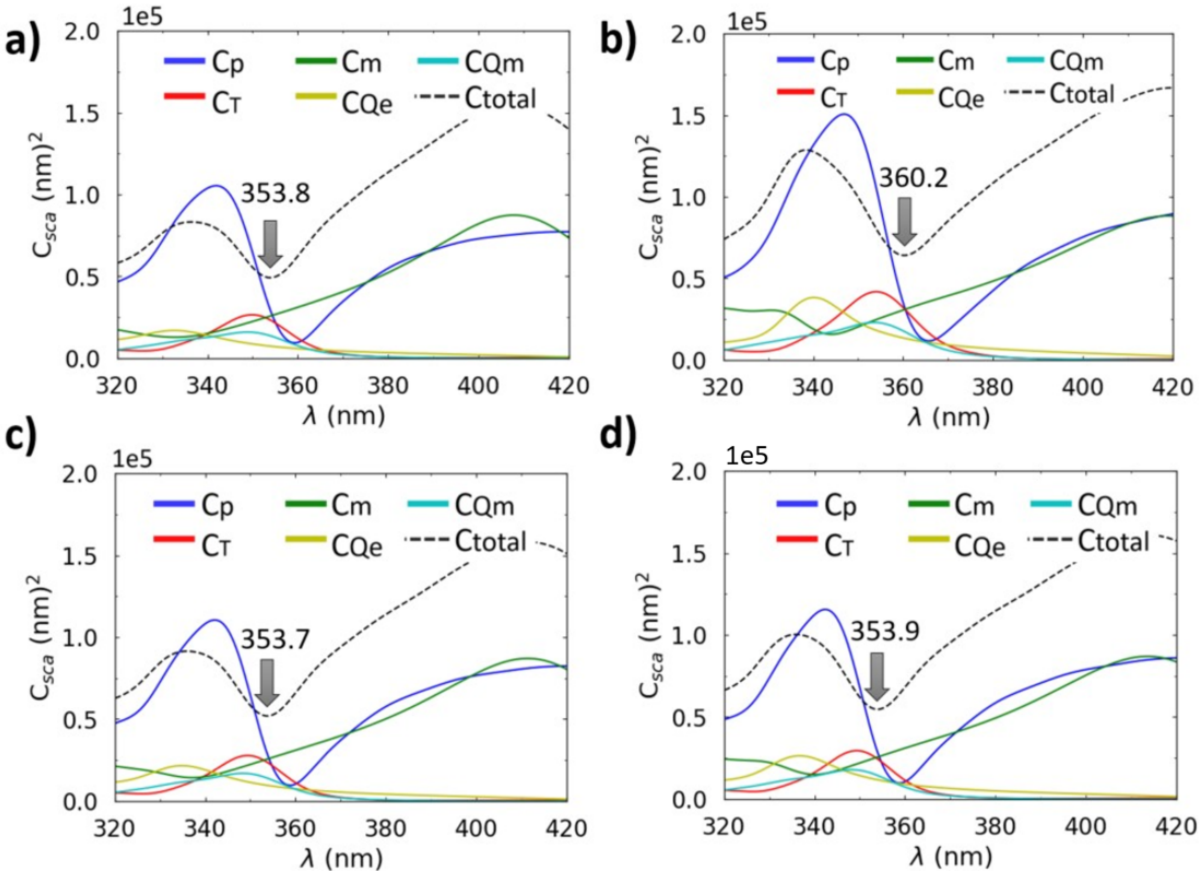


Figure 7.7: Contributions of various multipole moments: C_p (blue), C_T (red), C_m (green), C_{Qe} (yellow), C_{Qm} (cyan blue) and C_{total} (black) for different geometries that support anapole resonances within the UV-A region, using amorphous TiO_2: **a)** Nano-hexagon, **b)** Nano-octagon, **c)** Nano-decagon, **d)** Nano-ellipsoid. The arrow indicates the corresponding anapole state in λ (nm).

The field distributions for these structures exhibit the characteristics of an anapole state, with both electric and magnetic fields showing localized, non-radiating patterns. The confined electric fields within the structure, combined with the complementary magnetic field patterns, validate the results presented in the main article.

The impact of geometry on the multipolar contributions to scattering and the phase relationships between different dipole moments is significant, indicating that geometry dictates the appearance of anapole states. For instance, we observe distinct scattering and phase characteristics in the case of various TiO_2 nano-structures (Figs. 7.7 and 7.8). This finding suggests that the lack of edges in the nano-ellipsoid can be compensated by an increase in size (L = 200 nm, w = 180 nm, and h = 130 nm), emphasizing the significance of geometry in determining anapole states. Fig. 7.11 contains a visual representation comparing the nano-rectangle and the nano-ellipsoid, as well as the simulation results of the electric and magnetic fields of the anapole resonance for the different nano-structures. We have demonstrated so far different types of individual nano-structures supporting

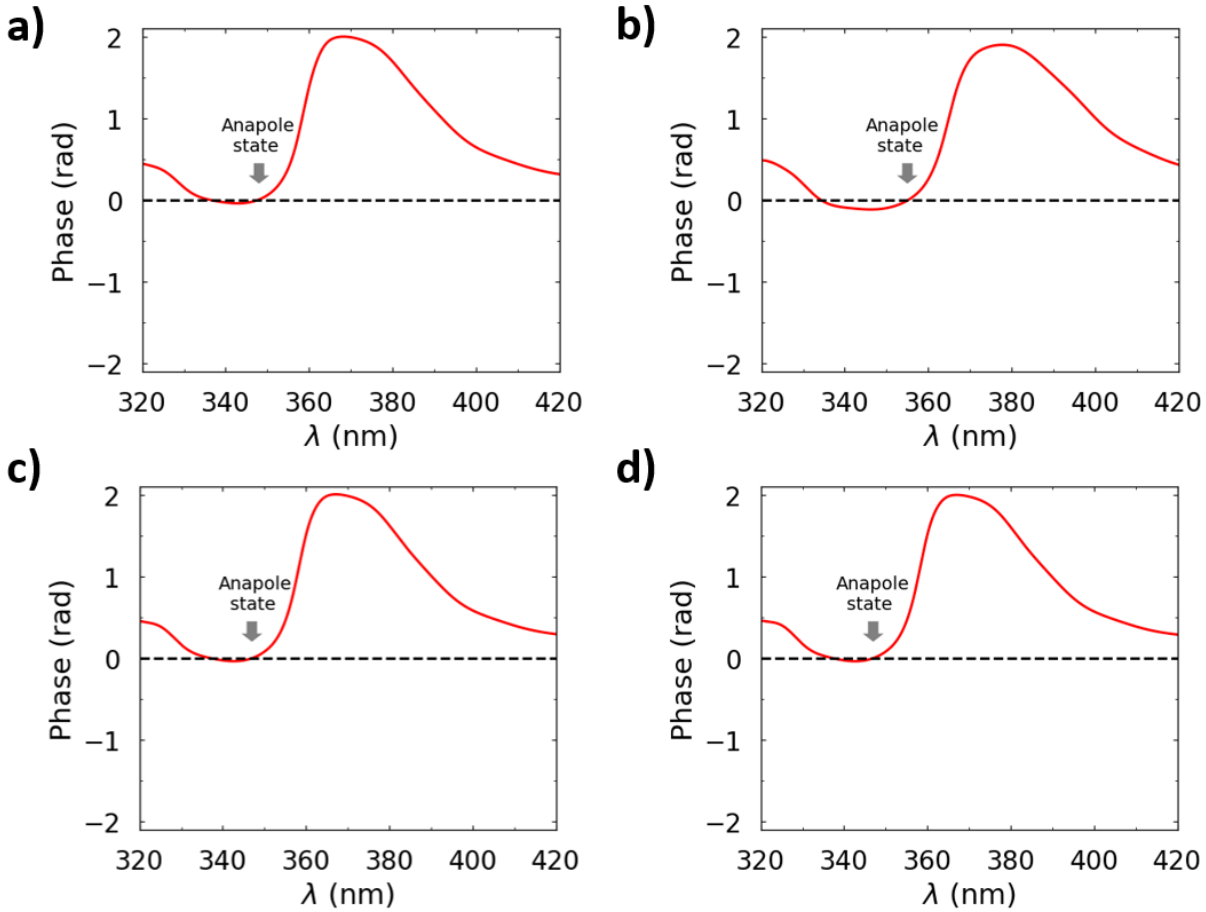


Figure 7.8: Phase difference between the electric and toroidal dipole moments for different geometries that support anapole resonances within the UV-A region, using amorphous TiO_2: **a)** Nano-hexagon,**b)** Nano-octagon, **c)** Nano-decagon, **d)** Nano-ellipsoid. The arrow indicates the anapole state.

anapole excitation in the UV. However, any experimental implementation would require the combination of many single nano-structures for the excitation of the anapoles to be detectable or performed over large areas. In order to demonstrate the feasibility of our single nano-structures designs we propose a metasurface arrangement based on realistic scales achievable via common nanofabrication techniques.

7.8 Comparison between nano-rectangle and nano-ellipsoid

Fig. 7.11 illustrates the comparison between nano-rectangles and nano-ellipsoids in terms of their ability to support anapole states. The top image compares the sizes of a nano-rectangle and a nano-ellipsoid, with dimensions of L = 190 nm, w = 160 nm, and h = 100 nm. The nano-ellipsoid is shown in dark blue and has similar dimensions but fails to support the anapole state. The bottom image illustrates the necessary increase in dimensions for the nano-ellipsoid to support the anapole state. By enlarging the nano-ellipsoid to dimensions L = 200 nm, w = 180 nm, and h = 130 nm, the anapole state is

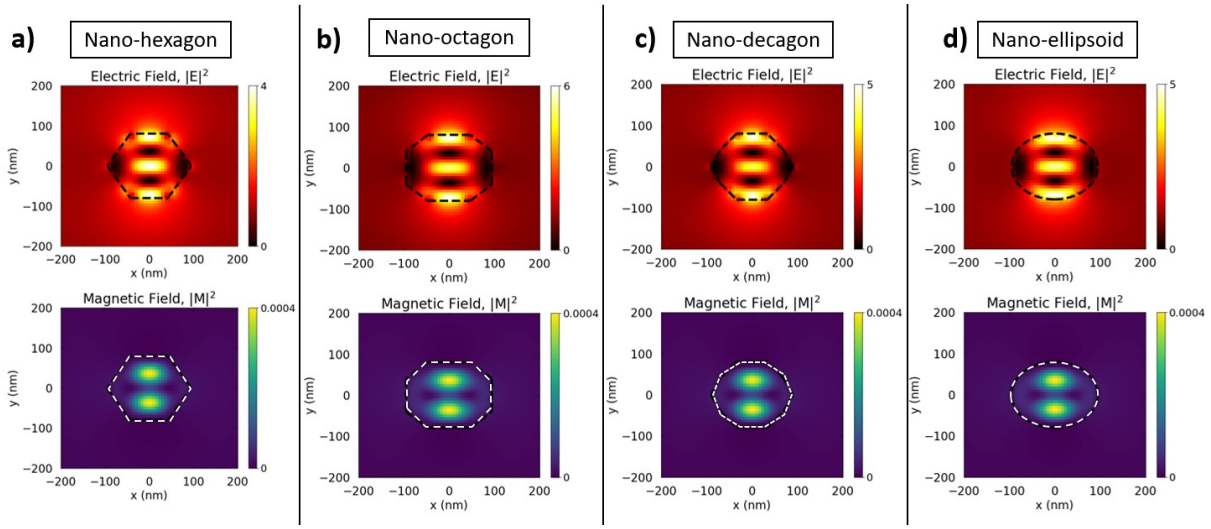


Figure 7.9: Simulation results of the anapole resonance for different nano-structures, illustrating how varying dimensions affect the resonance behavior. The wavelengths corresponding to the observed resonances are 353.8 nm, 360.2 nm, 353.7 nm, and 353.9 nm for nano-hexagon **a)**, nano-octagon **b)**, nano-decagon **c)** and nano-ellipsoid **d)** (with enlarged dimensions), respectively.

achieved.

The difficulty in fabricating nano-ellipsoids with such precise dimensional accuracy and uniformity is highlighted, making nano-rectangles more practical for experimental realization.

7.9 Metasurface

Fig. 7.12 illustrates a proposed metasurface formed by amorphous TiO_2 nano-rectangles (length L = 190 nm, width w = 160 nm and height h = 100 nm deposited atop a fused silica substrate 600 nm-thick), illuminated by UV light incident perpendicular to the metasurface plane. The refractive index data for fused silica is based on the Ref. [216]. The fused silica substrate is designed to be suspended in order to fulfill fabrication requirements on a possible experimental implementation. The presence of a low refractive index substrate will impact the final response of the metasurface, particularly in the excitation of the anapoles. Therefore, all the following calculations were performed taking into account a thin silica substrate. If the metasurface were embedded in a homogeneous medium, the refractive index contrast would change, potentially affecting the conditions for anapole excitation. The choice of utilizing nano-rectangles in the metasurface design is motivated by our goal to provide realistic and feasible designs. Firstly, results obtained for the nano-rectangle in previous simulations indicate its capability to excite anapoles effectively. Additionally, the nanofabrication techniques that would be required to produce the metasurfaces proposed here (e.g. e-beam lithography) are capable of producing more

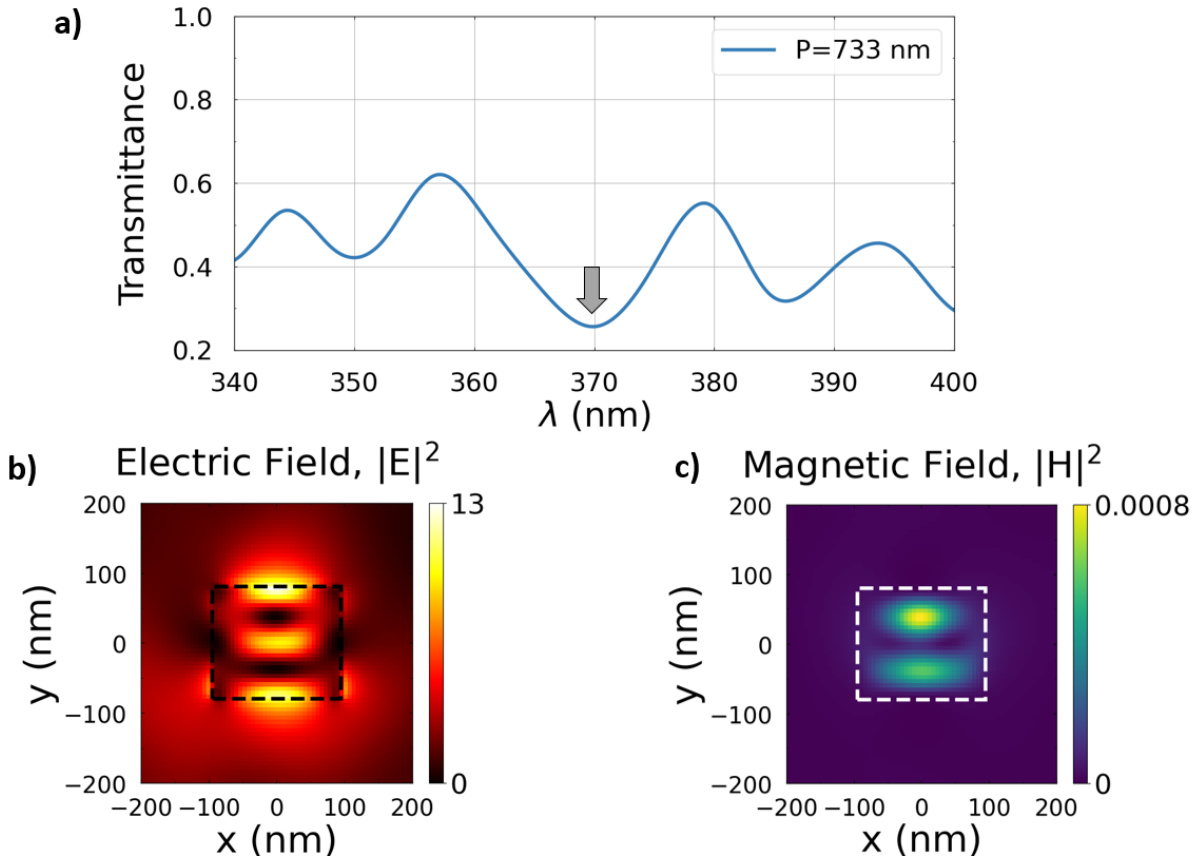


Figure 7.10: **a)** The transmittance spectrum shows a dip at 370 nm (indicated with an arrow), corresponding to the anapole state for a period of 733 nm (blue curve), indicating strong localization of the electromagnetic field. **b)** and **c)** Field intensity maps showing the electric and magnetic fields respectively, at the resonant anapole condition (P = 733 nm). The electric and magnetic fields are normalized relative to the source.

accurate square and rectangular features than round shapes. Hence, the practicality and ease of fabrication of the nano-rectangle structure, compared to other nano-structures, play a significant role in the decision-making process. All calculations are performed under the same illumination conditions as for the single configuration shown in Fig. 7.12 with the anapole excitation beam propagation normal to the metasurface and electric field polarization along the longest direction of the rectangular structures.

As can be observed in Fig. 7.10, we obtain anapole excitation at a wavelength of 370 nm for a periodicity (**P**) between individual meta-atoms of 733 nm. The anapole excitation is shown in Fig. 7.10 **a)** (blue curve) as a distinctly narrow dip in the transmittance spectrum characteristic of the anapole state. This significant reduction in scattering indicates that energy should be predominantly trapped near the nano-rectangle arrangement. Calculations for electric and magnetic field intensity at resonant wavelengths corroborate the anapole excitation as shown in Fig. 7.10 **b)** and **c)**, where strong field confinement within the nano-rectangle boundaries is observed. Fig. 7.13 contains the field intensity

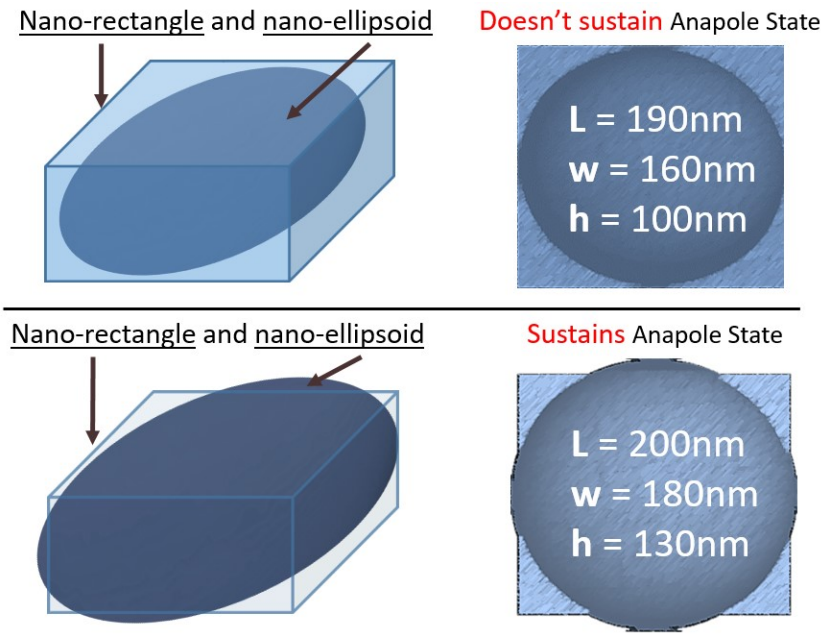


Figure 7.11: (**Top**) This image compares the sizes of a nano-ellipsoid and a nano-rectangle. The nano-rectangle, with dimensions L = 190 nm, w = 160 nm, and h = 100 nm, supports the anapole state. The nano-ellipsoid is shown in dark blue and has similar dimensions but does not support the anapole state. (**Bottom**) By enlarging the nano-ellipsoid to dimensions L = 200 nm, w = 180 nm, and h = 130 nm, the anapole state is achieved.

maps of the x-z plane cross-section of the nano-rectangle.

The anapole formed in the metasurface achieves a Q-factor of 34, approximately. This suggests moderate confinement between electromagnetic fields and radiation losses in our structure. This value may be attributed to the inherent dielectric losses of TiO_2 in the ultraviolet spectrum. We believe that the absorption characteristic of TiO_2 contributes to increased radiation losses, thus limiting the Q-factor. Further optimization of the metasurface could potentially enhance the Q-factor by reducing these losses.

It is important to note the role of the periodicity between individual meta-atoms in exciting the anapole state. To ensure that the resonance observed at 370 nm is indeed an anapole state, we conducted simulations for different periods beyond the UV range. Fig. 7.14 presents results for various periods, showing for instance that a period of 700 nm does not effectively excite anapoles. The excitation of the anapole at exactly 733 nm can be explained by taking into account the coupling conditions to in-plane modes of the metasurface.

According to the grating equation [217], a normal incidence beam will diffract into diffraction orders grazing to the grating surface for $\lambda = P/m$ with $m = 1, 2, \ldots$ and where m are the different diffracting orders and λ is the illuminating wavelength. In our case, the

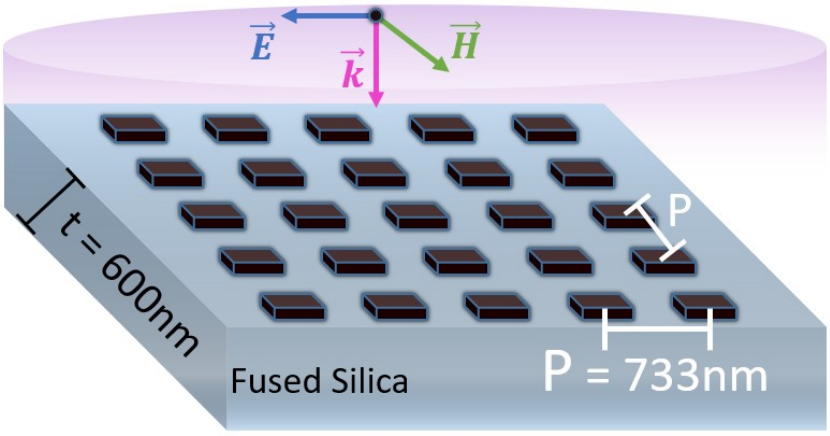


Figure 7.12: Metasurface of TiO_2 nano-rectangles deposited atop a fused silica substrate, illuminated by UV light incident perpendicular to the metasurface plane.

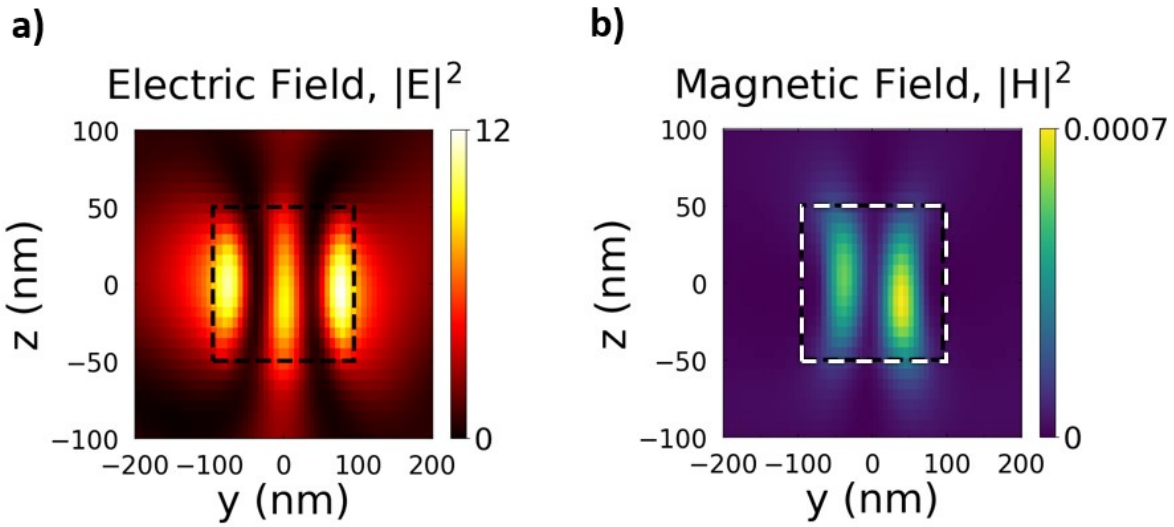


Figure 7.13: **a)** and **b)** Field intensity maps showing the electric and magnetic fields respectively, at the resonant anapole condition (P = 733 nm) in the x-z plane cross-section of the nano-rectangle.

anapole excitation occurs at $\lambda = 370\,\text{nm}$ (Fig. 7.10 **a)**) which corresponds to $\lambda \approx P/2 \approx 370\,\text{nm}$, indicating that the second-order diffraction (m=2) plays a significant role in this process. The presence of second-order diffraction contributes to the field confinement within the nano-rectangles. This diffraction enhances the coupling to the resonance, thereby promoting the strong field confinement that is characteristic of anapoles. A change in period for the same meta-atom dimensions (see Fig. 7.14) prevents efficient anapole excitation, as it opens a channel for de-confinement of the anapole field via diffraction into free space at $\lambda = 370\,\text{nm}$.

Fig. 7.13 shows both the electric field intensity map **a)** and the magnetic field intensity map **b)** in the x-z plane, showing strong localization within the nano-rectangle, thus confirming the anapole state with minimal external scattering.

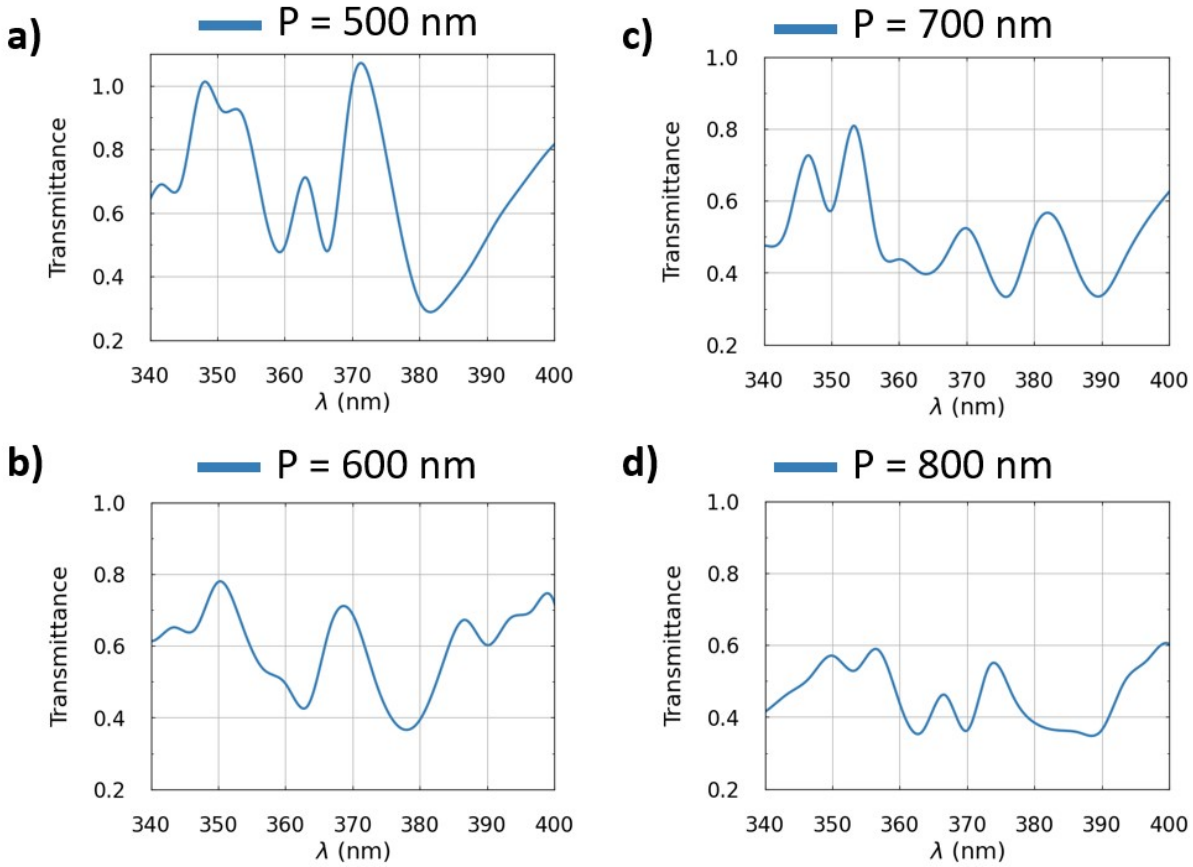


Figure 7.14: Transmittance spectrum for other periods: **a)** 500 nm, **b)** 600 nm, **c)** 700 nm and **d)** 800 nm.

Fig. 7.14 presents the transmittance spectra for different periodicities (**P**). While these spectra exhibit dips, they do not correspond to the pronounced dips associated with anapole states. The electromagnetic field intensity maps derived from simulations indicate that these dips correspond to different resonances. Specifically, the dips observed in these periodicities, although present in the same wavelength range, do not exhibit the characteristic field distribution of anapole states. This distinction helps clarify that not all low transmittance values are indicative of anapole states.

7.10 Conclusions

In this study, we demonstrated the excitation of electric anapole states in the UV range using amorphous TiO_2 nano-rectangles. By meticulously tailoring the geometry of these structures, we achieved a non-radiating anapole resonance at approximately 370 nm. Our findings indicate that the anapole resonance is highly sensitive to the geometric parameters of the nano-rectangles and the periodicity of the metasurface. The results emphasize the potential of TiO_2 nano-structures manipulating UV light, with promising applications in fields like nonlinear optics and photonic devices. The successful excitation of anapole states at short wavelengths opens new avenues for controlling light-matter interactions at the

nanoscale, contributing to the advancement of nanophotonics and optical metamaterials.

8 Conclusions and Discussion

8.1 Summary of main findings

This thesis set out to address a fundamental limitation of conventional photoionization detection (the fixed photon energy of discharge VUV lamps) by developing wavelength-tunable VUV photoionization for selective volatile organic compound (VOC) sensing. Two complementary directions were pursued: a table-top tunable VUV laser source based on nonlinear frequency conversion, and a nanophotonic system based on resonant TiO_2 nanostructures intended to enhance THG toward more compact UV/VUV sources. The main outcomes are summarized below.

A table-top tunable VUV source was demonstrated experimentally (Chapter 3) by combining a femtosecond Yb laser, an optical parametric amplifier with second-harmonic generation in a BBO crystal, and third-harmonic generation in suspended SiO_2 nanomembranes. This chain delivered quasi-monochromatic VUV light continuously tunable down to 120 nm, and was used to ionize representative VOCs (ethanolamine, eucalyptol) at saturated vapor pressure inside an in-house ionization chamber. To my knowledge, this is the first demonstration of THG VUV photoionization at atmospheric pressure with a tunable laser.

Building directly on this source, Chapter 4 introduced sample ionization potential spectroscopy (SIPS) and applied it to BTEX detection. By scanning the VUV pump wavelength across the molecular ionization thresholds of benzene, toluene, ethylbenzene, and the o- and p-xylene isomers, the work demonstrated molecular specific identification of individual BTEX components within a five component BTEX/N_2 mixture, with sensitivity reaching threshold-limit-value–time-weighted-average (TLV-TWA) levels relevant for occupational exposure. SIPS therefore enables real-time, molecule resolved VOC sensing without the laboratory infrastructure of GC–MS.

Chapter 5 reported the material study that the nanophotonic chapters build on. Comparing thermal ALD with three oxidants (H_2O, O_3, H_2O_2) at 200 °C, the work showed that O_3 produces films closest to stoichiometric TiO_2, with high refractive index, low extinction in the near-UV, and only minor anatase crystallization, whereas H_2O-grown films exhibit Ti^{3+}-related defect states ("black-TiO_2") that increase sub-bandgap absorption. AFM, XRD, and the combination of XPS, REELS, and UPS together provide a consistent picture in which oxidant chemistry, not just process temperature, sets the optical constants of the film.

Chapter 6 used these material parameters to validate energy velocity, $v_e = \langle \mathbf{S} \rangle / \langle u \rangle$, as the appropriate slow-light metric for TiO_2 in regimes where group velocity becomes ill-defined. Combining ellipsometric dispersion with Fourier-domain FDTD simulations of the Brillouin energy density, the analysis showed that the O_3-grown film reaches $v_e/c \approx 0.18$–0.21 near the UV band edge (380–400 nm). Backscattered THG spectra from the same films are consistent with, but do not by themselves establish, an association between this slow-light regime and an enhanced third-harmonic response: the strongest signal at 400 nm comes from a film for which the extinction coefficient could not be constrained well enough to extract v_e, and the contribution of the silicon substrate prevents extraction of an intrinsic film $\chi^{(3)}$ from these reflection geometries alone.

Finally, a peer-reviewed design study (Chapter 7) established that amorphous TiO_2 nano-rectangular cuboids on fused silica can support electric anapole resonances in the UV-A (350–380 nm), a regime where the combination of high refractive index and low extinction coefficient is hard to obtain. The multipole decomposition shows that the anapole condition is reached through destructive interference between electric and toroidal dipole moments, and the resonance wavelength is tunable through length-to-width aspect ratio. A 2D periodic arrangement (metasurface) preserves the resonance and produces a transmittance dip at the design wavelength. Among the geometries explored, the nano-rectangle remains the simplest fabrication target that supports the resonance, with comparable shapes (decagon, hexagon, ellipsoid) requiring more demanding fabrication for similar performance.

8.2 Comparative discussion of the two approaches

The photoionization sensing laser system (Chapters 3 and 4) is closer to a finished instrument. It is fully demonstrated experimentally, generates VUV continuously tunable to 120 nm, and is already coupled to a sensing modality (SIPS) capable of resolving individual VOCs at occupationally relevant concentrations. Its limitations are practical rather than physical: the system relies on a femtosecond OPA and BBO/nanomembrane chain that is table-top in scale rather than truly portable, and the SiO_2 nanomembranes have a finite damage threshold that bounds the achievable VUV pulse energy.

The nanophotonic approach (Chapters 5, 6, and 7) is at an earlier stage. Chapter 5 demonstrates that suitable TiO_2 films can be grown by ALD. Chapter 6 shows that the resulting material is in a slow-light regime, and establishes the energy velocity as a design indicator for identifying suitable operating wavelengths. Chapter 7 provides a verified design for a UV-A anapole metasurface operating in the same spectral region, where the strong dispersion of TiO_2 determines both the energy velocity and the resonant behaviour of the nanostructures. What this thesis does not yet provide is the integrated

demonstration: a fabricated TiO_2 anapole metasurface delivering measurably enhanced VUV through THG. The obstacle is fabrication: the metasurface requires lithographically defined cuboids of approximately $190 \times 160 \times 100$ nm, and the edge tolerances place the anapole condition close to the limit of what can be patterned reliably in 100 nm-thick TiO_2.

The two approaches should therefore be viewed as complementary. The laser source establishes the sensing concept with a system that exists today; the nanophotonic platform indicates a longer term path toward a compact source in which the bulky OPA/BBO system is replaced by a resonant thin film pumped by a simpler near-UV laser.

8.3 Outlook and future work

For the sensing platform, the priorities are extending the SIPS protocol beyond BTEX to broader VOCs, and reducing the optical component by replacing the OPA/BBO system with a more compact tunable UV-A source. Continuous monitoring trials in the IAQ contexts targeted by the LEARN project (schools and other occupied buildings) would test the SIPS protocol against the temporal dynamics for which it was designed.

For the nanophotonic design, the most informative next experiment is fabrication of the Chapter 7 anapole metasurface in O_3-grown ALD TiO_2. With substrate suppressed measurements and the energy velocity framework available, an intrinsic film $\chi^{(3)}$ can be extracted and the predicted enhancement at the anapole condition tested directly. Beyond the specific anapole geometry, the same material and the same metric should apply to other non-radiating modes (quasi-bound states in the continuum, for example) that may relax the lithographic tolerances at the price of a more complex resonance structure.

The anapole studied here is purely electric. A natural next step is the hybrid anapole, where electric and magnetic non-radiating contributions overlap, to test on the same UV-TiO_2 system whether it confines the field better or gives a stronger nonlinear response than the single-mode case treated in this thesis. The time domain is worth a separate look. In the simulations the electric and magnetic fields do not peak at the same instant, and that delay points to transient non-radiating behavior that steady state spectra cannot show (Section 7.6). Tracking the electric, magnetic, and toroidal dipole contributions in time, together with the radiated power, would show whether the structure supports a temporal or dynamic anapole, and whether that can be used to push the nonlinear conversion further.

Eventually, the two could be integrated (a TiO_2 resonator pumped by a compact near-UV laser, integrated directly with a photoionization system) would yield a wavelength tunable

VUV photoionization sensor small enough to deploy in real indoor environments. This thesis establishes the design, material, and metric needed to take on that integration.

Bibliography

[1] Rafael Vilarinho, Hedieh Mahmoodnia, Christian Maibohm, Diogo Poeta, José Fernandes, Carlos Calaza, Ana I. Gómez-Varela, Martín López-Garcia, Adelaide Miranda, and Pieter A. A. De Beule. Volatile organic compound photoionization utilizing tunable quasi-monochromatic vacuum ultraviolet down to 120 nm obtained by third harmonic generation from a suspended dielectric nanomembrane. *Opt. Express*, 33(12):26342–26355, Jun 2025.

[2] Rafael Vilarinho, Hedieh Mahmoodnia, Christian Maibohm, Diogo Poeta, Carlos Calaza, Susana Costa, Ana I. Gómez-Varela, Martín López-García, Adelaide Miranda, and Pieter A. A. De Beule. Real-time molecular specific BTEX detection at TLV-TWA concentration levels. *Environmental Technology & Innovation*, 42:104895, 2026.

[3] Aline Jolivet, Christophe Labbé, Cédric Frilay, Olivier Debieu, Philippe Marie, Bryan Horcholle, Franck Lemarié, Xavier Portier, Clara Grygiel, Sylvain Duprey, Wojciech Jadwisienczak, David Ingram, Mudit Upadhyay, Adrian David, Arnaud Fouchet, Ulrike Lüders, and Julien Cardin. Structural, optical, and electrical properties of tio2 thin films deposited by ald: Impact of the substrate, the deposited thickness and the deposition temperature. *Applied Surface Science*, 608:155214, 2023.

[4] P. J. Linstrom and W. G. Mallard. NIST chemistry WebBook, NIST standard reference database number 69. National Institute of Standards and Technology, Gaithersburg MD, 20899, 2001.

[5] Neil E. Klepeis, William C. Nelson, Wayne R. Ott, John P. Robinson, Andrew M. Tsang, Paul Switzer, Joseph V. Behar, Stephen C. Hern, and William H. Engelmann. The national human activity pattern survey (nhaps): A resource for assessing exposure to environmental pollutants. *Journal of Exposure Analysis and Environmental Epidemiology*, 11(3):231–252, 2001.

[6] Charikleia Karakosta, Stavros Vlachos, and Vasiliki Sideri. Improving indoor air quality in schools: the learn project approach. *Chemical Engineering Transactions*, 112:151–156, Oct. 2024.

[7] Csobod E, Annesi-Maesano I, Carrer P, Kephalopoulos S, Madureira J, Rudnai P, De Oliveira Fernandes E, Barrero J, Beregszászi T, Hyvärinen A, Moshammer H, Norback D, Páldy A, Pándics T, Sestini P, Stranger M, Taubel M, Varró MJ, Vaskovi

E, Ventura Silva G, and Viegi G. Sinphonie – schools indoor pollution and health observatory network in europe - final report. Technical guidance LB-NA-26738-EN-N, Publications Office of the European Union, Luxembourg (Luxembourg), 2014. (see pp. 4, 10–11).

[8] Sasan Sadrizadeh, Runming Yao, Feng Yuan, Hazim Awbi, William Bahnfleth, Yang Bi, Guangyu Cao, Cristiana Croitoru, Richard de Dear, Fariborz Haghighat, Prashant Kumar, Mojtaba Malayeri, Fuzhan Nasiri, Mathilde Ruud, Parastoo Sadeghian, Pawel Wargocki, Jing Xiong, Wei Yu, and Baizhan Li. Indoor air quality and health in schools: A critical review for developing the roadmap for the future school environment. *Journal of Building Engineering*, 57:104908, 2022.

[9] María D. Maeso-García, Francesc A. Esteve-Turrillas, and Jorge Verdú-Andrés. Applications of the photoionization detector (pid) in occupational hygiene. estimation of air changes per hour in premises with natural ventilation. *Chemosensors*, 9(12), 2021.

[10] Laurent Spinelle, Michel Gerboles, Gertjan Kok, Stefan Persijn, and Tilman Sauerwald. Review of portable and low-cost sensors for the ambient air monitoring of benzene and other volatile organic compounds. *Sensors*, 17(7), 2017.

[11] Ricardo Grosso Ferreira, Bernardo Brotas Carvalho, Luís Lemos Alves, Bruno Gonçalves, Victor Fernandez Villace, Lionel Marraffa, and Mário Lino da Silva. Vuv to ir emission spectroscopy and interferometry diagnostics for the european shock tube for high-enthalpy research. *Sensors*, 23(13), 2023.

[12] Judith Golda, Beatrix Biskup, Vincent Layes, Tristan Winzer, and Jan Benedikt. Vacuum ultraviolet spectroscopy of cold atmospheric pressure plasma jets. *Plasma Processes and Polymers*, 17(6):1900216, 2020.

[13] Kuniaki Konishi, Daisuke Akai, Yoshio Mita, Makoto Ishida, Junji Yumoto, and Makoto Kuwata-Gonokami. Tunable third harmonic generation in the vacuum ultraviolet region using dielectric nanomembranes. *APL Photonics*, 5(6):066103, 06 2020.

[14] Andrey Miroshnichenko, Andrey Evlyukhin, Ye Feng Yu, Reuben Bakker, A. Chipouline, Arseniy Kuznetsov, Boris Luk'yanchuk, Boris Chichkov, and Yuri Kivshar. Nonradiating anapole modes in dielectric nanoparticles. *Nature communications*, 6, 08 2015.

[15] Gustavo Grinblat, Yi Li, Michael Nielsen, Rupert Oulton, and Stefan Maier. En-

hanced third harmonic generation in single germanium nanodisks excited at the anapole mode. *Nano letters*, 16, 06 2016.

[16] Michael Semmlinger, Ming Zhang, Ming Lun Tseng, Tzu-Ting Huang, Jian Yang, Din Ping Tsai, Peter Nordlander, and Naomi J. Halas. Generating third harmonic vacuum ultraviolet light with a tio2 metasurface. *Nano Letters*, 19(12):8972–8978, 2019. PMID: 31693379.

[17] International Iberian Nanotechnology Laboratory. Learn project: Development of novel assessments for indoor air quality monitoring and impact on children's health. `https://www.learnproject-heu.eu`, 2022. Horizon Europe Grant Agreement No. 101057510.

[18] Lei Xu, Mohsen Rahmani, Khosro Kamali, Aristeidis Lamprianidis, Lavinia Ghirardini, Jürgen Sautter, Rocio Camacho Morales, Haitao Chen, Matthew Parry, Isabelle Staude, Guoquan Zhang, Dragomir Neshev, and Andrey Miroshnichenko. Boosting third-harmonic generation by a mirror-enhanced anapole resonator. *Light: Science and Applications*, 7:44, 07 2018.

[19] Guillermo Serrera, Javier González-Colsa, and Pablo Albella. Amplified linear and nonlinear chiral sensing assisted by anapole modes in hybrid metasurfaces. *Applied Physics Letters*, 124:251701, 06 2024.

[20] Maxim R. Shcherbakov, Dragomir N. Neshev, Ben Hopkins, Alexander S. Shorokhov, Isabelle Staude, Elizaveta V. Melik-Gaykazyan, Manuel Decker, Alexander A. Ezhov, Andrey E. Miroshnichenko, Igal Brener, Andrey A. Fedyanin, and Yuri S. Kivshar. Enhanced third-harmonic generation in silicon nanoparticles driven by magnetic response. *Nano Letters*, 14(11):6488–6492, 2014. PMID: 25322350.

[21] Lei Wang, Sergey Kruk, Lei Xu, Mohsen Rahmani, Daria Smirnova, Alexander Solntsev, Ivan Kravchenko, Dragomir Neshev, and Yuri Kivshar. Shaping the third-harmonic radiation from silicon nanodimers. *Nanoscale*, 9, 02 2017.

[22] H Guo, S.C Lee, L.Y Chan, and W.M Li. Risk assessment of exposure to volatile organic compounds in different indoor environments. *Environmental Research*, 94(1):57–66, 2004.

[23] Carolin Rösch, Tibor Kohajda, Stefan Röder, Martin von Bergen, and Uwe Schlink. Relationship between sources and patterns of vocs in indoor air. *Atmospheric Pollution Research*, 5(1):129–137, 2014.

[24] A.P. Jones. Indoor air quality and health. *Atmospheric Environment*, 33(28):4535–4564, 1999.

[25] Ib Andersen, G.R. Lundqvist, and L. Mølhave. Indoor air pollution due to chipboard used as a construction material. *Atmospheric Environment (1967)*, 9(12):1121–1127, 1975.

[26] E. Lebret, H.J. van de Wiel, H.P. Bos, D. Noij, and J.S.M. Boleij. Volatile organic compounds in dutch homes. *Environment International*, 12(1):323–332, 1986. Indoor Air Quality.

[27] Bernd Seifert, Wolfgang Mailahn, Christine Schulz, and Detlef Ullrich. Seasonal variation of concentrations of volatile organic compounds in selected german homes. *Environment International*, 15(1):397–408, 1989. Indoor Air Quality.

[28] Cecilia C. Chan, Leon Vainer, John W. Martin, and David T. Williams. Determination of organic contaminants in residential indoor air using an adsorption-thermal desorption technique. *Journal of the Air and Waste Management Association*, 40(1):62–67, 1990.

[29] L. Wallace, E. Pellizzari, and C. Wendel. Total volatile organic concentrations in 2700 personal, indoor; and outdoor air samples collected in the us epa team studies. *Indoor Air*, 1(4):465–477, 1991.

[30] Sara Bjerre Sørensen and Kasper Kristensen. Low-cost sensor-based investigation of co2 and volatile organic compounds in classrooms: Exploring dynamics, ventilation effects and perceived air quality relations. *Building and Environment*, 254:111369, 2024.

[31] W. A. McClenny. Future monitoring techniques for vocs. In H. J. Th. Bloemen and J. Burn, editors, *Chemistry and Analysis of Volatile Organic Compounds in the Environment*, pages 237–267. Springer Netherlands, Dordrecht, 1993.

[32] Wei Liu, Li Zhou, Wenting Yuan, Ling Ruan, Xinkai Wang, Yucong Guo, Zhouqing Xie, Qifan Liu, and Chen Wang. Tracking indoor volatile organic compounds with online mass spectrometry. *TrAC Trends in Analytical Chemistry*, 171:117514, 2024.

[33] Gung-Hwa Hong, Thi-Cuc Le, Guan-Yu Lin, Hung-Wen Cheng, Jhih-Yuan Yu, Racha Dejchanchaiwong, Perapong Tekasakul, and Chuen-Jinn Tsai. Long-term field calibration of low-cost metal oxide voc sensor: Meteorological and interference gas effects. *Atmospheric Environment*, 310:119955, 2023.

[34] R.A.E.S.H. Staff. *The PID Handbook: Theory and Applications of Direct-Reading Photoionization Detectors*. RAE Systems by Honeywell, 2014.

[35] Min Dang, Ruidong Liu, Fengshuo Dong, Bing Liu, and Keyong Hou. Vacuum ultraviolet photoionization on-line mass spectrometry: Instrumentation developments and applications. *TrAC Trends in Analytical Chemistry*, 149:116542, 2022.

[36] Xiaosu Ding, Hongbo Lu, Antonios Tasoglou, Amisha Shah, and Nusrat Jung. Real-time indoor sensing of volatile organic compounds during building disinfection events via photoionization detection and proton transfer reaction mass spectrometry. *Building and Environment*, 246:110953, 10 2023.

[37] Adelaide Miranda and Pieter A. A. De Beule. Atmospheric photoionization detector with improved photon efficiency: A proof of concept for application of a nanolayer thin-film electrode. *Sensors*, 21(22), 2021.

[38] Martin Lopez-Garcia, Juan Galisteo-López, Alvaro Blanco, Cefe Lopez, and Antonio García-Martín. High degree of optical tunability of self-assembled photonic-plasmonic crystals by filling fraction modification. *Advanced Functional Materials*, 20:4338–4343, 12 2010.

[39] Thomas Huser and James Chan. Raman spectroscopy for physiological investigations of tissues and cells. *Advanced Drug Delivery Reviews*, 89:57–70, 2015. Pharmaceutical applications of Raman spectroscopy – from diagnosis to therapeutics.

[40] Jiří Bílek, Petr Maršolek, Ondřej Bílek, and Pavel Buček. Field test of mini photoionization detector-based sensors—monitoring of volatile organic pollutants in ambient air. *Environments*, 9(4), 2022.

[41] IDEAL Cluster. Scientific publications. `https://web.archive.org/web/20250526210000/https://www.idealcluster.eu/projects/`, 2022. Accessed: 2025-05-26.

[42] Jennifer S. Brodbelt. Photodissociation mass spectrometry: new tools for characterization of biological molecules. *Chem. Soc. Rev.*, 43:2757–2783, 2014.

[43] Jennifer S. Brodbelt, Lindsay J. Morrison, and Inês Santos. Ultraviolet photodissociation mass spectrometry for analysis of biological molecules. *Chemical Reviews*, 120(7):3328–3380, 2020. PMID: 31851501.

[44] John P. O'Brien, Brittany D. Needham, Jeremy C. Henderson, Emily M. Nowicki, M. Stephen Trent, and Jennifer S. Brodbelt. 193 nm ultraviolet photodissociation

mass spectrometry for the structural elucidation of lipid a compounds in complex mixtures. *Analytical Chemistry*, 86(4):2138–2145, 2014. PMID: 24446701.

[45] James L. Stephenson, Matthew M. Booth, Joseph A. Shalosky, John R. Eyler, and Richard A. Yost. Infrared multiple photon dissociation in the quadrupole ion trap via a multipass optical arrangement. *Journal of the American Society for Mass Spectrometry*, 5(10):886–893, 1994. PMID: 24226234.

[46] H.P. Freund and George Neil. Free-electron lasers: Vacuum electronic generators of coherent radiation. *Proceedings of the IEEE*, 87:782–803, 06 1999.

[47] Enrico Allaria, Laura Badano, S. Bassanese, Flavio Capotondi, D. Castronovo, Paolo Cinquegrana, Miltcho Danailov, G. D'Auria, Alexander Demidovich, Raffaele De Monte, De Ninno Giovanni, S. Di Mitri, B. Diviacco, W. Fawley, Mario Ferianis, Eugenio Ferrari, Giulio Gaio, David Gauthier, Luca Giannessi, and Marco Zangrando. The fermi free-electron lasers. *Journal of Synchrotron Radiation*, 22:485–491, 04 2015.

[48] M. Rothschild and D. J. Ehrlich. A review of excimer laser projection lithography. *Journal of Vacuum Science and Technology B: Microelectronics Processing and Phenomena*, 6(1):1–17, 01 1988.

[49] Tobias Erhardt, C. Norris, Ruben Rittberger, Michael Shelley, Martin Kutzschbach, Linda Marko, Alexander Schmidt, and Wolfgang Müller. Rationale, design and initial performance of a dual-wavelength (157 and 193 nm) cryo-la-icp-ms/ms system. *Journal of Analytical Atomic Spectrometry*, 40, 01 2025.

[50] Lucie Jurkovičová, Ltaief Ben Ltaief, Andreas Roos, Ondřej Hort, Ondřej Finke, Martin Albrecht, M Z Hoque, E. Klimešová, Akgash Sundaralingam, Roman Antipenkov, Annika Grenfell, Alexandr Špaček, Wojciech Szuba, Maria Krikunova, M. Mudrich, Jaroslav Nejdl, and Jakob Andreasson. Bright continuously tunable vacuum ultraviolet source for ultrafast spectroscopy. *Communications Physics*, 7, 01 2024.

[51] R.W. Boyd. *Nonlinear Optics*. Elsevier Science, 2020.

[52] Marvin Fejer, G. Magel, Dieter Jundt, and Robert Byer. Quasi-phase-matched second harmonic generation: tuning and tolerances. *Quantum Electronics, IEEE Journal of*, 28:2631–2654, 12 1992.

[53] Reza Akbari and Arkady Major. Optical, spectral and phase-matching properties

of bibo, bbo and lbo crystals for optical parametric oscillation in the visible and near-infrared wavelength ranges. *Laser Physics*, 23:035401, 01 2013.

[54] Peter Gregorčič, Aleš Babnik, and Janez Možina. Interference effects at a dielectric plate applied as a high-power-laser attenuator. *Opt. Express*, 18(4):3871–3882, Feb 2010.

[55] Rocio Camacho-Morales, Mohsen Rahmani, Sergey Kruk, Lei Wang, Lei Xu, Daria A. Smirnova, Alexander S. Solntsev, Andrey Miroshnichenko, Hark Hoe Tan, Fouad Karouta, Shagufta Naureen, Kaushal Vora, Luca Carletti, Costantino De Angelis, Chennupati Jagadish, Yuri S. Kivshar, and Dragomir N. Neshev. Nonlinear generation of vector beams from algaas nanoantennas. *Nano Letters*, 16(11):7191–7197, 2016. PMID: 27797212.

[56] Meibao Qin, Feng Wu, Tingting Liu, Dandan Zhang, and Shuyuan Xiao. Enhanced third-harmonic generation and degenerate four-wave mixing in an all-dielectric metasurface via brillouin zone folding induced bound states in the continuum. *Phys. Rev. B*, 111:035414, Jan 2025.

[57] Jin Chen, Chun-Li Hu, Fang Kong, and Jiang-Gao Mao. High-performance second-harmonic-generation (shg) materials: New developments and new strategies. *Accounts of Chemical Research*, 54(12):2775–2783, 2021. PMID: 34043910.

[58] Lun Qu, Lu Bai, Chunyan Jin, Qiang Liu, Wei Wu, Bofeng Gao, Juntao Li, Wei Cai, Mengxin Ren, and Jingjun Xu. Giant second harmonic generation from membrane metasurfaces. *Nano Letters*, 22(23):9652–9657, 2022. PMID: 36445198.

[59] Rocio Camacho-Morales, Lei Xu, Haizhong Zhang, Son Tung Ha, Leonid Krivitsky, Arseniy I. Kuznetsov, Mohsen Rahmani, and Dragomir Neshev. Sum-frequency generation in high-q gap metasurfaces driven by leaky-wave guided modes. *Nano Letters*, 22(15):6141–6148, 2022. PMID: 35867018.

[60] Qingchen Yuan, Liang Fang, Hanlin Fang, Juntao Li, Tao Wang, Wanqi Jie, Jianlin Zhao, and Xuetao Gan. Second harmonic and sum-frequency generations from a silicon metasurface integrated with a two-dimensional material. *ACS Photonics*, 6(9):2252–2259, 2019.

[61] Jian Wei You, Zhihao Lan, and Nicolae C. Panoiu. Four-wave mixing of topological edge plasmons in graphene metasurfaces. *Science Advances*, 6(13):eaaz3910, 2020.

[62] Lei Xu, Daria A Smirnova, Rocio Camacho-Morales, Rifat Ahmmed Aoni,

Khosro Zangeneh Kamali, Marcus Cai, Cuifeng Ying, Ze Zheng, Andrey E Miroshnichenko, Dragomir N Neshev, and Mohsen Rahmani. Enhanced four-wave mixing from multi-resonant silicon dimer-hole membrane metasurfaces. *New Journal of Physics*, 24(3):035002, mar 2022.

[63] Anastasiia Zalogina, Luca Carletti, Anton Rudenko, Jerome V. Moloney, Aditya Tripathi, Hoo-Cheol Lee, Ilya Shadrivov, Hong-Gyu Park, Yuri Kivshar, and Sergey S. Kruk. High-harmonic generation from a subwavelength dielectric resonator. *Science Advances*, 9(17):eadg2655, 2023.

[64] Sebastian A. Schulz and et al. Roadmap on photonic metasurfaces. *Applied Physics Letters*, 124(26):260701, 06 2024.

[65] Nanfang yu and Federico Capasso. Flat optics with designer metasurfaces. *Nature materials*, 13:139–50, 01 2014.

[66] Arseniy I. Kuznetsov, Andrey E. Miroshnichenko, Mark L. Brongersma, Yuri S. Kivshar, and Boris Luk'yanchuk. Optically resonant dielectric nanostructures. *Science*, 354(6314):aag2472, 2016.

[67] Denis G. Baranov, Dmitry A. Zuev, Sergey I. Lepeshov, Oleg V. Kotov, Alexander E. Krasnok, Andrey B. Evlyukhin, and Boris N. Chichkov. All-dielectric nanophotonics: the quest for better materials and fabrication techniques. *Optica*, 4(7):814–825, Jul 2017.

[68] Yali Huang, Haixia Xu, Yanxin Lu, and Yihang Chen. All-dielectric metasurface for achieving perfect reflection at visible wavelengths. *The Journal of Physical Chemistry C*, 122(5):2990–2996, 2018.

[69] Borja García García, María Gabriela Fernández-Manteca, Dimitrios C. Zografopoulos, Celia Gómez-Galdós, Alain A. Ocampo-Sosa, Luis Rodríguez-Cobo, José Francisco Algorri, and Adolfo Cobo. Plasmonic and dielectric metasurfaces for enhanced spectroscopic techniques. *Biosensors*, 15(7), 2025.

[70] Simona Checcucci, Thomas Bottein, Jean-Benoit Claude, Tom Wood, Magali Putero, Luc Favre, M. Gurioli, Marco Abbarchi, and David Grosso. Titania-based spherical mie resonators elaborated by high-throughput aerosol spray: Single object investigation. *Advanced Functional Materials*, 28:1801958, 08 2018.

[71] Han-Don Um, Deokjae Choi, Amit Solanki, Emerald Huang, Kwanyong Seo, and Fawwaz Habbal. Dynamic selection of visible wavelengths using resonant tio2

nanostructures. *Nanophotonics*, 12(11):1995–2005, 2023.

[72] Javier Cambiasso, Gustavo Grinblat, Yi Li, Aliaksandra Rakovich, Emiliano Cortés, and Stefan A. Maier. Bridging the gap between dielectric nanophotonics and the visible regime with effectively lossless gallium phosphide antennas. *Nano Letters*, 17(2):1219–1225, 2017.

[73] Saman Jahani and Zubin Jacob. All-dielectric metamaterials. *Nature Nanotechnology*, 11:23–36, 01 2016.

[74] Craig F. Bohren and Donald R. Huffman. *Absorption and Scattering of Light by Small Particles*. John Wiley and Sons, 1998.

[75] Gustavo Grinblat, Yi Li, Michael Nielsen, Rupert Oulton, and Stefan Maier. Efficient third harmonic generation and nonlinear subwavelength imaging at a higher-order anapole mode in a single germanium nanodisk. *ACS Nano*, 11, 01 2017.

[76] Gustavo Coelho Rezende, Stéphane Le Calvé, Jürgen J. Brandner, and David Newport. Micro photoionization detectors. *Sensors and Actuators B: Chemical*, 287:86–94, 2019.

[77] Frank Gunzer, Sascha Krüger, and Jürgen Grotemeyer. Photoionization and photofragmentation in mass spectrometry with visible and uv lasers. *Mass Spectrometry Reviews*, 38(2):202–217, 2019.

[78] Jie Li, Jian Lu, Andrew Chew, Seunghwoi Han, Jialin Li, Yi Wu, He Wang, Shambhu Ghimire, and Zenghu Chang. Attosecond science based on high harmonic generation from gases and solids. *Nature Communications*, 11:2748, 06 2020.

[79] Kuang-Yu Yang, Ruggero Verre, Jérémy Butet, Chen Yan, Tomasz J. Antosiewicz, Mikael Käll, and Olivier J. F. Martin. Wavevector-selective nonlinear plasmonic metasurfaces. *Nano Letters*, 17(9):5258–5263, 2017. PMID: 28829601.

[80] Sergejs Boroviks, Andrei Kiselev, Karim Achouri, and Olivier J. F. Martin. Demonstration of a plasmonic nonlinear pseudodiode. *Nano Letters*, 23(8):3362–3368, 2023. PMID: 37043888.

[81] Sergejs Boroviks and Olivier J. F. Martin. Monocrystalline gold metasurface to control anisotropic second-harmonic generation. *Advanced Optical Materials*, 13(13):2402525, 2025.

[82] P.R. West, S. Ishii, G.V. Naik, N.K. Emani, V.M. Shalaev, and A. Boltasseva.

Searching for better plasmonic materials. *Laser and Photonics Reviews*, 4(6):795–808, 2010.

[83] M. Kuttge, E. J. R. Vesseur, J. Verhoeven, H. J. Lezec, H. A. Atwater, and A. Polman. Loss mechanisms of surface plasmon polaritons on gold probed by cathodoluminescence imaging spectroscopy. *Applied Physics Letters*, 93(11):113110, 09 2008.

[84] Janne-Petteri Niemelä, Giovanni Marin, and Maarit Karppinen. Titanium dioxide thin films by atomic layer deposition: a review. *Semiconductor Science and Technology*, 32(9):093005, aug 2017.

[85] Robert C. Devlin, Mohammadreza Khorasaninejad, Wei Ting Chen, Jaewon Oh, and Federico Capasso. Broadband high-efficiency dielectric metasurfaces for the visible spectrum. *Proceedings of the National Academy of Sciences*, 113(38):10473–10478, 2016.

[86] Jorge Vazquez, Tibor Suta, Balint Fodor, László Makai, Oscar Contreras, Amin Bahrami, Kornelius Nielsch, and Hugo Tiznado. Optical properties of tio2 grown by atomic layer deposition using various oxidizing agents: The ellipsometry analysis of absorption properties. *Advanced Materials Interfaces*, 11, 07 2024.

[87] Mina Shahmohammadi, Rajib Mukherjee, Christos Takoudis, and Urmila Diwekar. Optimal design of novel precursors materials for the atomic layer deposition using computer-aided molecular design. *Chemical Engineering Science*, page 116416, 01 2021.

[88] Yong-Wan Kim and Do-Heyoung Kim. Atomic layer deposition of tio2 from tetrakis-dimethylamido-titanium and ozone. *Korean Journal of Chemical Engineering*, 29, 07 2012.

[89] Sebastian Miszczak and Bozena Pietrzyk. Anatase–rutile transformation of tio2 sol–gel coatings deposited on different substrates. *Ceramics International*, 41, 07 2015.

[90] Nobuaki Negishi and Koji Takeuchi. Preparation of tio2 thin film photocatalysts by dip coating using a highly viscous solvent. *Journal of Sol-Gel Science and Technology*, 22:23–31, 09 2001.

[91] A. Sandell, Martin Andersson, Mikael Johansson, Patrik Karlsson, Y. Alfredsson, J. Schnadt, H. Siegbahn, and Per Uvdal. Metalorganic chemical vapor deposition of

anatase titanium dioxide on si: Modifying the interface by pre-oxidation. *Surface Science - SURFACE SCI*, 530:63–70, 04 2003.

[92] Seongchan Kang, Rodolphe Mauchauffé, Yong Sung You, and Se Moon. Insights into the role of plasma in atmospheric pressure chemical vapor deposition of titanium dioxide thin films. *Scientific Reports*, 8, 11 2018.

[93] Mohammad Rehan, Xiaojun Lai, and Girish M. Kale. Hydrothermal synthesis of titanium dioxide nanoparticles studied employing in situ energy dispersive x-ray diffraction. *CrystEngComm*, 13:3725–3732, 2011.

[94] Makiko Yamagishi, Shna Kuriki, Pung Song, and Yuzo Shigesato. Thin film tio2 photocatalyst deposited by reactive magnetron sputtering. *Thin Solid Films*, 442:227–231, 10 2003.

[95] Octavian Simionescu, Cosmin Romanitan, Oana Brincoveanu, Valentin Ion, Octavian Buiu, and Andrei Avram. Rf magnetron sputtering deposition of tio2 thin films in a small continuous oxygen flow rate. *Coatings*, 9:442, 07 2019.

[96] Eric Kumi Barimah, R Penhale-Jones, R Salimian, Hari Upadhyaya, Abul Hasnath, and G. Jose. Phase evolution, morphological, optical and electrical properties of femtosecond pulsed laser deposited tio2 thin films. *Scientific Reports*, 10, 06 2020.

[97] Walied Alarif, Yasser Shaban, Mohammed Orif, Mohammed Ghandourah, Adnan Turki, Hajer Alorfi, and Hermine Tadros. Green synthesis of tio2 nanoparticles using natural marine extracts for antifouling activity. *Marine Drugs*, 21:62, 01 2023.

[98] Vladimir Gerasik and Marek Stastna. Complex group velocity and energy transport in absorbing media. *Phys. Rev. E*, 81:056602, May 2010.

[99] J. Peatross, S. A. Glasgow, and M. Ware. Average energy flow of optical pulses in dispersive media. *Phys. Rev. Lett.*, 84:2370–2373, Mar 2000.

[100] A Kuzmich, Arthur Dogariu, L Wang, P Milonni, and Raymond Chiao. Signal velocity, causality, and quantum noise in superluminal light pulse propagation. *Physical review letters*, 86:3925–9, 05 2001.

[101] Carlton M. Caves. Quantum limits on noise in linear amplifiers. *Phys. Rev. D*, 26:1817–1839, Oct 1982.

[102] T. G. Philbin. Electromagnetic energy momentum in dispersive media. *Phys. Rev. A*, 83:013823, Jan 2011.

[103] Aaron Welters, Yehuda Avniel, and Steven G. Johnson. Speed-of-light limitations in passive linear media. *Phys. Rev. A*, 90:023847, Aug 2014.

[104] M. Scalora, M. A. Vincenti, D. de Ceglia, N. Akozbek, M. Ferrera, C. Rizza, A. Alù, N. Litchinitser, C. Cojocaru, and J. Trull. Extreme electrodynamics in time-varying media. *Phys. Rev. A*, 112:013502, Jul 2025. See also Supplementary Material, Eq. (S1).

[105] Kseniia V. Baryshnikova, Daria A. Smirnova, and Boris S. Luk'yanchuk. Optical anapoles: Concepts and applications. *Advanced Optical Materials*, 7(14):1801350, 2019.

[106] Jiepeng Wu, Fanwei Zhang, Qiang Li, Qianbin Feng, Yu Wu, and Lijun Wu. Strong field enhancement in individual ϕ-shaped dielectric nanostructures based on anapole mode resonances. *Opt. Express*, 28(1):570–579, Jan 2020.

[107] Evelyn Díaz-Escobar, Ángela I. Barreda, Amadeu Griol, and Alejandro Martínez. Experimental observation of higher-order anapoles in individual silicon disks under in-plane illumination. *Applied Physics Letters*, 121(20), 2022.

[108] Diogo Poeta, Rafael Vilarinho, Martín López-García, and Pieter A. A. De Beule. Optical anapoles excited by uv-a illumination. *Opt. Lett.*, 49(24):7198–7201, Dec 2024.

[109] Yunyue Zhang, Mingyu Sun, Min Gu, and Qiming Zhang. Enhanced third harmonic generation in nanospheres at the anapole mode under tightly focused radially polarized beams. *Opt. Express*, 33(1):75–84, Jan 2025.

[110] W. C. Haxton, C.-P. Liu, and M. J. Ramsey-Musolf. Nuclear anapole moments. *Physical Review C*, 65(4), March 2002.

[111] Sergey E. Svyakhovskiy, Vladimir V. Ternovski, and Michael I. Tribelsky. Anapole: Its birth, life, and death. *Opt. Express*, 27(17):23894–23904, Aug 2019.

[112] Isobel Falconer. Corpuscles, electrons and cathode rays: J.j. thomson and the 'discovery of the electron'. *The British Journal for the History of Science*, 20(3):241–276, 1987.

[113] Paul Ehrenfest. Ungleichförmige elektrizitätsbewegungen ohne magnet- und strahlungsfeld. *Physik Z*, 11:708–709, 1910. How do electrons move in the atom and what keeps them there. The London, Edinburgh, and Dublin Philosophical Magazine and Journal of Science.

[114] N. Bohr. I. on the constitution of atoms and molecules. *The London, Edinburgh, and Dublin Philosophical Magazine and Journal of Science*, 26(151):1–25, 1913.

[115] Ia B Zel'Dovich. Electromagnetic interaction with parity violation. *Sov. Phys. JETP*, 6(6):1184–1186, 1958.

[116] T. D. Lee and C. N. Yang. Question of parity conservation in weak interactions. *Phys. Rev.*, 104:254–258, Oct 1956.

[117] W. C. Haxton. Atomic parity violation and the nuclear anapole moment. *Science*, 275(5307):1753–1753, 1997.

[118] S. Esposito. Transition from majorana to weyl fermions and anapole moments. *arXiv preprint arXiv:1304.3669*, 2013.

[119] C. M. Wilson and D. E. Prober. Quasiparticle number fluctuations in superconductors. *Phys. Rev. B*, 69:094524, Mar 2004.

[120] A Yu Kitaev. Unpaired majorana fermions in quantum wires. *Physics-Uspekhi*, 44(10S):131, oct 2001.

[121] C. S. Wood, S. C. Bennett, D. Cho, B. P. Masterson, J. L. Roberts, C. E. Tanner, and C. E. Wieman. Measurement of parity nonconservation and an anapole moment in cesium. *Science*, 275(5307):1759–1763, 1997.

[122] Bertozzi Eugenio. Toward a history of explanation in science communication: the case of madame wu experiment on parity violation. *Journal of Science Communication*, 16:A10, 07 2017.

[123] Leon M. Lederman and Dick Teresi. *The God Particle: If the Universe is the Answer, What is the Question?* Houghton Mifflin Harcourt, 1993.

[124] Trevon Badloe, Jihae Lee, Junhwa Seong, and Junsuk Rho. Tunable metasurfaces: The path to fully active nanophotonics. *Advanced Photonics Research*, 2, 07 2021.

[125] Kirill Koshelev, Sergey Kruk, Elizaveta Melik-Gaykazyan, Jae-Hyuck Choi, Andrey Bogdanov, Hong-Gyu Park, and Yuri Kivshar. Subwavelength dielectric resonators for nonlinear nanophotonics. *Science*, 367(6475):288–292, 2020.

[126] Wei Liu and Yuri S. Kivshar. Generalized kerker effects in nanophotonics and meta-optics [invited]. *Opt. Express*, 26(10):13085–13105, May 2018.

[127] Carl Pfeiffer and A. Grbic. Metamaterial huygens' surfaces: Tailoring wave fronts with reflectionless sheets. *Physical review letters*, 110:197401, 05 2013.

[128] Zelin Hu, Linyun Long, Rongqiao Wan, Chen Zhang, Lei Zhang, Jianchang Yan, Huigao Duan, and Liancheng Wang. Ultrawide bandgap aln metasurfaces for ultraviolet focusing and routing. *Opt. Lett.*, 45(13):3466–3469, Jul 2020.

[129] X. H. Wang, D. P. West, N. B. McKeown, and T. A. King. Determining the cubic susceptibility $\chi^{(3)}$ of films or glasses by the Maker fringe method: a representative study of spin-coated films of copper phthalocyanine derivation. *Journal of the Optical Society of America B*, 15(7):1895–1903, 1998.

[130] Luis V. Rodríguez de Marcos, Juan I. Larruquert, and José A. Méndez et al. Self-consistent optical constants of mgf2, laf3, and cef3 films. *Opt. Mater. Express*, 7(3):989–1006, Mar 2017.

[131] Ansys Inc., Vancouver, Canada. *Ansys® Lumerical FDTD Solutions, Professional 2023*, 2023.

[132] Víctor M. Rivilla, Izaskun Jiménez-Serra, Jesús Martín-Pintado, Carlos Briones, Lucas F. Rodríguez-Almeida, Fernando Rico-Villas, Belén Tercero, Shaoshan Zeng, Laura Colzi, Pablo de Vicente, Sergio Martín, and Miguel A. Requena-Torres. Discovery in space of ethanolamine, the simplest phospholipid head group. *Proceedings of the National Academy of Sciences*, 118(22):e2101314118, 2021.

[133] R. Ramachandran, Milan Sil, Prasanta Gorai, J. K. Meka, Pavithraa Sundararajan, J.-I. Lo, S.-L. Chou, Y.-J. Wu, P. Janardhan, B.-M. Cheng, Anil Bhardwaj, Víctor M. Rivilla, N. J. Mason, B. Sivaraman, and Ankan Das. Experimental and computational study of ethanolamine ices under astrochemical conditions. *The Astrophysical Journal*, 975(2):181, nov 2024.

[134] Santiago Aparicio, Rafael Alcalde, María J. Dávila, Begoña García, and José M. Leal. Properties of 1,8-cineole: a thermophysical and theoretical study. *The Journal of Physical Chemistry B*, 111(12):3167–3177, 2007. PMID: 17388451.

[135] Cai Zimin, Jianqing Peng, Yi Chen, Ling Tao, Yan-Yan Zhang, Ling-Yun Fu, Qing-de Long, and Xiangchun Shen. 1,8-cineole: a review of source, biological activities, and application 1,8-cineole: a review of source, biological activities, and application. *Journal of Asian Natural Products Research*, 10 2020.

[136] J. Schuberth. Volatile organic compounds determined in pharmaceutical products

by full evaporation technique and capillary gas chromatography/ion-trap detection. *Analytical Chemistry*, 68(8):1317–1320, 1996.

[137] Dimitra L. Capone, Katryna Van Leeuwen, Dennis K. Taylor, David W. Jeffery, Kevin H. Pardon, Gordon M. Elsey, and Mark A. Sefton. Evolution and occurrence of 1,8-cineole (eucalyptol) in australian wine. *Journal of Agricultural and Food Chemistry*, 59(3):953–959, 2011. PMID: 21204528.

[138] Seiko Goto, Hinako Suzuki, Toshinori Nakagawa, and Kuniyoshi Shimizu. The effect of eucalyptol on nursing home residents. *Scientific Reports*, 10:3996, 03 2020.

[139] G. A. Eiceman, Z. Karpas, and Jr. H. H. Hill. *Ion Mobility Spectrometry*. CRC Press, 3rd edition, 2013.

[140] Jr. Jellison, G. E. and F. A. Modine. Parameterization of the optical functions of amorphous materials in the interband region. *Applied Physics Letters*, 69(3):371–373, 07 1996.

[141] S.C. Siah, B. Hoex, and A.G. Aberle. Accurate characterization of thin films on rough surfaces by spectroscopic ellipsometry. *Thin Solid Films*, 545:451–457, 2013.

[142] Elizabeth Hedl, Željko Samec, Vesna Blažek Bregović, Alexander Bergmann, and Jordi Sancho-Parramon. A coating design to rule them all: asymmetric fabry–perot resonators with diverse optical response based on au films with high-temperature sensitivity. *Opt. Lett.*, 49(15):4102–4105, Aug 2024.

[143] Imrana Kabir, L. Sheppard, Riyadh Shamiri, Pramod Koshy, Rong Liu, W. Joe, Anh Le, X. Lu, W.-F Chen, and Charles Sorrell. Contamination of tio2 thin films spin coated on borosilicate and rutile substrates. *Journal of Materials Science*, 55, 03 2020.

[144] Hal Amick, Michael Gendreau, Todd Busch, and Colin Gordon. Evolving criteria for research facilities: vibration. *Proceedings of SPIE, The International Society for Optical Engineering*, 5933:16–28, 08 2005.

[145] Adelaide Miranda, Marco Martins, and Pieter A. A. De Beule. Simultaneous differential spinning disk fluorescence optical sectioning microscopy and nanomechanical mapping atomic force microscopy. *Review of Scientific Instruments*, 86(9):093705, 09 2015.

[146] Meijiao Huang, Liang Guo, and Fengyi Jiang. Regime map of the effective medium approximation modelling of micro-rough surfaces in ellipsometry. *Sensors*, 24:1242,

02 2024.

[147] Maarten Vos, Sean W. King, and Benjamin L. French. Measurement of the band gap by reflection electron energy loss spectroscopy. *Journal of Electron Spectroscopy and Related Phenomena*, 212:74–80, 2016.

[148] Marco Piccardo, Chi-Kang Li, Yuh-Renn Wu, James S. Speck, Bastien Bonef, Robert M. Farrell, Marcel Filoche, Lucio Martinelli, Jacques Peretti, and Claude Weisbuch. Localization landscape theory of disorder in semiconductors. ii. urbach tails of disordered quantum well layers. *Phys. Rev. B*, 95:144205, Apr 2017.

[149] Xiaobo Chen, Lei Liu, Peter Y. Yu, and Samuel S. Mao. Increasing solar absorption for photocatalysis with black hydrogenated titanium dioxide nanocrystals. *Science*, 331(6018):746–750, 2011.

[150] Stefan Wendt, Renald Schaub, Jesper Matthiesen, Ebbe Vestergaard, Erik Wahlstrom, M Rasmussen, Peter Thostrup, Luis Molina, Erik Lægsgaard, I Stensgaard, and Flemming Besenbacher. Oxygen vacancies on tio2(110) and their interaction with h2o and o2: A combined high-resolution stm and dft study. *Surface Science*, 598:226–245, 12 2005.

[151] Thomas F. Krauss. Why do we need slow light? *Nature Photonics*, 2(8):448–450, August 2008.

[152] Dmitry Budker and Michael Romalis. Optical magnetometry. *Nature Physics*, 3, 12 2006.

[153] Toshihiko Baba. Slow light in photonic crystals. *Nature Photonics*, 2:465–473, 08 2008.

[154] Shuang Zhang, Dentcho Genov, Yuan Wang, Ming Liu, and Xiang Zhang. Plasmon-induced transparency in metamaterials. *Physical review letters*, 101:047401, 08 2008.

[155] Robert W. Boyd. Material slow light and structural slow light: similarities and differences for nonlinear optics [Invited]. *Journal of the Optical Society of America B*, 28(12):A38, December 2011.

[156] H Dachraoui, T Auguste, A Helmstedt, P Bartz, M Michelswirth, N Mueller, W Pfeiffer, P Salieres, and U Heinzmann. Interplay between absorption, dispersion and refraction in high-order harmonic generation. *Journal of Physics B: Atomic, Molecular and Optical Physics*, 42(17):175402, aug 2009.

[157] George C. Sherman and Kurt Edmund Oughstun. Energy-velocity description of pulse propagation in absorbing, dispersive dielectrics. *J. Opt. Soc. Am. B*, 12(2):229–247, Feb 1995.

[158] Christopher C. Evans, Jonathan D. B. Bradley, Erwin A. Martí-Paname no, and Eric Mazur. Mixed two- and three-photon absorption in bulk rutile (tio2) around 800 nm. *Opt. Express*, 20(3):3118–3128, Jan 2012.

[159] R.M. Joseph and A. Taflove. Fdtd maxwell's equations models for nonlinear electrodynamics and optics. *IEEE Transactions on Antennas and Propagation*, 45(3):364–374, 1997.

[160] Winston Frias, Andrei Smolyakov, and Akira Hirose. Non-local energy transport in tunneling and plasmonic structures. *Opt. Express*, 19(16):15281–15296, Aug 2011.

[161] Maibelin Rosales, Tamara Zoltan, Ciro Yadarola, Edgar Mosquera, Francisco Gracia, and Andreina García. The influence of the morphology of 1d tio2 nanostructures on photogeneration of reactive oxygen species and enhanced photocatalytic activity. *Journal of Molecular Liquids*, 281:59–69, 2019.

[162] Feng Zhou, Chunjie Yan, Qi Sun, and Sridhar Komarneni. Tio2/sepiolite nanocomposites doped with rare earth ions: Preparation, characterization and visible light photocatalytic activity. *Microporous and Mesoporous Materials*, 274:25–32, 2019.

[163] Wael H.M. Abdelraheem, Meghshyam K. Patil, Mallikarjuna N. Nadagouda, and Dionysios D. Dionysiou. Hydrothermal synthesis of photoactive nitrogen- and boron-codoped tio2 nanoparticles for the treatment of bisphenol a in wastewater: Synthesis, photocatalytic activity, degradation byproducts and reaction pathways. *Applied Catalysis B: Environmental*, 241:598–611, 2019.

[164] Junli Li, Min Zhang, Zhongjie Guan, Qiuye Li, Chunqing He, and Jianjun Yang. Synergistic effect of surface and bulk single-electron-trapped oxygen vacancy of tio2 in the photocatalytic reduction of co2. *Applied Catalysis B: Environmental*, 206:300–307, 2017.

[165] Thomas R. Gordon, Matteo Cargnello, Taejong Paik, Filippo Mangolini, Ralph T. Weber, Paolo Fornasiero, and Christopher B. Murray. Nonaqueous synthesis of tio2 nanocrystals using tif4 to engineer morphology, oxygen vacancy concentration, and photocatalytic activity. *Journal of the American Chemical Society*, 134(15):6751–6761, 2012. PMID: 22444667.

[166] Gayan W. C. Kumarage, Hadjer Hakkoum, and Elisabetta Comini. Recent advancements in tio2 nanostructures: Sustainable synthesis and gas sensing. *Nanomaterials*, 13(8), 2023.

[167] Maria-Anna Gatou, Athanasia Syrrakou, Nefeli Lagopati, and Evangelia A. Pavlatou. Photocatalytic tio2-based nanostructures as a promising material for diverse environmental applications: A review. *Reactions*, 5(1):135–194, 2024.

[168] Qingyao Wang, Huili Li, Xueling Yu, Yao Jia, Yuan Chang, and Shanmin Gao. Morphology regulated bi2wo6 nanoparticles on tio2 nanotubes by solvothermal sb3+ doping as effective photocatalysts for wastewater treatment. *Electrochimica Acta*, 330:135167, 2020.

[169] Arun Jayaseelan, Nachiappan Senthilnathan, Goutham Rangarajan, Ram Prasath Alagappan, Gopinath Kannappan Panchamoorthy, and Eric Lichtfouse. Synthesis and application of titanium dioxide photocatalysis for energy, decontamination and viral disinfection: a review. *Environmental Chemistry Letters*, 08 2022.

[170] Di Jiang, Haifeng Yuan, Zhen Liu, Yuke Chen, Yangyang Li, Xiaoli Zhang, Guobin Xue, Hong Liu, Xiaoyan Liu, Lili Zhao, and Weijia Zhou. Defect-anchored single-atom-layer pt clusters on tio2-x/ti for efficient hydrogen evolution via photothermal reforming plastics. *Applied Catalysis B: Environmental*, 339:123081, 2023.

[171] Christopher C. Evans, Chengyu Liu, and Jin Suntivich. Low-loss titanium dioxide waveguides and resonators using a dielectric lift-off fabrication process. *Opt. Express*, 23(9):11160–11169, May 2015.

[172] Jacek Nizioł, Magdalena Zięba, Maciej Śniechowski, Ewa Gondek, Wojciech Pakieła, and Paweł Karasiński. Demonstration of the optical isotropy of tio2 thin films prepared by the sol–gel method. *Materials*, 17(14), 2024.

[173] Oili M.E. Ylivaara, Andreas Langner, Xuwen Liu, Dieter Schneider, Jaakko Julin, Kai Arstila, Sakari Sintonen, Saima Ali, Harri Lipsanen, Timo Sajavaara, Simo-Pekka Hannula, and Riikka L. Puurunen. Mechanical and optical properties of as-grown and thermally annealed titanium dioxide from titanium tetrachloride and water by atomic layer deposition. *Thin Solid Films*, 732:138758, 2021.

[174] Dipto Saha, Rohini Ajimsha, Karan Rajiv, Chandrachur Mukherjee, Mukul Gupta, Pankaj Misra, and L. Kukreja. Spectroscopic ellipsometry characterization of amorphous and crystalline tio2 thin films grown by atomic layer deposition at different temperatures. *Applied Surface Science*, 315:116–123, 10 2014.

[175] Denis Baranov, R. Verre, and Pawel Karpinski. Anapole-enhanced intrinsic raman scattering from silicon nanodisks. *ACS Photonics*, 5, 05 2018.

[176] Juan Sebastian Totero Gongora, Andrey Miroshnichenko, Yuri Kivshar, and Andrea Fratalocchi. Anapole nanolasers for mode-locking and ultrafast pulse generation. *Nature Communications*, 8:15535, 05 2017.

[177] Evangelia Takou, Anna C. Tasolamprou, Odysseas Tsilipakos, and Eleftherios N. Economou. Dynamic anapole in metasurfaces made of sculptured cylinders. *Phys. Rev. B*, 100:085431, Aug 2019.

[178] Aristeidis G. Lamprianidis and Andrey E. Miroshnichenko. Excitation of nonradiating magnetic anapole states with azimuthally polarized vector beams. *Beilstein Journal of Nanotechnology*, 9:1478–1490, 2018.

[179] Alexey Basharin, Vitaly Chuguevsky, Nikita Volsky, M. Kafesaki, and E. Economou. Extremely high q-factor metamaterials due to anapole excitation. *Physical Review B*, 95, 07 2016.

[180] José Francisco Algorri, Dimitrios C. Zografopoulos, Antonio Ferraro, Braulio García-Cámara, Ricardo Vergaz, Romeo Beccherelli, and José Manuel Sánchez-Pena. Anapole modes in hollow nanocuboid dielectric metasurfaces for refractometric sensing. *Nanomaterials*, 9(1), 2019.

[181] Jinyong Hu, Wangdi Bai, Chuxuan Tan, Yiming Li, Qi Lin, and Lingling Wang. Highly electric field enhancement induced by anapole modes coupling in the hybrid dielectric–metal nanoantenna. *Optics Communications*, 511:127987, 2022.

[182] Polina Kapitanova, Esmaeel Zanganeh, Nikita Pavlov, Mingzhao Song, Pavel Belov, Andrey Evlyukhin, and Andrey Miroshnichenko. Seeing the unseen: Experimental observation of magnetic anapole state inside a high-index dielectric particle. *Annalen der Physik*, 532(12):2000293, 2020.

[183] Evangelia Takou, A. Tasolamprou, Odysseas Tsilipakos, Zacharias Viskadourakis, M. Kafesaki, George Kenanakis, and E. Economou. Anapole tolerance to dissipation losses in thermally tunable water-based metasurfaces. *Physical Review Applied*, 15, 01 2021.

[184] Susanta Kumar Das, Christoph Schwanke, Andreas Pfuch, Wolfgang Seeber, Martin Bock, Günter Steinmeyer, Thomas Elsaesser, and Ruediger Grunwald. Highly efficient thg in tio2 nanolayers for third-order pulse characterization. *Opt. Express*,

19(18):16985–16995, Aug 2011.

[185] Christopher Taudt. *Polymer Characterization*, pages 89–121. Springer Fachmedien Wiesbaden, Wiesbaden, 2022. See Chapter 4, Eq. (4.38).

[186] R Ruppin. Electromagnetic energy density in a dispersive and absorptive material. *Physics Letters A*, 299(2):309–312, 2002.

[187] Pi-Gang Luan. Power loss and electromagnetic energy density in a dispersive metamaterial medium. *Phys. Rev. E*, 80:046601, Oct 2009.

[188] Frederico Nunes, Thiago Vasconcelos, Marcel Bezerra, and John Weiner. Electromagnetic energy density in dispersive and dissipative media. *Journal of the Optical Society of America B*, 28:1544–1552, 05 2011.

[189] Ardavan F. Oskooi, David Roundy, Mihai Ibanescu, Peter Bermel, J.D. Joannopoulos, and Steven G. Johnson. Meep: A flexible free-software package for electromagnetic simulations by the fdtd method. *Computer Physics Communications*, 181(3):687–702, 2010.

[190] Luis V. Rodríguez de Marcos, Juan I. Larruquert, José A. Méndez, and José A. Aznárez. Self-consistent optical constants of sio2 and ta2o5 films. *Opt. Mater. Express*, 6(11):3622–3637, Nov 2016.

[191] D. E. Aspnes and A. A. Studna. Dielectric functions and optical parameters of si, ge, gap, gaas, gasb, inp, inas, and insb from 1.5 to 6.0 ev. *Phys. Rev. B*, 27:985–1009, Jan 1983.

[192] Robert W. Boyd. Chapter 2 - wave-equation description of nonlinear optical interactions. In Robert W. Boyd, editor, *Nonlinear Optics (Third Edition)*, pages 69–133. Academic Press, Burlington, third edition edition, 2008.

[193] Safa O. Kasap. *Optoelectronics and Photonics: Principles and Practices*. Pearson Education, Boston, 2 edition, 2013. Ch. 1: p. 31; p. 33 (Fig. 1.8); p. 34 for SiO2.

[194] Chunyan Jin, Ben Liu, Zhongxiang Lei, and J. Sun. Structure and photoluminescence of the tio2 films grown by atomic layer deposition using tetrakis-dimethylamino titanium and ozone. *Nanoscale research letters*, 10:95, 12 2015.

[195] Robert Adair, L. L. Chase, and Stephen A. Payne. Nonlinear refractive index of optical crystals. *Phys. Rev. B*, 39:3337–3350, Feb 1989.

[196] Tadanori Hashimoto, Toshinobu Yoko, and Sumio Sakka. Sol–gel preparation and third-order nonlinear optical properties of TiO_2 thin films. *Bulletin of the Chemical Society of Japan*, 67(3):653–660, 03 1994.

[197] Hua Long, Aiping Chen, Guang Yang, Yuhua Li, and Peixiang Lu. Third-order optical nonlinearities in anatase and rutile TiO_2 thin films. *Thin Solid Films*, 517:5601–5604, 08 2009.

[198] Milton Kerker, Herman W. Chew, PETER J. Mcnulty, Josip P. Kratohvil, D. D. Cooke, Michael J. Sculley, and M. P. Lee. Light scattering and fluorescence by small particles having internal structure. *The journal of histochemistry and cytochemistry : official journal of the Histochemistry Society*, 27:250–263, 1979.

[199] Tolga Ergin, Nicolas Stenger, Patrice Brenner, John B. Pendry, and Martin Wegener. Three-dimensional invisibility cloak at optical wavelengths. *Science*, 328(5976):337–339, 2010.

[200] Reza Masoudian Saadabad, Lujun Huang, and Andrey B. Evlyukhin et al. Multi-faceted anapole: from physics to applications. *Opt. Mater. Express*, 12(5):1817–1837, May 2022.

[201] Hooman Barati Sedeh and Natalia M. Litchinitser. Singular optics empowered by engineered optical materials. *Nanophotonics*, 12(14):2687–2716, 2023.

[202] Kirill Koshelev, Gael Favraud, Andrey Bogdanov, Yuri Kivshar, and Andrea Fratalocchi. Nonradiating photonics with resonant dielectric nanostructures. *Nanophotonics*, 8(5):725–745, 2019.

[203] Evelyn Díaz-Escobar, Thomas Bauer, Elena Pinilla-Cienfuegos, Ángela I Barreda, Amadeu Griol, L Kuipers, and Alejandro Martínez. Radiationless anapole states in on-chip photonics. *Light: Science and Applications*, 10:204, 2021.

[204] Andrey B. Evlyukhin, Tim Fischer, and Carsten Reinhardt et al. Optical theorem and multipole scattering of light by arbitrarily shaped nanoparticles. *Phys. Rev. B*, 94:205434, Nov 2016.

[205] Konstantin Litvinenko, Nguyen Le, B. Redlich, C. Pidgeon, N.V. Abrosimov, Yu.M. Andreev, Zhiming Huang, and Ben Murdin. The multi-photon induced fano effect. *Nature Communications*, 12, 01 2021.

[206] Lei Wei, Zheng Xi, and Nandini Bhattacharya et al. Excitation of the radiationless anapole mode. *Optica*, 3(8):799–802, Aug 2016.

[207] Kwang-Hyon Kim and Wi-Song Rim. Anapole resonances facilitated by high-index contrast between substrate and dielectric nanodisk enhance vacuum ultraviolet generation. *ACS Photonics*, 5(12):4769–4775, 2018.

[208] Kwang-Hyon Kim. Low-index dielectric metasurfaces supported by metallic substrates for efficient second-harmonic generation in the blue-ultraviolet range. *Phys. Chem. Chem. Phys.*, 22:7300–7305, 2020.

[209] Tatsuki Hinamoto and Minoru Fujii. Menp: an open-source matlab implementation of multipole expansion for nanophotonics. *OSA Continuum*, 4(5):1640–1648, May 2021.

[210] Shi-Qiang Li and Kenneth B. Crozier. Origin of the anapole condition as revealed by a simple expansion beyond the toroidal multipole. *Phys. Rev. B*, 97:245423, Jun 2018.

[211] Yang Li, Guanqing Zhang, Yutao Tang, Xuecai Zhang, Wenfeng Cai, Yanjun Liu, Tun Cao, and Guixin Li. Third harmonic generation from the gold/amorphous silicon hybrid metasurface. *Nanophotonics*, 11(10):2245–2251, 2022.

[212] Adrià Canós Valero, Egor Gurvitz, Fedor Benimetskiy, Dmitry Pidgayko, Anton Samusev, Andrey Evlyukhin, Vjaceslavs Bobrovs, Dmitrii Redka, Michael Tribelsky, Mohsen Rahmani, Khosro Kamali, Alexander Pavlov, Andrey Miroshnichenko, and Alexander Shalin. Theory, observation, and ultrafast response of the hybrid anapole regime in light scattering. *Laser and Photonics Reviews*, 15:2100114, 08 2021.

[213] Mohammad Mirmoosa, Grigorii Ptitcyn, R. Fleury, and Sergei Tretiakov. Instantaneous radiation from time-varying electric and magnetic dipoles. *Physical Review A*, 102(1), 7 2020.

[214] Manuela Jakob, Haim Levanon, and Prashant V. Kamat. Charge distribution between uv-irradiated tio2 and gold nanoparticles: Determination of shift in the fermi level. *Nano Letters*, 3(3):353–358, 2003.

[215] Ichiro Tanabe and Yukihiro Ozaki. Far- and deep-ultraviolet spectroscopic investigations for titanium dioxide: electronic absorption, rayleigh scattering, and raman spectroscopy. *J. Mater. Chem. C*, 4:7706–7717, 2016.

[216] I. H. Malitson. Interspecimen comparison of the refractive index of fused silica∗,†. *J. Opt. Soc. Am.*, 55(10):1205–1209, Oct 1965.

[217] Jiaqi Quan, Qingjia Zhou, Yanyan Cao, and Yadong Xu. Understanding of wood

anomalies in metallic gratings from phase gradient metasurfaces. *Frontiers in Materials*, 9:901794, 04 2022.

[218] Ilpo Niskanen, Viviane Forsberg, Daniel Zakrisson, Salim Reza, Magnus Hummelgård, Britta Andres, Igor Fedorov, Terhi Suopajärvi, Henrikki Liimatainen, and Göran Thungström. Determination of nanoparticle size using rayleigh approximation and mie theory. *Chemical Engineering Science*, 201:222–229, 2019.

[219] Tatyana Ivanova, Antoaneta Harizanova, Tatyana Koutzarova, and Raphael Closset. Crystallization and optical behaviour of nanocomposite sol-gel tio2:ag films. *Molecules*, 29(21), 2024.

[220] R. Asahi, Y. Taga, W. Mannstadt, and A. J. Freeman. Electronic and optical properties of anatase tio_2. *Phys. Rev. B*, 61:7459–7465, Mar 2000.

A List of Publications

A list of publications arising from this thesis.

Peer-reviewed journal articles

1. **D. Poeta**, R. Vilarinho, M. López-García, and P. De Beule, "Optical anapoles excited by UV-A illumination," *Optics Letters* **49**(24), 7198–7201 (2024). DOI: 10.1364/OL.536925.

2. R. Vilarinho, H. Mahmoodnia, C. Maibohm, **D. Poeta**, J. Fernandes, C. Calaza, A. I. Gómez-Varela, M. López-García, A. Miranda, and P. De Beule, "Volatile organic compound photoionization utilizing tunable quasi-monochromatic vacuum ultraviolet down to 120 nm obtained by third harmonic generation from a suspended dielectric nanomembrane," *Optics Express* **33**(12), 26342 (2025). DOI: 10.1364/OE.564577.

3. R. Vilarinho, H. Mahmoodnia, C. Maibohm, **D. Poeta**, C. Calaza, S. Costa, A. I. Gómez-Varela, M. López-García, A. Miranda, and P. De Beule, "Real-time molecular specific BTEX detection at TLV-TWA concentration levels," *Environmental Technology & Innovation* **42**, 104895 (2026). DOI: 10.1016/j.eti.2026.104895.

Manuscripts under review

4. **D. Poeta** *et al.*, "Energy velocity as a measure for the nonlinear response of TiO_2 in the ultraviolet," *manuscript under review* (2026).

B Conference Contributions

This appendix reproduces the posters and presentations through which parts of this work were shown during its development, together with a collaboration in which ALD TiO_2 films grown for Chapter 5 were supplied to an external group.

The material reproduced here shows the state of the work at the time of each presentation.

- Three Minute Thesis (3MT) competition, INL, Braga, Portugal, March 2024. Oral presentation, *Anapoles Excited by Ultraviolet Illumination.*
- 1st Symposium of the Early-Stage Researchers Association (ESRA), INL, Braga, Portugal, 19–20 November 2024. Poster: Diogo Poeta, Rafael Vilarinho, Martín López-García, Pieter A. A. De Beule, *Optical Anapoles Excited by UV-A Illumination.*
- SPIE Optics + Optoelectronics 2025, Metamaterials XV, Prague, Czech Republic, 7–10 April 2025. Poster: Diogo Poeta, Rafael Vilarinho, Hedieh Mahmoodnia, Carlos Calaza, Helder Fonseca, Adelaide Miranda, Pieter A. A. De Beule, *TiO_2/SiO_2 metasurfaces for the enhancement of THG with pump wavelengths in the UV-A.*
- Thin-film collaboration, UCCS, 2025. ALD TiO_2 films grown with O_3 as oxidant were provided to Valerie Thorns as a reference material for a PhD research on thin films.
- European Researchers' Night 2025 Photo Contest, entry *TiO_2 Reflections: Towards a Greener Future.*

C Acronyms

AFM	atomic force microscopy
ALD	atomic layer deposition
BBO	β-barium borate
BTEX	benzene, toluene, ethylbenzene and xylenes
CVD	chemical vapor deposition
DAQ	data acquisition
ED	electric dipole
FDTD	finite-difference time-domain
FWHM	full width at half maximum
GC–MS	gas chromatography–mass spectrometry
HHG	high-harmonic generation
IAQ	indoor air quality
IE	ionization energy
IP	ionization potential
IR	infrared
LBO	lithium triborate
MD	magnetic dipole
MEEP	MIT Electromagnetic Equation Propagation
MFC	mass flow controller
MS/MS	tandem mass spectrometry
MSE	mean squared error
NIR	near-infrared
OEL	occupational exposure limit
OPA	optical parametric amplifier
PCB	printed circuit board
PECVD	plasma-enhanced chemical vapor deposition
PID	photoionization detector
PML	perfectly matched layer
REELS	reflection electron energy-loss spectroscopy
RMS	root mean square
SHG	second-harmonic generation
SIPS	sample ionization potential spectroscopy
SVP	saturated vapor pressure
TD	toroidal dipole

TDMAT	tetrakis(dimethylamido)titanium
TE	transverse electric
THG	third-harmonic generation
TLV-TWA	threshold limit value, time-weighted average
TMA	trimethylaluminium
TTIP	titanium tetraisopropoxide
TVOC	total volatile organic compounds
TWA	time-weighted average
UPS	ultraviolet photoelectron spectroscopy
UV	ultraviolet
UV-A	ultraviolet-A
VOC	volatile organic compound
VUV	vacuum ultraviolet
XPS	X-ray photoelectron spectroscopy
XRD	X-ray diffraction

D Materials and Methods

D.1 Materials and methods

This appendix documents the experimental and computational work behind the main chapters. Section D.1.1 covers the laboratory development of the photoionization detection (PID) platform: the vapor generation rigs, the sapphire window filtering, the electronic acquisition chain, and the operational lessons from running it. Section D.2 covers the simulations referenced from the main chapters.

D.1.1 Photoionization detection platform

The platform was built and commissioned in stages. The first stage was an experimental prototype together with a characterization of the IVC102U switched integrator front end. Then came LabVIEW control of the mass flow controllers (MFCs), the sapphire window selectivity tests, several PCB revisions with their risk assessments, and finally a multi-lamp configuration intended to cover a wider range of ionization potentials.

Vapor generation

Validating a PID requires reproducible vapor concentrations with controlled dilution. Two rigs were developed for this. Test rig A produces a carrier gas stream of nitrogen or air enriched with vapor from a liquid compound held in a side-arm flask. MFCs then dilute the enriched stream so that the concentration can be set repeatably. The gas line consists of a carrier gas source, one MFC per compound (further compounds are added as parallel channels), the side-arm flask with the target compound, a PID sensor at the flask outlet, and DAQ hardware run from LabVIEW. Tubing and compression fittings connect the gas path, and the MFC control and PID readout signals go to the DAQ and are logged on the acquisition computer. A benchtop build of this layout was the experimental basis for the first PID validation. Repeatability depended heavily on leak-tight connections and on careful handling of the seals, fittings, and flask interface.

Temperature dependent vapor pressures were estimated with the Antoine relation:

$$\log_{10} P = A - \frac{B}{C + T}, \tag{D.1}$$

where P is the vapor pressure, T is the temperature, and A, B and C are compound specific Antoine parameters. These parameters depend on the unit system; the set used

in this appendix is for P in bar and T in K.

Table D.1: Antoine parameters and SVP values for VOCs (as used in this work).

	Antoine parameters			Temperature and SVP values		
Compound	A	B	C	T (K)	T (°C)	SVP (kPa)
Benzo[a]pyrene	Data not available for any temperature range					
Phenanthrene	4.51922	2428.448	-70.95	373.00	99.85	0.03
Naphthalene	4.27117	1831.571	-61.329	353.48	80.33	1.00
p-xylene	4.14553	1474.403	-55.377	293.15	20.00	0.88
Styrene	4.05930	1459.909	-59.551	305.60	32.45	1.34
o-xylene	4.93755	1901.373	-26.268	293.15	20.00	0.65
Ethylbenzene	4.07488	1419.315	-60.539	329.74	56.59	6.35
Toluene	4.23679	1426.448	-45.957	293.15	20.00	2.93
Benzene	4.01814	1203.835	-53.226	293.15	20.00	10.00
Acetaldehyde	3.68639	822.894	-69.899	293.15	20.00	100.10
Formaldehyde	4.28176	959.430	-29.758	250.86	-22.29	87.60

Sapphire window filtering

The selectivity of a fixed energy PID lamp is limited: every mixture component with an ionization potential below the delivered photon energy adds to the measured ion current. A sapphire (Al_2O_3) window was tested as an optical filter in front of a xenon PID lamp, to attenuate the shorter wavelengths and lower the effective photon energy reaching the sampled gas. On a single lamp platform this suppression of the higher energy part of the emission spectrum is what makes threshold comparisons possible.

Sapphire transmission at thicknesses that were not tabulated was estimated with Beer–Lambert attenuation:

$$I(z) = I_0 \exp(-\mu z), \qquad T(z) = \frac{I(z)}{I_0} = \exp(-\mu z), \tag{D.2}$$

where I_0 is the incident intensity, $I(z)$ the intensity transmitted through a thickness z, μ an effective attenuation coefficient, and $T(z)$ the transmission. The coefficient was taken from the slope between the two tabulated points at $z = 2\,\text{mm}$ and $z = 4\,\text{mm}$,

$$\mu = \frac{1}{z_2 - z_1} \ln\left[\frac{T(z_1)}{T(z_2)}\right] = \frac{1}{2} \ln\left(\frac{11}{1.1}\right) \approx 1.15/\text{mm}. \tag{D.3}$$

The two point slope is used here rather than a fit to either point on its own, because the reflection loss at the two sapphire surfaces is independent of thickness and cancels in the ratio, whereas a single point fit folds it into μ and biases the coefficient low. On the

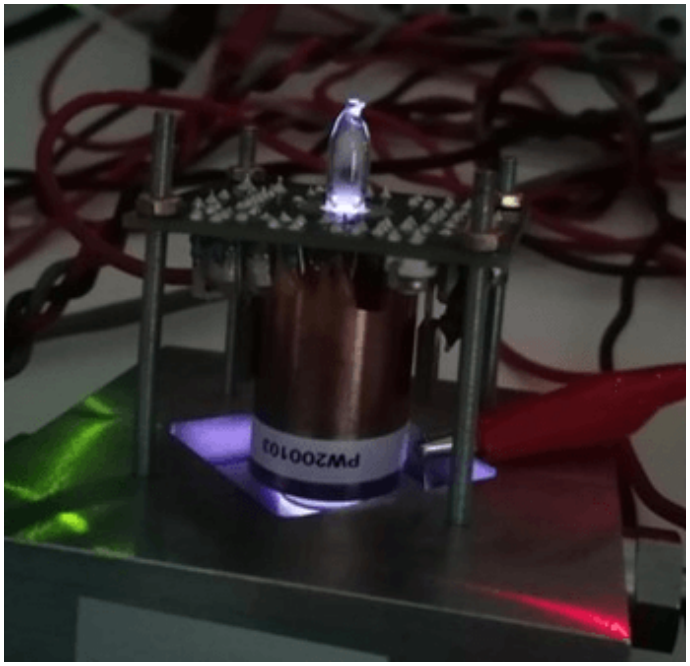


Figure D.1: Krypton photoionization lamp (typical emission lines near 10.0 and 10.6 eV) in our designed chamber.

same basis the interpolated value for the 3 mm window is the geometric mean of the two bracketing transmissions,

$$T(3\,\mathrm{mm}) = \sqrt{T(2\,\mathrm{mm})\,T(4\,\mathrm{mm})} \approx 3.5\%. \tag{D.4}$$

Table D.2: Transmission values for sapphire windows and interpolated estimate at 3 mm thickness.

Thickness z (mm)	Transmission T (%)
2	11
3	≈ 3.5
4	≈ 1.1

Four measurements were run to confirm that the recorded signal came from photoionization. The first had the valve closed with no airflow. The second used airflow at 100 mL/min with no compound. The third added ethanolamine to the side-arm flask at the same flow rate, and the fourth repeated it with the sapphire window in place.

The no-flow measurement sat at about -1.40 V, and this offset was subtracted from the other three traces. Without the sapphire window, ethanolamine gave a ramp like response as the compound ionized and the resulting current was integrated. With the window in place the voltage stayed flat under the same protocol, since sapphire blocks the short wavelength end of the lamp emission.

Acquisition chain

The IVC102U is a precision switched integrator transimpedance amplifier. It turns a small input current into an output voltage by integrating over a set period. The integrating capacitance was $C_{\text{int}} = 100\,\text{pF}$. For an integration time T_{int} and capacitance C_{int}, the output is

$$V_0 = -\frac{I_{\text{in}}\, T_{\text{int}}}{C_{\text{int}}}. \tag{D.5}$$

Aiming for $|V_0| = 5\,\text{V}$ with $T_{\text{int}} = 200\,\text{ms}$ and $C_{\text{int}} = 100\,\text{pF}$ gives

$$|I_{\text{in}}| = \frac{|V_0|\, C_{\text{int}}}{T_{\text{int}}} \approx 2.5\,\text{nA}. \tag{D.6}$$

The IVC102U needs digital timing control for reset, integrate, and hold. The LabVIEW implementation reproduced a datasheet style timing sequence with two digital control signals, S1 and S2, and sampled V_0 during a set acquisition window. A typical cycle used a 200 ms integrate window inside a 400 ms period, followed by reset and acquisition. The integrator was characterized by injecting controlled currents at the nA scale from a source meter and recording the output on an oscilloscope. Saturation time dropped as the injected current grew, which is the behavior an integrator should give. The slope of the output against the injected current was then used as a stability indicator for later PID operation.

For trace analysis work with controlled dilution and LabVIEW integration, the MFC had to run nitrogen at 1 bar over a 0–200 mL/min range at 20 °C, accept 0–5 V analog control from LabVIEW, and fit to 1/4" Swagelok fittings.

Multi-lamp configuration

To detect compounds across a range of ionization potentials, four PID lamps were assembled in series: a xenon 9.6 eV lamp, a Hamamatsu lamp, a xenon 8.4 eV stage (obtained through sapphire window filtering), and a krypton 10.6 eV lamp. The target compound goes into a side-arm flask and is carried through under MFC controlled flow while the lamp series runs and LabVIEW records the voltage responses. Each lamp delivers a different photon energy and therefore imposes a different ionization threshold, so an ionization or no-ionization result maps onto a range of compound ionization potentials.

Table D.3: Five "shadow compounds" selected for SVP validation (Antoine parameters, conditions, SVP, ionization potential and threshold wavelength).

Compound	A	B	C	T (K)	T (°C)	SVP (kPa)	IP (eV)	λ (nm)
Ethanolamine	4.29252	1408.873	-116.093	338.60	65.45	0.91	8.96	138.3752
Isopropanol	4.57795	1221.423	-87.474	395.10	121.95	405.00	10.17	121.9116
Butanol	4.54607	1351.555	-93.340	295.80	22.65	0.74	9.99	124.1083
Acetone	4.42448	1312.253	-32.445	259.16	-13.99	4.33	9.70	127.8186
Methyl salicylate	4.02455	1585.874	-99.173	352.00	78.85	0.56	7.65	162.0708

Operation, maintenance and risks

Several practical problems came up repeatedly during commissioning and measurement. Minor leaks in the gas system destabilized the dilution and the baseline. The lamp behaved intermittently when the electrical contact between the lamp and the PCB contact pads was unreliable. The sapphire windows picked up contamination and needed periodic cleaning, and some of that contamination seemed to trace back to the elastomer O-rings used to seal the component interfaces. Troubleshooting sometimes meant reseating or handling the lamp, so a strict no-touch routine was not always possible and cleaning and inspection became part of normal operation.

The cleaning procedure for PID lamps and sapphire windows was:

1. turn off the power supply and let the lamp cool;
2. gently take the lamp out of the holder or grid;
3. remove dust with a clean, lint-free tissue;
4. if needed, wipe again with a little isopropyl alcohol, using acetone only where it was compatible with the surrounding materials;
5. avoid touching the lamp or window surfaces with bare hands, to keep skin oils off them.

Conductive materials such as copper should be kept off and away from the lamp and window surfaces during operation, since they can create unwanted discharge paths and add contamination.

Most of the risk sat in the electronics. The DAQ hardware, the LabVIEW installation and the PCBs are old enough that any of them can fail mid run, and a damaged supply or cable can short. All of it was inspected on a schedule and replaced rather than repaired. Lamp windows and PCB surfaces also collect contamination that degrades performance, so cleaning was part of the same routine.

On the gas side the concern was a sudden nitrogen release, which can lift a stopper and empty the flask. Valves were opened slowly, stoppers were clamped, and the flask vented through a relief line so that it could not over pressurize. Spilled liquid is a slip hazard as well as a VOC source, so spills were marked and cleaned up at once.

Useful next steps include software updates, replacing the DAQ hardware after LabVIEW updates, working out the limits on multi-PID acquisition, keeping a record of lamp lifetime and usage, and pinning down the concentration ranges the laboratory can reach, including how they compare with toxicity relevant ranges.

One practical constraint during this work was the changing availability of certain PID lamp types and form factors while Heraeus Noblelight operations moved under Excelitas Technologies, which affected procurement and replacement options.

At an intermediate stage of the project we considered a dedicated enclosure to transport the prototype and operate it on site. A preliminary AutoCAD concept was drawn up, with a multilayer box, integrated wheels, and a cooling stage to keep the instrument stable during transport and measurement. Building it would have required a full mechanical design and fabrication effort beyond the timeline of the project, so the in-house enclosure was not pursued. Transport and environmental control were instead handed to an external company specialized in such systems, which kept the optical and sensing work as the priority.

D.2 Simulations

D.2.1 Simulation setup

All simulations were run with the finite-difference time-domain (FDTD) solver in Ansys Lumerical. The nanorectangle was modeled as a free-standing structure in vacuum, illuminated by a broadband plane wave source, and the simulation region was terminated with perfectly matched layers on all boundaries. The automatic non-uniform mesh of the solver was used, with the mesh refined over the structure until the resonance positions no longer shifted appreciably. Optical constants were taken from the Lumerical material database and from tabulated n and k values imported as sampled data.

D.2.2 Field behavior in metallic and dielectric nanorectangles

To study the electromagnetic field behavior in nanorectangles, we ran simulations on four materials: silver (Ag), gold (Au), a tabulated low index TiO_2 dataset, and amorphous TiO_2. The nanorectangle dimensions were fixed at 200 nm long, 150 nm wide, and 100 nm thick, and the field was simulated across a range of wavelengths. One field monitor sat

5 nm from the surface and a second one sat at the center of the nanorectangle.

The incident source was a plane wave propagating perpendicular to the nanorectangle. The electric field (**E**) was polarized along the length of the rectangle, which excites polarization dependent phenomena such as anapole states. The wave vector (**k**) stayed orthogonal to the surface of the rectangle, and the magnetic field (**H**) was perpendicular to both **E** and **k** (Fig. D.2). This geometry keeps interference effects small and makes the resonance and field distribution features easier to read.

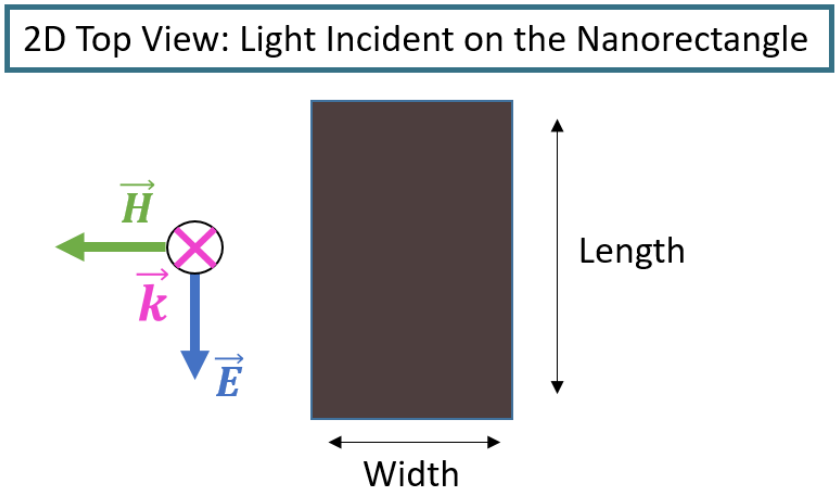


Figure D.2: 2D top view illustrating the interaction of light with the rectangular nanostructure. The plane wave propagates with the wave vector (**k**) along the z-axis (normal to the surface). The electric field (**E**) is aligned parallel to the rectangle's length (x-axis), while the magnetic field (**H**) is aligned parallel to its width (y-axis). Both (**E**) and (**H**) are orthogonal to the propagation vector (**k**).

Fig. D.3 shows the dispersion profiles of the materials and how light interacts with the metallic and the dielectric ones. The optical data for Ag and Au came from the Lumerical material database.

The refractive index (n) and extinction coefficient (k) of the low index TiO_2 dataset give $n = 2.4$ and $k = 0$ at a wavelength of 370 nm. Three TiO_2 datasets are used in this work, and their refractive indices order as anatase > amorphous > low index tabulated data. The amorphous TiO_2 of Jolivet et al. [3] therefore has a considerably higher index than the tabulated dataset used here, while remaining below anatase over the range of interest.

Aspect ratio has a known effect on scattering and resonance. Structures with a low aspect ratio scatter much like spheres, with dominant dipolar resonances. At higher aspect ratios the resonances split into longitudinal modes along the length of the nanostructure and transverse modes across its width. This split produces more complex scattering, with a field localization that changes the spectral response.

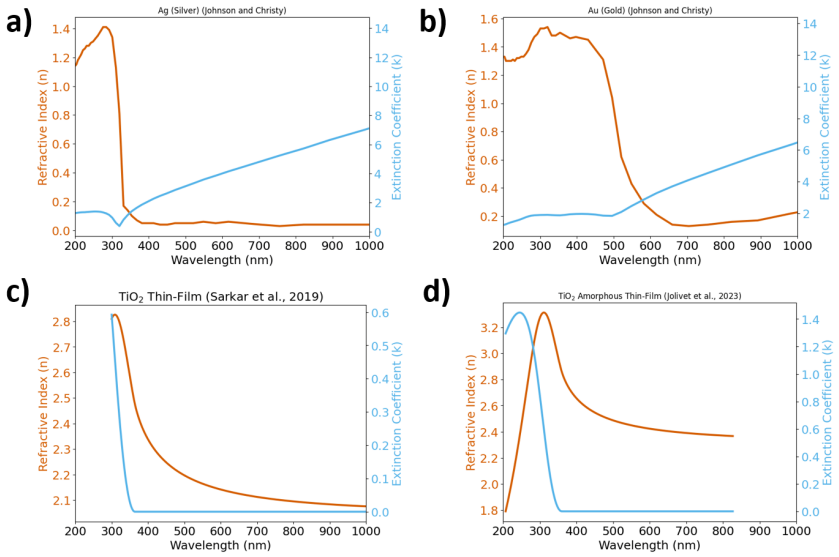


Figure D.3: Dispersion profile of **a)** Ag, **b)** Au, **c)** low index TiO_2 and **d)** amorphous TiO_2 [3] refractive indices (n) and extinctions (k).

Transverse mode resonances are especially sensitive to the width of the nanostructure. When the resonance positions shift strongly with width but barely move with thickness or length, the oscillation is a transverse mode. This separation matters for understanding the mode dynamics and for designing structures with a target optical response.

At dimensions well below the wavelength, the nanorectangles scatter weakly and absorb most of the light, acting as oscillating dipoles with nearly uniform field distributions. The absorption cross-section scales with particle volume and scattering scales with the square of the volume, so absorption dominates.

As the nanorectangle grows, higher order multipole contributions appear and the scattering moves from the Rayleigh regime into the Mie regime. In the Mie regime the scattering patterns get more complex, with several resonances tied to distinct oscillation modes. Scattering efficiency rises, and under some conditions the radiation patterns become anisotropic [218].

Electric field intensity distributions were evaluated at 370, 450, and 520 nm to look at the external field localization and the resonance characteristics. Figs. D.4, D.5, and D.6 show these distributions with a monitor 5 nm outside the nanorectangle surface. The behavior depends on the material.

For metallic structures like Ag and Au, the field localizes strongly at the corners of the nanorectangles. This comes from plasmonic effects, and Ag confines the field more tightly because it has lower intrinsic damping. Dielectric materials like TiO_2 give a more uniform field distribution, which is the signature of Mie type resonances. Amorphous TiO_2 gives

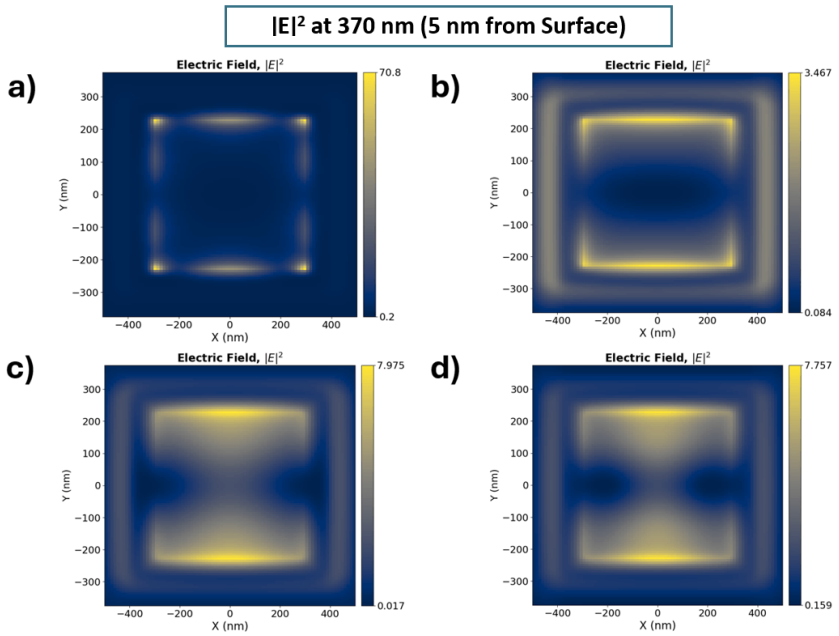


Figure D.4: Electric field intensity distribution at 370 nm λ. **a)** Ag, **b)** Au, **c)** low index TiO_2, **d)** TiO_2 amorphous. The monitor is positioned 5 nm from the nanorectangle surface, and the fields are simulated under plane wave excitation with the electric field (**E**) aligned along the rectangle's length.

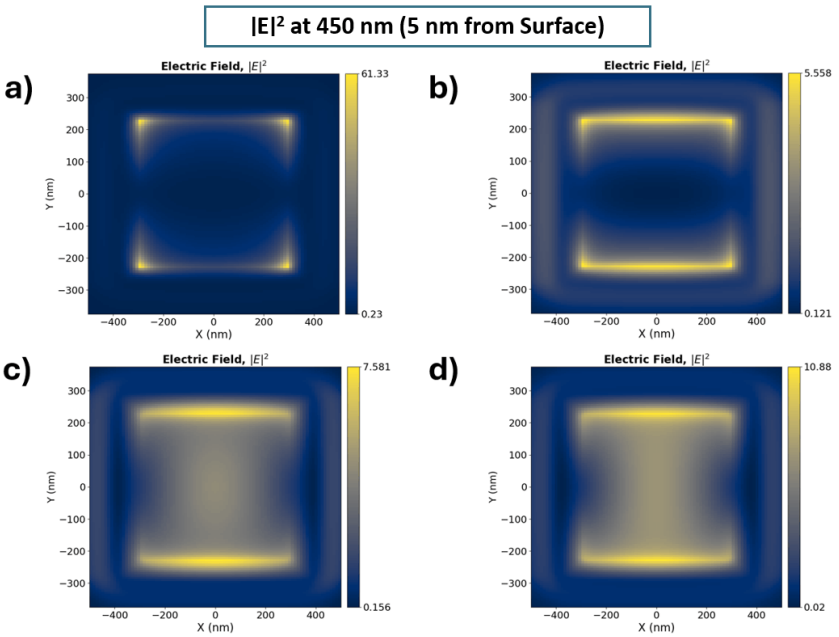


Figure D.5: Electric field intensity distribution at 450 nm λ. **a)** Ag, **b)** Au, **c)** low index TiO_2, **d)** TiO_2 amorphous. The monitor is positioned 5 nm from the nanorectangle surface, and the fields are simulated under plane wave excitation with the electric field (**E**) aligned along the rectangle's length.

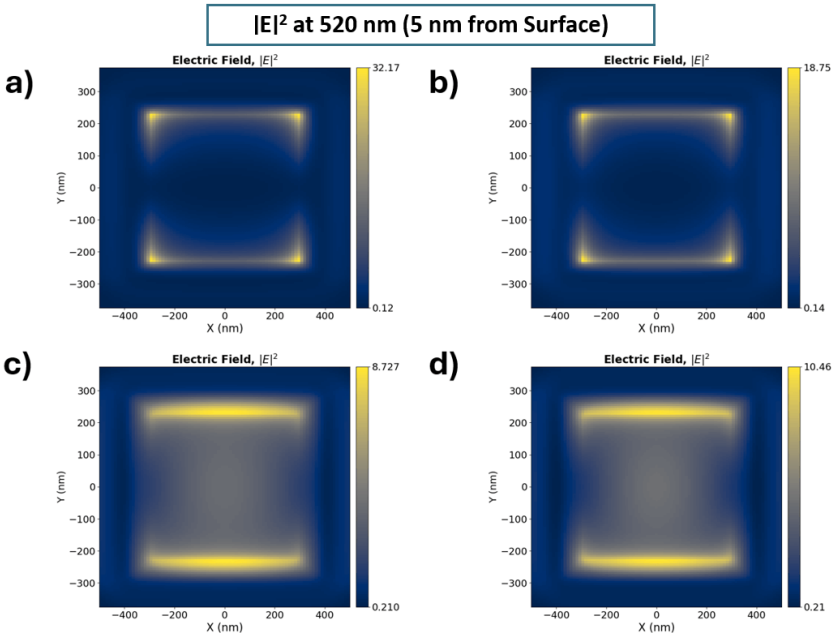


Figure D.6: Electric field intensity distribution at 520 nm λ. **a)** Ag, **b)** Au, **c)** low index TiO_2, **d)** TiO_2 amorphous. The monitor is positioned 5 nm from the nanorectangle surface, and the fields are simulated under plane wave excitation with the electric field (**E**) aligned along the rectangle's length.

the smoothest and least localized fields, and this reflects its weaker scattering compared with the crystalline phase.

Figs. D.7, D.8, and D.9 show the field distributions inside the nanorectangle, with the monitor at its center. These maps bring out the confined resonances in the TiO_2 materials and the contrast between the plasmonic and the dielectric systems. In amorphous TiO_2 the internal field is evenly spread, which supports the nonradiating anapole states. The material properties govern the scattering and absorption behavior, and here they favor the localized resonances of interest.

The field distributions show the basic difference between the plasmonic materials (silver, gold) and the dielectric one (TiO_2). In silver and gold the strong corner field enhancements come from surface plasmon resonances, which do not favor anapole excitation. In TiO_2 the field spreads through the structure, which confines the resonance inside it and allows the nonradiating anapole modes.

The anapole state behaves differently during transient processes, where it can switch between a non-radiative dark state sustained by continuous wave irradiation and a radiative mode that interacts with its surroundings. An anapole cannot exist as a stand-alone excitation [111], so it depends on continuous external irradiation.

A near-zero phase relationship, close to zero radians, goes together with the cancellation

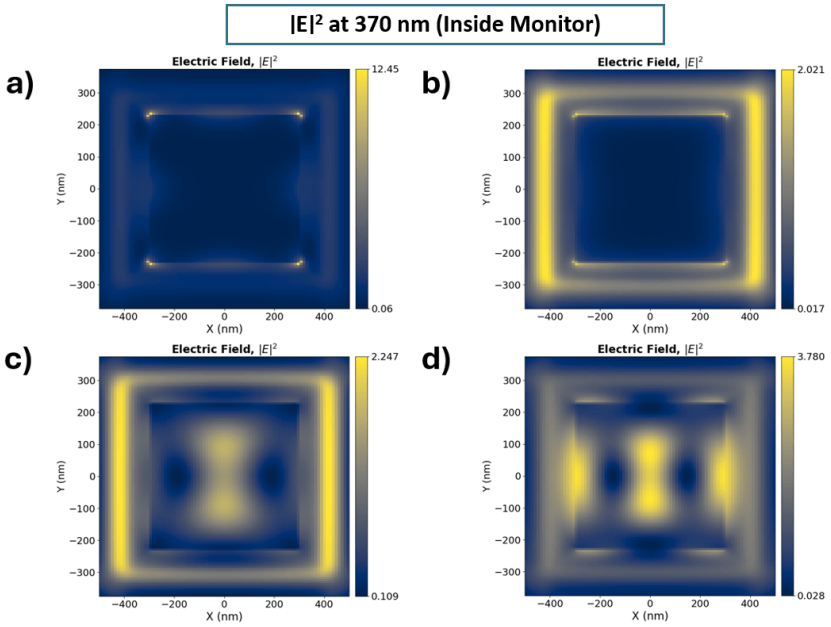


Figure D.7: Electric field intensity distribution at 370 nm λ. **a)** Ag, **b)** Au, **c)** low index TiO_2, **d)** TiO_2 amorphous. The monitor is positioned at the geometric center of the nanorectangle. The fields are simulated under plane wave excitation with the electric field (**E**) aligned along the rectangle's length.

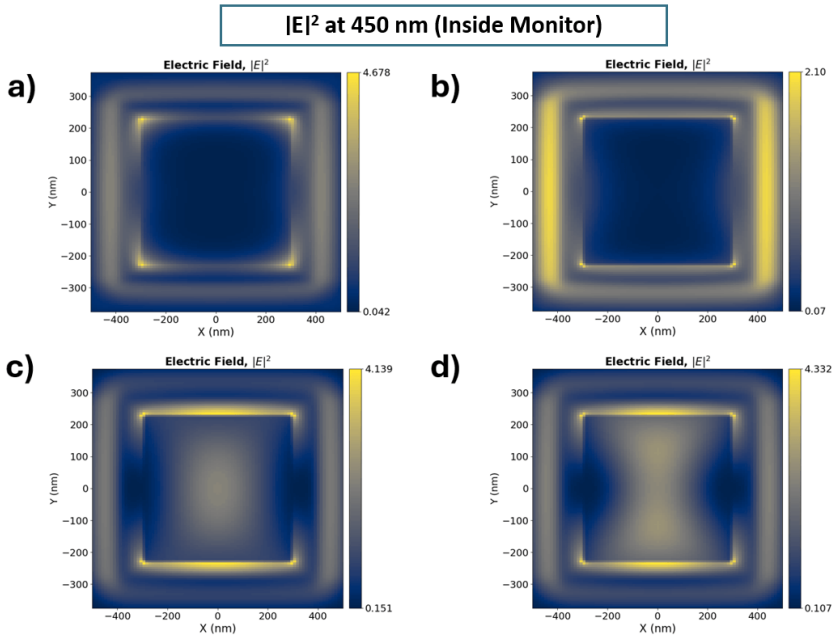


Figure D.8: Electric field intensity distribution at 450 nm λ. **a)** Ag, **b)** Au, **c)** low index TiO_2, **d)** TiO_2 amorphous. The monitor is positioned at the geometric center of the nanorectangle. The fields are simulated under plane wave excitation with the electric field (**E**) aligned along the rectangle's length.

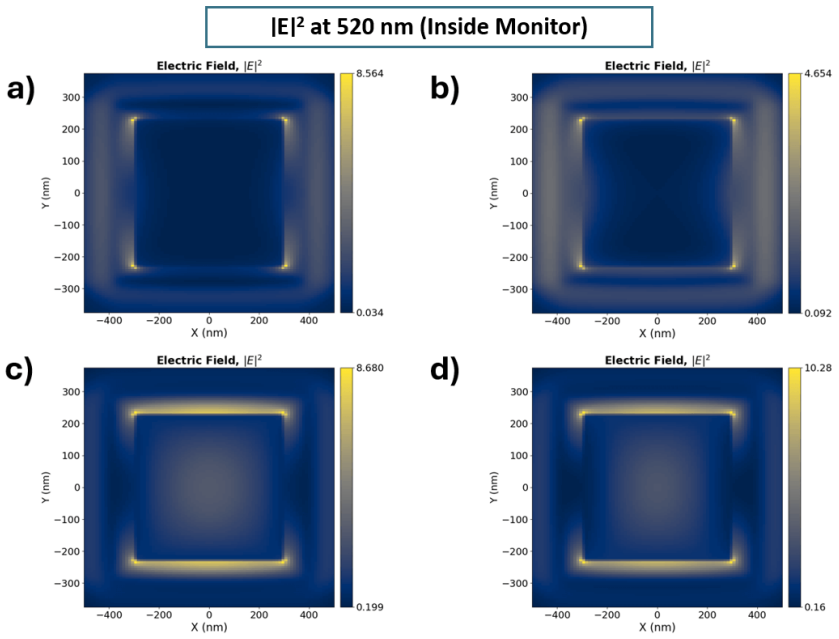


Figure D.9: Electric field intensity distribution at 520 nm λ. **a)** Ag, **b)** Au, **c)** low index TiO_2, **d)** TiO_2 amorphous. The monitor is positioned at the geometric center of the nanorectangle. The fields are simulated under plane wave excitation with the electric field (**E**) aligned along the rectangle's length.

conditions the anapole state needs. This phase matching reduces the radiation in certain directions, so the total scattering drops measurably.

D.2.3 Anatase and amorphous TiO_2

We compare anatase and amorphous TiO_2 across their optical properties, scattering behavior, and field distributions, shown in Figs. 7.2 and D.10. Both have suitable optical properties in the ultraviolet (UV), but their differences affect how nanostructures for anapole excitation are designed and how they perform.

Fig. D.10 shows the refractive index (n) and extinction coefficient (k) for the two phases. Anatase has a consistently higher refractive index than amorphous TiO_2 because it is denser and crystalline. Amorphous TiO_2 has a wider bandgap (3.3–3.5 eV against 3.2 eV for anatase) and works at lower deposition temperatures, which makes it easier to fabricate [3].

The scattering cross-section and the multipolar decomposition bring out further differences. In Fig. 7.2 **c)** the scattering dip for amorphous TiO_2, which is the mark of the anapole state, sits at 366.5 nm. For anatase TiO_2 the corresponding dip is at 373.3 nm, in Fig. D.11 **a)**. This wavelength shift comes from the different refractive index and extinction coefficient of the two phases, which changes their scattering and absorption.

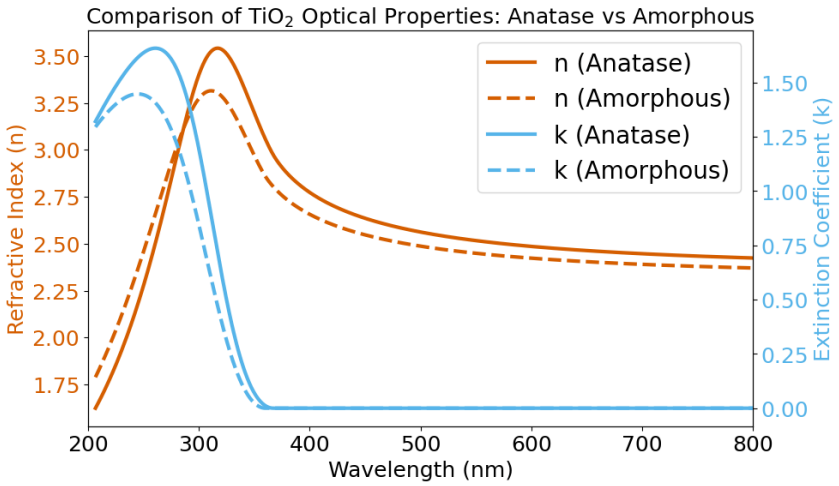


Figure D.10: Comparison of optical properties of TiO_2 in anatase and amorphous phases. The refractive index n (orange) and extinction coefficient k (blue) as functions of wavelength. Solid lines represent the anatase phase, while dashed lines represent the amorphous phase. The refractive index (n) and extinction coefficient (k) are higher for the anatase phase at shorter wavelengths, while differences diminish as the wavelength increases.

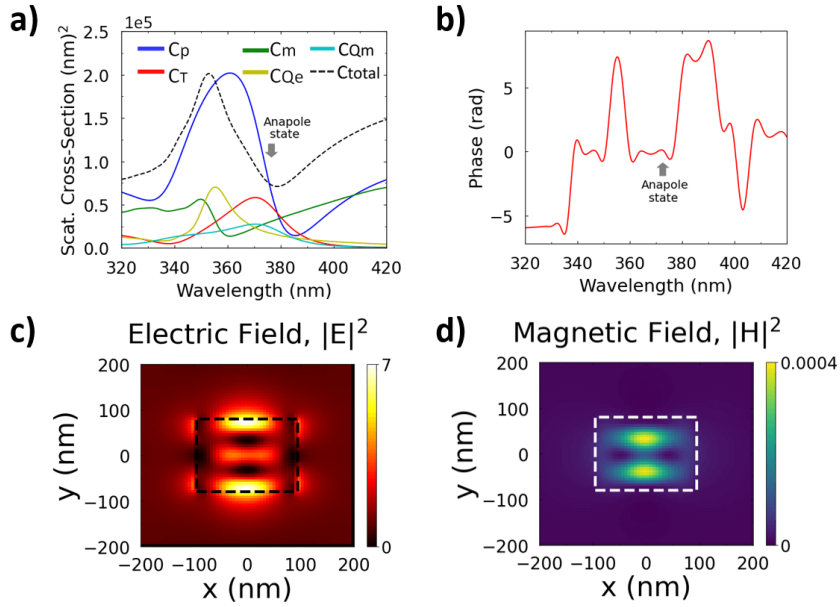


Figure D.11: Study for anatase TiO_2. **a)** Contributions of various multipole moments: C_p (blue), C_T (red), C_m (green), C_{Qe} (yellow), C_{Qm} (cyan blue) and total contribution C_{total} (black). **b)** Phase of electric dipole moment (blue) and toroidal dipole moment (red). **c)** Electric and **d)** magnetic field intensities in the free-standing nano-rectangle structure. The electric and magnetic fields are normalized relative to the source.

The two cases differ mainly in the phase behavior and how it relates to the optical properties of the material. Anatase, being crystalline, has a higher refractive index than the amorphous phase. That higher index gives a stronger light-matter interaction, which pushes the anapole state to a longer wavelength in anatase, consistent with the 6.8 nm offset between the two scattering dips. The crystalline structure also shapes the dispersion of the optical properties, so the phase modulates more sharply than in the smoother amorphous phase [219].

These results show how crystallinity and optical properties such as refractive index and dispersion affect anapole excitation. The sharper, more oscillatory phase in anatase follows from its ordered lattice and the anisotropic optical response that comes with it [220], whereas the amorphous material is more isotropic and less ordered.

E Statutory Declaration

I hereby declare that this thesis is my own work and that, to the best of my knowledge and belief, it contains no material previously published or written by another person, nor material which to a substantial extent has been accepted for the award of any other degree or diploma at any university or other institution of tertiary education, except where due acknowledgement is made in the text.

I further declare that I have adhered to the principles of good scientific practice and that all sources and aids used have been properly cited.

Place, Date	Signature

F Acknowledgements

To my family. To my supervisors, Dr. Pieter de Beule and Prof. Thomas Huser. To everyone who supported me from afar: INL (Portugal), Tyndall Institute (Ireland), University of Bath (England), University of Lille (France) and Bielefeld University (Germany), friends in Spain, and to Michael Scalora (U.S. Army DEVCOM Aviation & Missile Center) whose generous advice helped me in the final stage of this work.

Thank you.